

2021 ESC Guidelines on cardiovascular disease prevention in clinical practice

Developed by the Task Force for cardiovascular disease prevention in clinical practice with representatives of the European Society of Cardiology and 12 medical societies

With the special contribution of the European Association of Preventive Cardiology (EAPC)

Authors/Task Force Members: Frank L.J. Visseren* (Chairperson) (Netherlands), François Mach* (Chairperson) (Switzerland), Yvo M. Smulders[†] (Task Force Coordinator) (Netherlands), David Carballo[†] (Task Force Coordinator) (Switzerland), Konstantinos C. Koskinas (Switzerland), Maria Bäck (Sweden), Athanase Benetos⁸ (France), Alessandro Biffi^{7,10} (Italy), José-Manuel Boavida⁹ (Portugal), Davide Capodanno (Italy), Bernard Cosyns (Belgium), Carolyn Crawford (Northern Ireland), Constantinos H. Davos (Greece), Ileana Desormais (France), Emanuele Di Angelantonio (United Kingdom), Oscar H. Franco (Switzerland), Sigrun Halvorsen (Norway), F. D. Richard Hobbs¹³ (United Kingdom), Monika Hollander (Netherlands), Ewa A. Jankowska (Poland), Matthias Michal¹¹ (Germany), Simona Sacco⁶ (Italy), Naveed Sattar (United Kingdom), Lale Tokgozoglu² (Turkey), Serena Tonstad (Norway), Konstantinos P. Tsioufis⁵ (Greece), Ineke van Dis³ (Netherlands), Isabelle C. van Gelder (Netherlands), Christoph Wanner⁴ (Germany), Bryan Williams (United Kingdom), ESC Scientific Document Group

* Corresponding authors: The two chairpersons contributed equally to the document. Frank Visseren, Department of Vascular Medicine, University Medical Center Utrecht, Heidelberglaan 100, 3584 CX Utrecht, Netherlands. Tel: +31 (0)88 7557324, E-mail: F.L.J.Visseren@umcutrecht.nl. François Mach, Cardiology Department, Geneva University Hospital, Perret-Gentil 4, 1211 Geneva, Switzerland. Tel: +41 (0)22 372 71 92, E-mail: francois.mach@hcuge.ch. [†] The two task force coordinators contributed equally to the document.

Author/Task Force Member affiliations: listed in Author information.

ESC Clinical Practice Guidelines Committee (CPG): listed in the Appendix.

ESC subspecialty communities having participated in the development of this document. Associations: Association of Cardiovascular Nursing & Allied Professions (ACNAP), European Association of Cardiovascular Imaging (EACVI), European Association of Preventive Cardiology (EAPC), European Heart Rhythm Association (EHRA), Heart Failure Association (HFA). **Councils:** Council on Valvular Heart Disease. **Working Groups:** Aorta and Peripheral Vascular Diseases, Atherosclerosis and Vascular Biology, Cardiovascular Pharmacotherapy.

Patient Forum

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Abbreviations and acronyms

%HR _{max}	Percentage of maximum heart rate
ABC	Atrial fibrillation Better Care
ABI	Ankle brachial index
ABPM	Ambulatory blood pressure monitoring
ACCORD	Action to Control Cardiovascular Risk in Diabetes
ACE	Angiotensin-converting enzyme
ACR	Albumin-to-creatinine ratio
ACS	Acute coronary syndromes
ADA	American Diabetes Association
ADVANCE	Action in Diabetes and Vascular Disease: preterAx and diamicroN-MR Controlled Evaluation
AF	Atrial fibrillation
AMI	Acute myocardial infarction
ARB	Angiotensin receptor blocker
ARNI	Angiotensin receptor neprilysin inhibitor
ASCEND	A Study of Cardiovascular Events in Diabetes
ASCVD	Atherosclerotic cardiovascular disease
<i>b.i.d.</i>	Bis in die (twice a day)
BMI	Body mass index
BP	Blood pressure
b.p.m.	Beats per minute
CAC	Coronary artery calcium
CAD	Coronary artery disease
CANTOS	Canakinumab Antiinflammatory Thrombosis Outcome Study
CCB	Calcium channel blocker
CCS	Chronic coronary syndromes

CCTA	Contrast computed tomography angiography	HR	Hazard ratio
CHD	Coronary heart disease	IL	Interleukin
CI	Confidence interval	IMPROVE-IT	Improved Reduction of Outcomes: Vytorin Efficacy International Trial
CKD	Chronic kidney disease	IMT	Intima-media thickness
CKD-EPI	Chronic Kidney Disease Epidemiology	INVEST	INternational VErapamil-SR/Trandolapril STudy
COLCOT	Colchicine Cardiovascular Outcomes Trial	LDL	Low-density lipoprotein
COMPASS	Cardiovascular Outcomes for People Using Anticoagulation Strategies	LDL-C	Low-density lipoprotein cholesterol
COPD	Chronic obstructive pulmonary disease	LDLR	Low-density lipoprotein receptor
CR	Cardiac rehabilitation	LEAD	Lower extremity artery disease
CTA	Computed tomography angiography	LIFE-CVD	LIFETIME-perspective CardioVascular Disease
CV	Cardiovascular	LoDoCo	Low-dose colchicine
CVD	Cardiovascular disease	LV	Left ventricular/ventricle
DAPA-CKD	Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease	LVEF	Left ventricular ejection fraction
DAPT	Dual antiplatelet therapy	MACE	Major adverse cardiovascular events
DASH	Dietary Approaches to Stop Hypertension	MET	Metabolic equivalent of task
DBP	Diastolic blood pressure	mHealth	Mobile device-based healthcare
DCCT	Diabetes Control and Complications Trial	MRA	Mineralocorticoid receptor antagonist
DIAL	Diabetes lifetime-perspective prediction	MUFA	Monounsaturated fatty acid
DM	Diabetes mellitus	N/A	Not applicable
e-cigarettes	Electronic cigarettes	NAFLD	Non-alcoholic fatty liver disease
EAPC	European Association of Preventive Cardiology	NRT	Nicotine-replacement therapy
EAS	European Atherosclerosis Society	NYHA	New York Heart Association
EASD	European Association for the Study of Diabetes	<i>o.d.</i>	<i>Omni die</i> (once a day)
EBCR	Exercise-based cardiac rehabilitation	OARS	Open-ended questions, Affirmation, Reflecting listening, and Summarizing
ECG	Electrocardiographic/electrocardiogram	OR	Odds ratio
ED	Erectile dysfunction	OSA	Obstructive sleep apnoea
eGFR	Estimated glomerular filtration rate	PA	Physical activity
EORP	EURObservational Research Programme	PAD	Peripheral artery disease
EPIC	European Prospective Investigation into Cancer and Nutrition	PAP	Positive airway pressure
ESC	European Society of Cardiology	PCI	Percutaneous coronary intervention
ESH	European Society of Hypertension	PCSK9	Proprotein convertase subtilisin/kexin type 9
ESVS	European Society for Vascular Surgery	PM	Particulate matter
EU	European Union	PM _{2.5}	Particulate matter <2.5 µm
EUROASPIRE	European Action on Secondary and Primary Prevention by Intervention to Reduce Events	PUFA	Polyunsaturated fatty acid
EuroHeart	European Unified Registries On Heart Care Evaluation and Randomized Trials	QI	Quality indicator
EXPERT	EXercise Prescription in Everyday practice & Rehabilitation Training	RAAS	Renin-angiotensin-aldosterone system
FEV1	Forced expiratory volume in 1 second	RAS	Renin-angiotensin system
FH	Familial hypercholesterolaemia	RCT	Randomized controlled trial
FITT	Frequency, intensity, time duration, and type of exercise	REDUCE-IT	Reduction of Cardiovascular Events with Icosapent Ethyl – Intervention Trial
GFR	Glomerular filtration rate	REWIND	Researching Cardiovascular Events With a Weekly Incretin in Diabetes
GLP-1RA	Glucagon-like peptide-1 receptor agonist	RPE	Rating of perceived exertion
HbA1c	Glycated haemoglobin	RR	Relative risk
HBPM	Home blood pressure monitoring	SAVOR-TIMI 53	Saxagliptin Assessment of Vascular Outcomes Recorded in Patients with Diabetes Mellitus Thrombolysis in Myocardial Infarction
HDL-C	High-density lipoprotein cholesterol	SBP	Systolic blood pressure
HF	Heart failure	SCORE	Systemic Coronary Risk Estimation
HFpEF	Heart failure with preserved ejection fraction	SCORE2	Systemic Coronary Risk Estimation 2
HFrEF	Heart failure with reduced ejection fraction	SCORE2-OP	Systematic Coronary Risk Estimation 2-Older Persons
HIV	Human immunodeficiency virus	SCOT-HEART	Scottish Computed Tomography of the Heart
HMOD	Hypertension-mediated organ damage	SGLT2	Sodium-glucose cotransporter 2

SHARP	Study of Heart and Renal Protection
SMART	Secondary Manifestations of Arterial Disease
SMART	Specific, Measurable, Achievable, Realistic, Timely
SMART-REACH	Secondary Manifestations of Arterial Disease-Reduction of Atherothrombosis for Continued Health
SNRI	Serotonin-noradrenaline reuptake inhibitor
SPRINT	Systolic Blood Pressure Intervention Trial
SSRI	Selective serotonin reuptake inhibitor
STAREE	STAtin Therapy for Reducing Events in the Elderly
STRENGTH	Long-Term Outcomes Study to Assess Statin Residual Risk with Epanova in High Cardiovascular Risk Patients with Hypertriglyceridemia
SUPRIM	Secondary Prevention in Uppsala Primary Health Care project
SWITCHD	Stockholm Women's Intervention Trial for Coronary Heart Disease
TIA	Transient ischaemic attack
TNF	Tumour necrosis factor
TOD	Target organ damage
UK	United Kingdom
UKPDS	UK Prospective Diabetes Study
VADT	Veterans Affairs Diabetes Trial
VITAL	Vitamin D and Omega-3 Trial
VO ₂	Oxygen consumption
WHO	World Health Organization

1. Preamble

Guidelines summarize and evaluate available evidence with the aim of assisting health professionals in proposing the best management strategies for an individual patient with a given condition. Guidelines and their recommendations should facilitate decision making of health professionals in their daily practice. However, the final decisions concerning an individual patient must be made by the responsible health professional(s) in consultation with the patient and caregiver as appropriate.

A great number of guidelines have been issued in recent years by the European Society of Cardiology (ESC), as well as by other societies and organizations. Because of their impact on clinical practice, quality criteria for the development of guidelines have been established in order to make all decisions transparent to the user. The recommendations for formulating and issuing ESC Guidelines can be found on the ESC website (<https://www.escardio.org/Guidelines>). The ESC Guidelines represent the official position of the ESC on a given topic and are regularly updated.

In addition to the publication of Clinical Practice Guidelines, the ESC carries out the EURObservational Research Programme of international registries of cardiovascular diseases and interventions which are essential to assess diagnostic/therapeutic processes, use of resources and adherence to guidelines. These registries aim at

providing a better understanding of medical practice in Europe and around the world, based on high-quality data collected during routine clinical practice.

Furthermore, the ESC has developed and embedded in this document a set of quality indicators (QIs), which are tools to evaluate the level of implementation of the guidelines and may be used by the ESC, hospitals, healthcare providers and professionals to measure clinical practice as well as used in educational programmes, alongside the key messages from the guidelines, to improve quality of care and clinical outcomes.

The Members of this Task Force were selected by the ESC, including representation from its relevant ESC sub-specialty groups, in order to represent professionals involved with the medical care of patients with this pathology. Selected experts in the field undertook a comprehensive review of the published evidence for management of a given condition according to ESC Clinical Practice Guidelines Committee (CPG) policy. A critical evaluation of diagnostic and therapeutic procedures was performed, including assessment of the risk–benefit ratio. The level of evidence and the strength of the recommendation of particular management options were weighed and graded according to pre-defined scales, as outlined below.

The experts of the writing and reviewing panels provided declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. Their declarations of interest were reviewed according to the ESC declaration of interest rules and can be found on the ESC website (<http://www.escardio.org/guidelines>) and have been compiled in a report and published in a supplementary document simultaneously to the guidelines.

This process ensures transparency and prevents potential biases in the development and review processes. Any changes in declarations of interest that arise during the writing period were notified to the ESC and updated. The Task Force received its entire financial support from the ESC without any involvement from the healthcare industry.

The ESC CPG supervises and coordinates the preparation of new guidelines. The Committee is also responsible for the endorsement process of these guidelines. The ESC Guidelines undergo extensive review by the CPG and external experts. After appropriate revisions the guidelines are signed-off by all the experts involved in the Task Force. The finalized document is signed-off by the CPG for publication in the European Heart Journal. The guidelines were developed after careful consideration of the scientific and medical knowledge and the evidence available at the time of their dating.

The task of developing ESC Guidelines also includes the creation of educational tools and implementation programmes for the recommendations including condensed pocket guideline versions, summary slides, summary cards for non-specialists and an electronic version for digital applications (smartphones, etc.). These versions are abridged and thus, for more detailed information, the user should always access to the full text version of the guidelines, which is freely available via the ESC website and hosted on the EHJ website. The National Cardiac Societies of the ESC are encouraged to endorse, adopt, translate and implement all ESC Guidelines. Implementation

Table 1 Classes of recommendations

Classes of recommendations

Definition		Wording to use
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/ efficacy of the given treatment or procedure.	
Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

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Table 2 Levels of evidence

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

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programmes are needed because it has been shown that the outcome of disease may be favourably influenced by the thorough application of clinical recommendations.

Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgment, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies. However, the ESC Guidelines do not override in any way whatsoever the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient or the patient's caregiver where appropriate and/or

necessary. It is also the health professional's responsibility to verify the rules and regulations applicable in each country to drugs and devices at the time of prescription.

2. Introduction

Atherosclerotic cardiovascular (CV) disease (ASCVD) incidence and mortality rates are declining in many countries in Europe, but it is still a major cause of morbidity and mortality. Over the past few decades, major ASCVD risk factors have been identified. The most

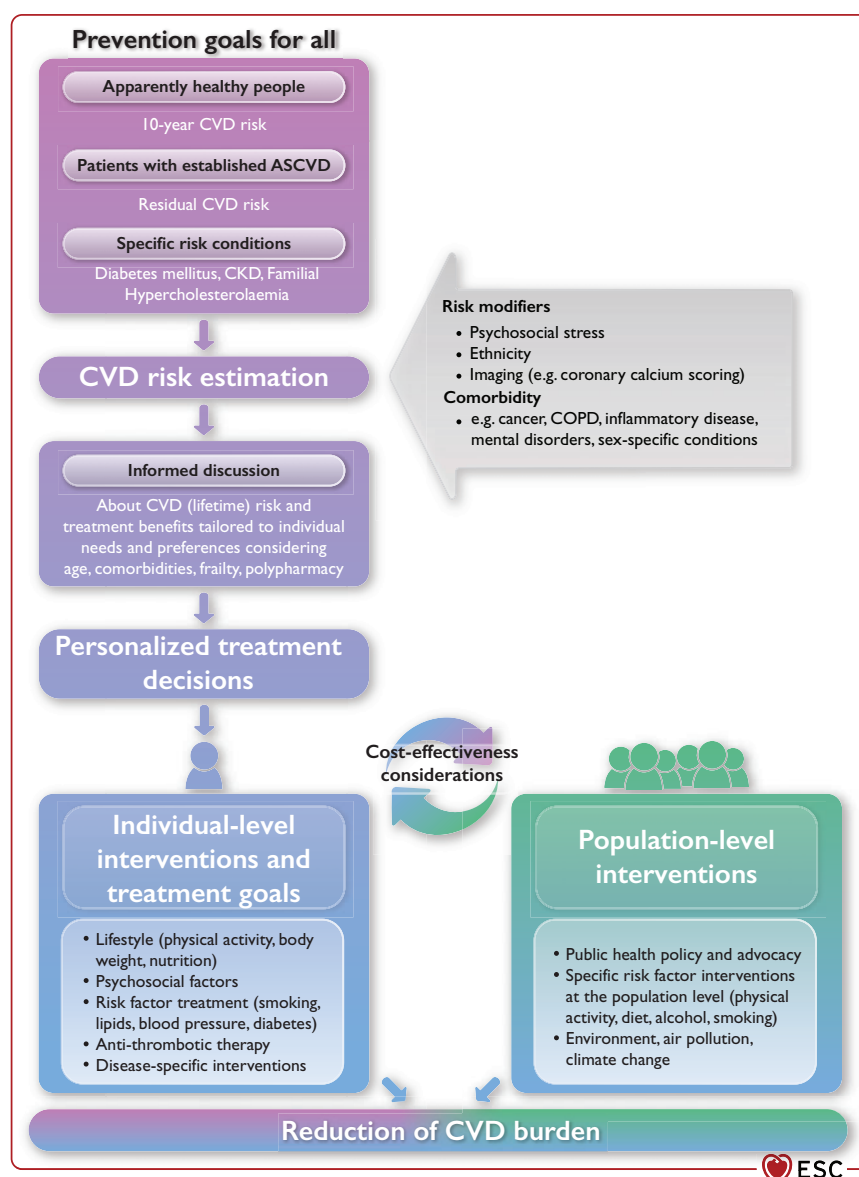


Figure 1 Central Illustration. ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; CVD = cardiovascular disease

important way to prevent ASCVD is to promote a healthy lifestyle throughout life, especially not smoking. Effective and safe risk factor treatments have been developed, and most drugs are now generic and available at low costs. Nevertheless, the prevalence of unhealthy lifestyle is still high, and ASCVD risk factors are often poorly treated, even in patients considered to be at high (residual) CVD risk.¹ Prevention of CV events by reducing CVD risk is the topic of these guidelines.

2.1. Definition and rationale

The present guidelines have been developed to support healthcare professionals in their efforts to reduce the burden of ASCVD in both individual patients, as well as at a population level. The previous European Guidelines on CVD prevention in clinical practice were published in 2016.² Recent developments in prediction of

cardiovascular disease (CVD) risk and treatment benefit, as well as novel treatments and treatment goals, necessitated new, up-to-date guidelines. The current guidelines on CVD prevention in clinical practice concentrate principally but not exclusively on the risk factors, risk classification, and prevention of ASCVD.

The current guidelines provide recommendations on ASCVD prevention to support shared decision-making by the patient and their healthcare professional based on individual patient characteristics. Special considerations have been given to differences in age, sex and gender, life expectancy, risk factor profiles, ethnic, and geographic differences. Estimating CVD risk not only in apparently healthy subjects, but also in older persons and in patients with established ASCVD or diabetes mellitus (DM), provides information for tailored intervention on an individual level. Treatment goals can be individualized in a stepwise approach. 'Residual' CVD risk is defined as the risk

estimated after initial lifestyle changes and risk factor treatment, and is mostly used in patients with established ASCVD. For younger apparently healthy subjects, lifetime CVD risk estimates are available to support treatment decisions, replacing 10-year risk algorithms that consistently estimate low 10-year risk even in the presence of high risk factor levels. In an ageing population, treatment decisions require a specific CVD risk score that takes competing non-CVD risk into account, as well as specific low-density lipoprotein cholesterol (LDL-C) and blood pressure (BP) treatment considerations. Estimating lifetime benefit in individual patients of smoking cessation, LDL-C lowering, and BP lowering provides opportunities to communicate benefit of treatment in an easy-to-understand way. Personalized treatment decisions using CVD risk estimations and a stepwise approach to treatment is more complex than a more general one-size-fits-all prevention strategy, but reflects the diversity in patients and patient characteristics in clinical practice.

Regarding LDL-C, BP, and glycaemic control in patients with DM, goals and targets remain as recommended in recent European Society of Cardiology (ESC) Guidelines.^{3–5} These prevention guidelines propose a new, stepwise approach to treatment intensification as a tool to help physicians and patients pursue these targets in a way that fits patient profile and preferences. Of note, however, new evidence and/or new consensus may have resulted in some differences with these recent domain-specific ESC Guidelines. New evidence on antithrombotic treatment regimens for ASCVD prevention is also presented. Sex-specific aspects are included.

ASCVD prevention needs an integrated, interdisciplinary approach including input from several disciplines and areas of expertise. We must work together in a patient- and family-centred way to address each of the core components of prevention and rehabilitation, including lifestyle modification, psychosocial factors, risk factor treatment, and social determinants (Central Illustration).

2.2. Development

The Task Force chairs and members were appointed by the ESC Clinical Practice Guidelines Committee (CPG). Each member of the Task Force was assigned specific writing tasks, which were reviewed by other (sub)section writers, the section coordinators, and the chairs. The text was developed over 11 months, during which the Task Force members met collectively on three occasions and corresponded intensively between meetings. The review panel consisted of experts selected by all the scientific societies that were involved in the development of these guidelines, not only the ESC.

2.3. Cost-effectiveness

The Task Force acknowledge the fact that healthcare budgets are, in many circumstances, limited and thus that certain recommendations and goals may not always be attainable. However, the current guidelines do not provide cost-effectiveness analyses. Large national and regional differences in budgets and costs associated with both interventions and diseases/events preclude valid universal cost-effectiveness analyses. However, some recommendations clearly have financial implications, either in terms of costs for individual patients and/or in terms of budget impact. Some of these recommendations pertain to diagnosis (e.g. large-scale use of expensive imaging tests such as computed tomography), others to interventions (e.g.

expensive drugs, such as novel lipid-lowering or anti-diabetic drugs). For such recommendations, it is inappropriate to 'unconditionally' implement them without first considering cost-effectiveness in a national or regional context or, ideally, to perform formal cost-effectiveness analyses with country-specific input parameters and cost-effectiveness thresholds.

2.4. What is new?

New recommendations, and new and revised concepts, are presented in Table 3.

3. Risk factors and clinical conditions

3.1. Target population for assessing cardiovascular disease risk

CVD risk assessment or screening can be done opportunistically or systematically. Opportunistic screening, which means screening without a predefined strategy, is done when a person presents for some other reason. Systematic screening can be done in the general population as part of a formal screening programme, with call and recall of patients, or in targeted subpopulations such as subjects with type 2 DM, or family history of premature CVD. Systematic screening results in improvements in risk factors, but has no effect on CVD outcomes.^{6–9} Opportunistic screening for ASCVD risk factors, such as BP or lipids, is effective at increasing detection rates and is recommended, although a beneficial effect on clinical outcome is uncertain.¹⁰

Structured national programmes aiming to identify undocumented ASCVD risk factors in adults over 40 years of age without DM or ASCVD and treat them have shown better risk factor control, but there are conflicting results as to clinical outcomes.^{11,12} A high-risk strategy of inviting the population predicted to be at the highest risk according to an integrated risk score would be equally effective at preventing new cases of CVD and have potential cost savings.¹³ One large trial of mobile ultrasound screening for aortic aneurysm, peripheral artery disease (PAD), and hypertension in males aged 65–74 years showed a 7% mortality reduction at 5 years.¹⁴

A common criticism of screening in general is the potential that false positive and false negative results may cause harm. However, evidence on CVD screening shows that those who participate do not report mental distress.^{15–18}

Systematic CVD risk assessment in the general population (adult men >40 and women >50 years of age) with no known CV risk factors appears not cost-effective in reducing subsequent vascular events and premature death, at least in short-term follow-up, but does increase detection of CV risk factors. Risk assessment is not a one-time event; it should be repeated, for example, every 5 years, although there are no empirical data to guide intervals.

3.2. Risk factors and risk classification

3.2.1. Risk factors

The main causal and modifiable ASCVD risk factors are blood apolipoprotein-B-containing lipoproteins [of which low-density lipoprotein (LDL) is most abundant], high BP, cigarette smoking, and DM.

Table 3 What is new

New or revised	Recommendations in 2013 version	Class	Recommendations in 2021 version	Class
Risk factors and clinical conditions – section 3				
New			In apparently healthy people <70 years of age without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorders, estimation of 10-year fatal and nonfatal CVD risk with SCORE2 is recommended.	I
New			In apparently healthy people ≥70 years of age without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorder, estimation of 10-year fatal and nonfatal CVD risk with SCORE2-OP is recommended.	I
New			Patients with established ASCVD and/or DM and/or moderate-to-severe renal disease and/or genetic/rarer lipid or BP disorders are to be considered at high or very high CVD risk.	I
New			A stepwise treatment-intensification approach aiming at intensive risk factor treatment is recommended for apparently healthy people at high or very high ASCVD risk, as well as patients with established ASCVD and/or DM, with consideration of CVD risk, treatment benefit of risk factors, risk modifiers, comorbidities, and patient preferences.	I
New			Treatment of ASCVD risk factors is recommended in apparently healthy people without DM, CKD, genetic/rarer lipid or BP disorders who are at very high CVD risk (SCORE2 ≥7.5% for age under 50; SCORE2 ≥10% for age 50–69; SCORE2-OP ≥15% for age ≥70).	I
New			An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended.	I
New			It is recommended that mental disorders with either significant functional impairment or decreased use of healthcare systems be considered as influencing	I
New			Treatment of ASCVD risk factors should be considered in apparently healthy people without DM, CKD, genetic/rarer lipid, or BP disorders who are at high CVD risk (SCORE2 2.5 to <7.5% for age under 50; SCORE2 5 to <10% for age 50–69; SCORE2-OP 7.5 to <15% for age ≥70 years), taking ASCVD risk modifiers, lifetime risk and treatment benefit, and patient preferences into account.	IIa
New			In apparently healthy people, after estimation of 10-year fatal and non-fatal CVD risk, lifetime risk and treatment benefit, risk modifiers, frailty, polypharmacy, and patient preferences should be considered.	IIa
New			Presence of migraine with aura should be considered in CVD risk assessment.	IIa
New			Assessment of CVD risk should be considered in men with ED.	IIa
New			In women with a history of premature or stillbirth, periodic screening for hypertension and DM may be considered.	IIb
New			Assessment of total CVD risk may be considered in adults with chronic inflammatory conditions.	IIb
New			Avoidance of combined hormonal contraceptives may be considered in women with migraine with aura.	IIb
Risk factors and interventions at the individual level – section 4				
New			It is recommended to reduce sedentary time to engage in at least light activity throughout the day to reduce all-cause and CV mortality and morbidity.	I

Continued

Table 3 Continued

New or revised	Recommendations in 2013 version	Class	Recommendations in 2021 version	Class
New			It is recommended to adopt a Mediterranean or similar diet to lower risk of CVD.	I
New			It is recommended to restrict alcohol consumption to a maximum of 100 g per week.	I
New			It is recommended to eat fish, preferably fatty, at least once a week and restrict (processed) meat.	I
New			Patients with mental disorders need intensified attention and support to improve adherence to lifestyle changes and drug treatment.	I
New			Smoking cessation is recommended regardless of weight gain, as weight gain does not lessen the ASCVD benefits of cessation.	I
New			In patients with established ASCVD, lipid-lowering treatment with an ultimate LDL-C goal of <1.4 mmol/L (55 mg/dL) and a ≥50% reduction of LDL-C vs. baseline is recommended.	I
New			For secondary prevention patients not achieving their goals on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor is recommended.	I
New			In patients with type 2 DM at very high risk (e.g. with established ASCVD and/or severe TOD), intensive lipid-lowering therapy, ultimately aiming at ≥50% LDL-C reduction and an LDL-C of <1.4 mmol/L (<55 mg/dL) is recommended.	I
New			In patients with type 2 DM >40 years of age at high risk, lipid-lowering treatment with an ultimate LDL-C goal of ≥50% LDL-C reduction and an LDL-C of <1.8 mmol/L (70 mg/dL) is recommended.	I
New			It is recommended that the first objective of treatment is to lower BP to <140/90 mmHg in all patients, and that subsequent BP targets are tailored to age and specific comorbidities.	I
New			In treated patients aged 18–69 years, it is recommended that SBP should ultimately be lowered to a target range of 120–130 mmHg in most patients.	I
New			In treated patients aged ≥70 years, it is recommended that SBP should generally be targeted to <140 and down to 130 mmHg if tolerated.	I
New			In all treated patients, DBP is recommended to be lowered to <80 mmHg.	I
New			In persons with type 2 DM and ASCVD, the use of a GLP-1RA or SGLT2 inhibitor with proven outcome benefits is recommended to reduce CV and/or cardiorenal outcomes.	I
New			In patients with type 2 DM and CKD, the use of an SGLT2 inhibitor is recommended to improve CVD and/or cardiorenal outcomes.	I
New			In patients with type 2 DM and HFrEF, use of an SGLT2 inhibitor with proven outcome benefits is recommended to lessen HF hospitalizations and CV death.	I
New			Participation in a medically supervised, structured, comprehensive, multidisciplinary EBCR and prevention programme for patients after ASCVD events and/or revascularization, and for patients with HF (mainly HFrEF), is recommended to improve patient outcomes.	I

Continued

Table 3 Continued

New or revised	Recommendations in 2013 version	Class	Recommendations in 2021 version	Class
New			Lifestyle interventions, such as group or individual education, behaviour-change techniques, telephone counselling, and use of consumer-based wearable activity trackers, should be considered to increase PA participation.	Ila
New			Bariatric surgery for obese high-risk individuals should be considered when lifestyle change does not result in maintained weight loss.	Ila
New			ASCVD patients with stress should be considered for referral to psychotherapeutic stress management to improve CVD outcomes and reduce stress symptoms.	Ila
New			Patients with CHD and moderate-to-severe major depression should be considered for antidepressive treatment with an SSRI.	Ila
New			An ultimate LDL-C goal of <1.4 mmol/L (55 mg/dL) and LDL-C reduction of ≥50% from baseline should be considered in apparently healthy persons <70 years at very high risk.	Ila
New			An ultimate LDL-C goal of <1.8 mmol/L (70 mg/dL) and LDL-C reduction of ≥50% from baseline should be considered in apparently healthy persons <70 years at high risk.	Ila
New			For those motivated to try, considerable weight loss with use of low-calorie diets followed by food reintroduction and weight-maintenance phases early after diagnosis can lead to DM remission and should be considered.	Ila
New			In patients with type 2 DM and TOD, the use of an SGLT2 inhibitor or GLP-1RA with proven outcome benefits may be considered to reduce future CVD and total mortality.	Ilb
New			For primary prevention patients at very high risk, but without FH, if the LDL-C goal is not achieved on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor may be considered.	Ilb
New			In high-risk (or above) patients with triglycerides >1.5 mmol/L (135 mg/dL) despite statin treatment and lifestyle measures, n-3 PUFAs (icosapent ethyl 2 X 2 g/day) may be considered in combination with a statin.	Ilb
New			Initiation of statin treatment for primary prevention in older people aged ≥70 may be considered, if at high risk or above.	Ilb
New			Statin therapy may be considered in persons aged ≤40 years with type 1 or type 2 DM with evidence of TOD and/or an LDL-C level >2.6 mmol/L (100 mg/dL), as long as pregnancy is not being planned.	Ilb
New			In patients with DM at high or very high CVD risk, low-dose aspirin may be considered for primary prevention in the absence of clear contraindications.	Ilb
New			Home-based CR, telehealth, and mHealth interventions may be considered to increase patient participation and long-term adherence to healthy behaviours.	Ilb
New			In patients with HF and major depression, SSRIs, SNRIs, and tricyclic antidepressants are not recommended.	III
New			In patients with dialysis-dependent CKD who are free of ASCVD, commencing statin therapy is not recommended.	III

Continued

Table 3 Continued

New or revised	Recommendations in 2013 version	Class	Recommendations in 2021 version	Class
Policy interventions at the population level – section 5				
New			Putting in place measures to reduce air pollution, including reducing PM emission and gaseous pollutants, reducing the use of fossil fuels, and limiting carbon dioxide emissions, are recommended to reduce CVD mortality and morbidity.	I
Risk management of disease-specific cardiovascular disease – section 6				
New			It is recommended that patients with HF are enrolled in a comprehensive CR programme to reduce the risk of HF hospitalization and death.	I
New			It is recommended to screen patients with HF for both CV and non-CV comorbidities which, if present, should be treated, provided safe and effective interventions exist, not only to alleviate symptoms but also to improve prognosis.	I
New			In patients with a cerebrovascular event, improvement of lifestyle factors in addition to appropriate pharmacological management is recommended.	I
New			Identification and management of risk factors and concomitant diseases are recommended to be an integral part of treatment in patients with AF.	I
New			Adding a second antithrombotic drug (a P2Y ₁₂ inhibitor or low-dose rivaroxaban) to aspirin for long-term secondary prevention should be considered in patients with a high risk of ischaemic events and without high bleeding risk.	IIa
New			In patients with DM and chronic symptomatic LEAD without high bleeding risk, a combination of low-dose rivaroxaban (2.5 mg b.i.d.) and aspirin (100 mg o.d.) may be considered.	IIb
			Adding a second antithrombotic drug to aspirin for long-term secondary prevention may be considered in patients with a moderate risk of ischaemic events and without a high bleeding risk.	IIb
Risk factors and clinical conditions – section 3				
Revised	ABI may be considered as a risk modifier in CVD risk assessment.	IIb	The routine collection of other potential modifiers, such as genetic risk scores, circulating or urinary biomarkers, or vascular tests or imaging methods (other than CAC scoring or carotid ultrasound for plaque determination), is not recommended.	III
Risk factors and interventions at the individual level – section 4				
Revised	Drug treatment should be considered in patients with grade 1 or 2 hypertension who are at high CVD risk.	IIa	For grade 1 hypertension, treatment initiation based on absolute CVD risk, estimated lifetime benefit, and the presence of HMOD is recommended.	I
Revised	In patients with type 2 DM and CVD, use of an SGLT2 inhibitor should be considered early in the course of the disease to reduce CVD and total mortality.	IIa	In persons with type 2 DM and ASCVD, the use of a GLP-1RA or SGLT2 inhibitor with proven outcome benefits is recommended to reduce CV and/or cardiorenal outcomes.	I

ABI = ankle brachial index; AF = atrial fibrillation; ASCVD = atherosclerotic cardiovascular disease; b.i.d. = *bis in die* (twice a day); BP = blood pressure; CAC = coronary artery calcium; CHD = coronary heart disease; CKD = chronic kidney disease; CR = cardiac rehabilitation; CV = cardiovascular; CVD = cardiovascular disease; DM = diabetes mellitus; EBCR = exercise-based cardiac rehabilitation; ED = erectile dysfunction; FH = familial hypercholesterolaemia; GLP-1RA = glucagon-like peptide-1 receptor agonist; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; HMOD = hypertension-mediated organ damage; LDL-C = low-density lipoprotein cholesterol; LEAD = lower extremity artery disease; mHealth = mobile device-based healthcare; o.d. = *omni die* (once a day); PA = physical activity; PCSK9 = proprotein convertase subtilisin/kexin type 9; PM = particulate matter; PUFA = polyunsaturated fatty acid; SBP = systolic blood pressure; SCORE2 = Systematic Coronary Risk Estimation 2; SCORE2-OP = Systematic Coronary Risk Estimation 2-Older Persons; SGLT2 = sodium-glucose cotransporter 2; SNRI = serotonin-noradrenaline reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; TOD = target organ damage.

New sections**Section 3**

- 3.2.2 Sex and gender and their impact on health
- 3.2.3 Atherosclerotic cardiovascular disease risk classification
 - 3.2.3.1 A stepwise approach to risk factor treatment and treatment intensification
 - 3.2.3.2 Risk estimation in apparently healthy people
 - 3.2.3.3 Translating atherosclerotic cardiovascular disease risk to treatment thresholds
 - 3.2.3.4 Risk estimation and risk factor treatment in apparently healthy people 50–69 years of age
 - 3.2.3.5 Risk estimation and risk factor treatment estimation in apparently healthy people ≥ 70 years of age
 - 3.2.3.6 Risk estimation and risk factor treatment in apparently healthy people < 50 years of age
 - 3.2.3.7 Risk estimation and risk factor treatment in patients with established atherosclerotic cardiovascular disease
- 3.2.4 Communication of cardiovascular disease risk
- 3.3.1 Psychosocial factors
- 3.3.4 Frailty
- 3.3.8 Environmental exposure
- 3.4 Clinical conditions
 - 3.4.2 Atrial fibrillation
 - 3.4.3 Heart failure
 - 3.4.5 Chronic obstructive pulmonary disease
 - 3.4.6 Inflammatory conditions
 - 3.4.7 Infections (human immunodeficiency virus, influenza, periodontitis)
 - 3.4.8 Migraine
 - 3.4.9 Sleep disorders and obstructive sleep apnoea
 - 3.4.10 Mental disorders
 - 3.4.11 Non-alcoholic fatty liver disease
 - 3.4.12 Sex-specific conditions

Section 4

- 4.10 Anti-inflammatory treatment

New /revised concepts**Section 3**

- SCORE2 and SCORE2-OP risk charts for fatal and non-fatal (myocardial infarction, stroke) ASCVD
- Estimating 10-year total CVD risk in apparently healthy people 50–69 years of age
- Estimating lifetime risk in apparently healthy people < 50 years of age
- Estimating 10-year total CVD risk in apparently healthy people ≥ 70 years of age
- Cut-offs of 10-year CVD risk, based on SCORE2/SCORE2-OP, to define low–moderate risk, high risk, and very high risk for apparently healthy people in different age groups (< 50 , 50–69, and ≥ 70 years)
- Estimating 10-year CVD risk in patients with established CVD and/or DM
- Lifetime benefit of stopping smoking, reducing LDL-C, or lowering SBP ([sections 3 and 4](#))
- A stepwise approach to attaining ultimate treatment goals ([sections 3 and 4](#))
- Communication of CVD risk and benefit of treatment to patients in an understandable way
- Stepwise approach to risk factor treatment and treatment intensification

Section 4

- Explicitly addressing cost-effectiveness (on a loco-regional or national level) before implementing some recommendations
- Non-fasting lipid measurement ([section 4.6.1.1](#))
- A stepwise approach to attaining treatment goals ([sections 3 and 4](#))
- Anti-inflammatory treatment for very-high-risk patients

Section 5

- Taking into consideration population level interventions to mitigate the effects of pollution on CVD health

Section 6

- Risk management of disease-specific CVD. This section addresses CVD prevention when certain underlying diseases are present and aims to provide guidance on how to prevent the worsening of existing, or the development of further, comorbidities that could increase the overall risk of CVD
- Subsections include: 6.1 Coronary artery disease; 6.2 Heart failure; 6.3 Cerebrovascular disease; 6.4 Lower extremity artery disease; 6.5 Chronic kidney disease; 6.6 Atrial fibrillation; 6.7 Multimorbidity

ASCVD = atherosclerotic cardiovascular disease; CVD = cardiovascular disease; DM = diabetes mellitus; LDL-C = low-density lipoprotein cholesterol; SBP = systolic blood pressure; SCORE2 = Systematic Coronary Risk Estimation 2; SCORE2-OP = Systematic Coronary Risk Estimation 2-Older Persons.

Recommendations for CVD risk assessment

Recommendations	Class ^a	Level ^b
Systematic global CVD risk assessment is recommended in individuals with any major vascular risk factor (i.e. family history of premature CVD, FH, CVD risk factors such as smoking, arterial hypertension, DM, raised lipid level, obesity, or comorbidities increasing CVD risk).	I	C
Systematic or opportunistic CV risk assessment in the general population in men >40 years of age and in women >50 years of age or postmenopausal with no known ASCVD risk factors may be considered. ⁹	IIb	C
In those individuals who have undergone CVD risk assessment in the context of opportunistic screening, a repetition of screening after 5 years (or sooner if risk was close to treatment thresholds) may be considered.	IIb	C
Opportunistic screening of BP in adults at risk for the development of hypertension, such as those who are overweight or with a known family history of hypertension, should be considered. ¹⁹	IIa	B
Systematic CVD risk assessment in men <40 years of age and women <50 years of age with no known CV risk factors is not recommended. ⁹	III	C

ASCVD = atherosclerotic cardiovascular disease; BP = blood pressure; CV = cardiovascular; CVD = cardiovascular disease; DM = diabetes mellitus; FH = familial hypercholesterolaemia.

^aClass of recommendation.

^bLevel of evidence.

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Another important risk factor is adiposity, which increases CVD risk via both major conventional risk factors and other mechanisms. In addition to these, there are many other relevant risk factors, modifiers, and clinical conditions, which are addressed under risk modifiers and clinical conditions (sections 3.3 and 3.4).

3.2.1.1 Cholesterol

The causal role of LDL-C, and other apo-B-containing lipoproteins, in the development of ASCVD is demonstrated beyond any doubt by genetic, observational, and interventional studies.²⁰ The key attributes of LDL-C as a risk factor for ASCVD are:

- Prolonged lower LDL-C is associated with lower risk of ASCVD throughout the range studied, and the results of randomized controlled trials (RCTs) indicate that lowering LDL-C safely reduces CVD risk even at low LDL-C levels [e.g. LDL-C <1.4 mmol/L (55 mg/dL)].²⁰
- The relative reduction in CVD risk is proportional to the absolute size of the change in LDL-C, irrespective of the drug(s) used to achieve such change.²¹
- The absolute benefit of lowering LDL-C depends on the absolute risk of ASCVD and the absolute reduction in LDL-C, so

even a small absolute reduction in LDL-C may be beneficial in a high- or very-high-risk patient.²²

- Non-high-density lipoprotein cholesterol (HDL-C) encompasses all atherogenic (apo-B-containing) lipoproteins, and is calculated as: total cholesterol – HDL-C = non-HDL-C. The relationship between non-HDL-C and CV risk is at least as strong as the relationship with LDL-C. Non-HDL-C levels contain, in essence, the same information as a measurement of apo-B plasma concentration.^{23,24} Non-HDL-C is used as an input in the Systemic Coronary Risk Estimation 2 (SCORE2) and SCORE2-Older Persons (SCORE2-OP) risk algorithms.

HDL-C is inversely associated with CVD risk. Very high HDL-C levels may signal an increased CVD risk. There is, however, no evidence from Mendelian randomization studies, or randomized trials of cholesteryl ester transfer protein inhibitors, that raising plasma HDL-C reduces CVD risk.^{25–28} HDL-C is nonetheless a useful biomarker to refine risk estimation using the SCORE2 algorithms. The SCORE2 algorithm cannot be used for patients with a genetic lipid disorder, such as familial hypercholesterolaemia (FH). Specific LDL-C thresholds and targets are recommended irrespective of estimated CV risk for patients with FH or other rare/genetic lipid disorders.

3.2.1.2 Blood pressure

Longitudinal studies, genetic epidemiological studies, and RCTs have shown that raised BP is a major cause of both ASCVD and non-atherosclerotic CVD [particularly heart failure (HF)], accounting for 9.4 million deaths and 7% of global disability adjusted life-years.²⁹ Elevated BP is a risk factor for the development of coronary artery disease (CAD), HF, cerebrovascular disease, lower extremity arterial disease (LEAD), chronic kidney disease (CKD), and atrial fibrillation (AF). The risk of death from either CAD or stroke increases linearly from BP levels as low as 90 mmHg systolic and 75 mmHg diastolic upwards.^{30,31} The absolute benefit of reducing systolic BP (SBP) depends on absolute risk and the absolute reduction in SBP, except that lower limits of SBP are imposed by tolerability and safety considerations. Management is determined by the category of hypertension (optimal, normal, high-normal, stages 1 to 3, and isolated systolic hypertension), defined according to seated office BP, ambulatory BP monitoring (ABPM), or home BP average values (see section 4.7). Evidence suggests that lifetime BP evolution differs in women compared to men, potentially resulting in an increased CVD risk at lower BP thresholds.^{32–34} The SCORE2 algorithm cannot be used for patients with secondary causes and rarer forms of hypertension, such as primary hyperaldosteronism.

3.2.1.3 Cigarette smoking

Cigarette smoking is responsible for 50% of all avoidable deaths in smokers, with half of these due to ASCVD. A lifetime smoker has a 50% probability of dying due to smoking, and on average will lose 10 years of life.³⁵ The CVD risk in smokers <50 years of age is five-fold higher than in non-smokers.³⁶ Prolonged smoking is more hazardous for women than for men.³⁷ Worldwide, after high SBP, smoking is the leading risk factor for disability adjusted life-years.³⁸ Second-hand smoke is associated with an increase in CVD risk.³⁹ Some smokeless tobacco is also associated with increased risk of CVD.⁴⁰

3.2.1.4 Diabetes mellitus

Type 1 DM, type 2 DM, and prediabetes are independent risk factors for ASCVD, increasing risk of ASCVD by about two-fold, depending on the population and therapeutic control.⁴¹ Women with type 2 DM appear to have a particularly higher risk for stroke.⁴² Patients with type 2 DM are likely to have multiple ASCVD risk factors (including dyslipidaemia and hypertension), each of which mediates an increase in risk of both ASCVD and non-ASCVD.

3.2.1.5 Adiposity

Over recent decades, body mass index (BMI)—measured as weight (in kg) divided by squared height (in m²)—has increased substantially worldwide in children, adolescents, and adults.⁴³ Mendelian randomization analyses suggest a linear relation between BMI and mortality in non-smokers and a J-shaped relation in ever-smokers.⁴⁴ All-cause mortality is lowest at a BMI of 20–25 kg/m² in apparently healthy people, with a J-shaped or U-shaped relation.^{45,46} In HF patients, there is evidence for an obesity paradox, with lower mortality risk in patients with higher BMI. A meta-analysis concluded that both BMI and waist circumference are similarly, strongly, and continuously associated with ASCVD and type 2 DM.⁴⁷

3.2.2. Sex and gender and their impact on health

The current prevention guidelines recognize the importance of integrating sex, gender, and gender identity considerations into the risk assessment and clinical management of individuals and populations. These guidelines also acknowledge the complexity of the inter-relationship between these concepts and CV, as well as psychological, health. There is, at present, no official ESC position on the specific terminology to be used. According to the World Health Organization (WHO), sex 'refers to the different biological and physiological characteristics of females, males, and intersex persons, such as chromosomes, hormones and reproductive organs'.⁴⁸

This is to be distinguished from gender, which 'refers to the characteristics of women, men, girls and boys that are socially constructed. This includes norms, behaviours and roles associated with being a woman, man, girl or boy, as well as relationships with each other. As a social construct, gender varies from society to society and can change over time'.⁴⁸ The Global Health 50/50 definition further states that gender refers 'to the socially constructed norms that impose and determine roles, relationships, and positional power for all people across their lifetime'.⁴⁹

Where evidence exists on the risk modifying effect of sex or where sex-specific clinical conditions and clinical management strategies exist, this has been included in these guidelines.⁵⁰ The influence of gender on an individual's experience and access to healthcare is paramount.⁵⁰ The specific health concerns related to gender are thus also acknowledged in these prevention Guidelines.

Epigenetic effects of social constructs appear to condition the translation of biological sex into disease pathophysiology. Furthermore, social constructs can also be determinants of health access, healthcare utilization, disease perception, decision-making, and perhaps therapeutic response,⁵⁰ including in the field of CVD and ASCVD prevention. Research is ongoing, but gaps in evidence remain and this has also been recognized in the guidelines.

Examples of specific topics regarding physiological, pathological, and clinical differences related to sex and gender that have been

studied include left ventricular (LV) ejection fraction (LVEF), adverse drug reactions, trends in ASCVD risk factors and awareness, sex disparities in the management of and outcomes after acute coronary syndromes (ACS).^{51–58} Furthermore, CVD health after menopause transition, pregnancy disorders, and gynaecologic conditions have recently been reviewed.⁵⁹

3.2.3. Cardiovascular disease risk classification

The current guidelines on CVD prevention in clinical practice concentrate principally, but not exclusively, on risk and prevention of ASCVD. This includes risk factors, risk prediction, risk modifiers, as well as clinical conditions that often increase the likelihood of ASCVD.

Identifying patients who will benefit most from ASCVD risk factor treatment is central to ASCVD prevention efforts. In general, the higher the absolute CVD risk, the higher the absolute benefit of risk factor treatment, and thus the lower the *number needed to treat* to prevent one CVD event during a period of time.^{60,61} With this in mind, the estimation of CVD risk remains the cornerstone of these guidelines and thus appears at the forefront of the proposed management schemes, which are summarized in flowcharts.

Age is the major driver of CVD risk. Women below 50 years and men below 40 years of age are almost invariably at low 10-year CVD risk, but may have unfavourable modifiable risk factors that sharply increase their longer-term CVD risk. Conversely, men over 65 years and women over 75 years of age are almost always at high 10-year CVD risk. Only between the ages of 55 and 75 years in women and 40 and 65 years in men does the 10-year CVD risk vary around commonly used thresholds for intervention. The age categories <50, 50–69, and ≥70 years should be used with common sense and flexibility. Different age ranges may be considered for men and women and may differ according to geographic region. Uncertainty around risk estimations should also be considered.

CVD risk can also be assessed in patients with type 2 DM and in patients with established ASCVD. The populations or patient groups in whom CVD risk needs to be considered are summarized and presented in Table 4. Lifetime CVD risk estimation is available for various groups of patients, and enables estimation of lifetime benefit from preventive interventions such as smoking cessation (see section 4.5.1), lipid-lowering (see section 4.6.2.1), and BP treatment (see section 4.7.5.2). Lifetime risk and benefit estimation may be used for communication in the shared decision-making process, together with consideration of comorbidities, frailty, patient preferences for initiating (STEP 1) and intensifying (STEP 2) risk factor treatment (Figure 2).

3.2.3.1 A stepwise approach to risk factor treatment and treatment intensification

As explained before, targets and goals for LDL-C, BP, and glycaemic control in DM remain as recommended in recent ESC Guidelines.^{3–5} These guidelines propose a stepwise approach to treatment intensification as a tool to help physicians and patients pursue these targets in a way that fits patient profiles and preferences. This principle (outlined in Figure 2, using the example of a stepwise approach) is not conceptually novel, but rather reflects routine clinical practice, in which treatment strategies are initiated and then intensified, both as part of a shared decision-making process involving healthcare professionals and patients.

Table 4 Patient categories and associated cardiovascular disease risk.

Patient category	Subgroups	Risk categories	CVD risk and therapy benefit estimation
Apparently healthy persons			
Persons without established ASCVD, diabetes mellitus, CKD, Familial Hypercholesterolemia	<50 years	Low- to high-risk	10-year CVD risk estimation (SCORE2). Lifetime risk and benefit estimation of risk factor treatment (e.g. with the LIFE-CVD lifetime model) to facilitate the communication of CVD risk and treatment benefits.
	50–69 years	Low- to very high-risk	10-year CVD risk estimation (SCORE2). Lifetime benefit estimation of risk factor treatment (e.g. with the LIFE-CVD lifetime model) to facilitate the communication of treatment benefits.
	≥70 years	Low- to very high-risk	10-year CVD risk estimation (SCORE2-OP). Lifetime benefit estimation of risk factor treatment (e.g. with the LIFE-CVD lifetime model) to facilitate the communication of treatment benefits.
Patients with CKD			
CKD without diabetes or ASCVD	Moderate CKD (eGFR 30–44 mL/min/1.73 m ² and ACR <30 or eGFR 45–59 mL/min/1.73 m ² and ACR 30–300 or eGFR ≥60 mL/min/1.73 m ² and ACR >300)	High-risk	N/A
	Severe CKD (eGFR <30 mL/min/1.73 m ² or eGFR 30–44 mL/min/1.73 m ² and ACR >30)	Very high-risk	N/A
Familial Hypercholesterolemia			
Associated with markedly elevated cholesterol levels	N/A	High-risk	N/A
Patients with type 2 diabetes mellitus			
Patients with type 1 DM above 40 years of age may also be classified according to these criteria	Patients with well controlled short-standing DM (e.g. <10 years), no evidence of TOD and no additional ASCVD risk factors	Moderate-risk	N/A
	Patients with DM without ASCVD and/or severe TOD, and not fulfilling the moderate risk criteria.	High-risk	Residual 10-year CVD risk estimation after general prevention goals (e.g. with the ADVANCE risk score or DIAL model). Consider lifetime CVD risk and benefit estimation of risk factor treatment (e.g. DIAL model).
	Patients with DM with established ASCVD and/or severe TOD: ^{87, 93–95} • eGFR <45 mL/min/1.73 m ² irrespective of albuminuria • eGFR 45–59 mL/min/1.73 m ² and microalbuminuria (ACR 30–300 mg/g) • Proteinuria (ACR >300 mg/g) • Presence of microvascular disease in at least 3 different sites (e.g. microalbuminuria plus retinopathy plus neuropathy)	Very high-risk	Residual 10-year CVD risk estimation after general prevention goals (e.g. with the SMART risk score for established CVD or with the ADVANCE risk score or with the DIAL model). Consider lifetime CVD risk and benefit estimation of risk factor treatment (e.g. DIAL model).
Patients with established ASCVD			
Documented ASCVD, clinical or unequivocal on imaging. Documented clinical ASCVD includes previous AMI, ACS, coronary revascularization and other arterial revascularization procedures, stroke and TIA, aortic aneurysm and PAD. Unequivocally documented ASCVD on imaging includes plaque on coronary angiography or carotid ultrasound or on CTA. It does NOT include some increase in continuous imaging parameters such as intima-media thickness of the carotid artery.	N/A	Very high-risk	Residual CVD risk estimation after general prevention goals (e.g. 10-year risk with the SMART risk score for patients with established CVD or 1- or 2-year risk with EUROASPIRE risk score for patients with CHD). Consider lifetime CVD risk and benefit estimation of risk factor treatment (e.g. SMART-REACH model; or DIAL model if diabetes).

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ACR = albumin-to-creatinine ratio: (to convert mg/g to mg/mmol: divide by 10); ACS = acute coronary syndromes; ADVANCE = Action in Diabetes and Vascular disease: preterAx and diamicroN-MR Controlled Evaluation; AMI = acute myocardial infarction; ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; CTA = computed tomography angiography; CV = cardiovascular; CVD = cardiovascular disease; DIAL = Diabetes lifetime-perspective prediction; DM = diabetes mellitus; FH = familial hypercholesterolaemia; eGFR = estimated glomerular filtration rate; IMT = intima-media thickness; LIFE-CVD = LIFETIME-perspective CardioVascular Disease; N/A = not applicable; PAD = peripheral artery disease; REACH = Reduction of Atherothrombosis for Continued Health; SBP = systolic blood pressure; SCORE = Systematic Coronary Risk Estimation; SMART = Secondary Manifestations of Arterial Disease; TIA = transient ischaemic attack.

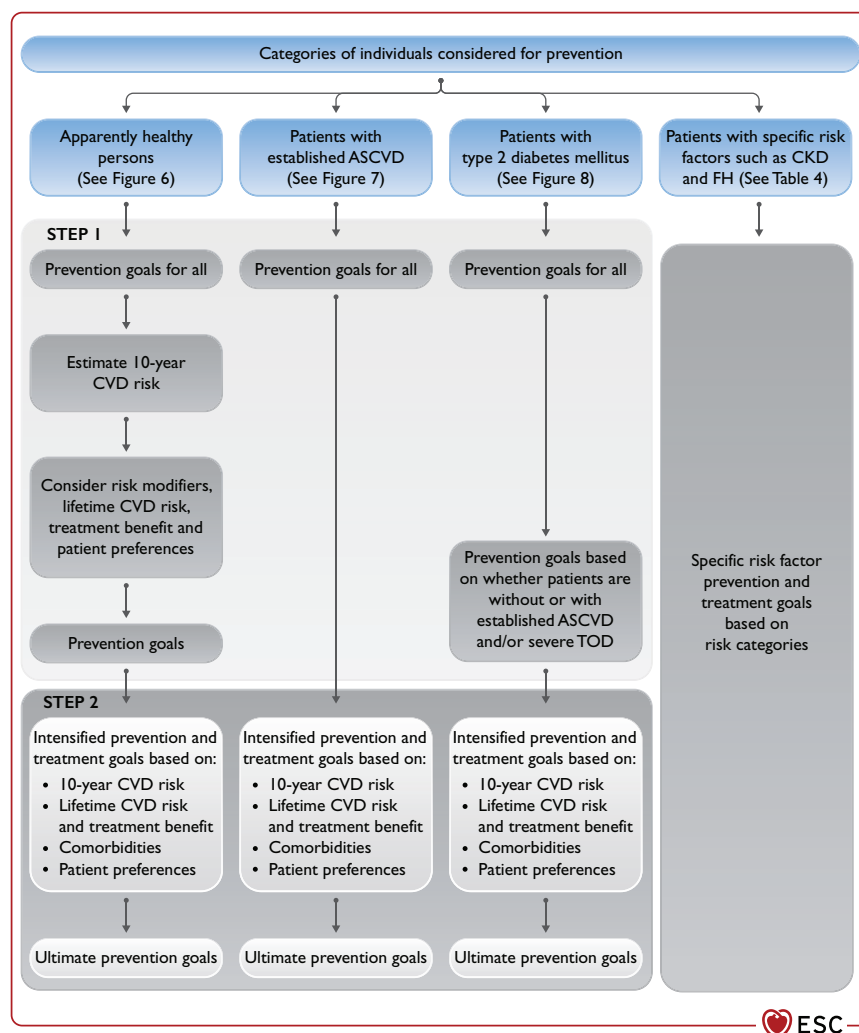


Figure 2 Examples of a stepwise approach to risk stratification and treatment options. ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; DM = diabetes mellitus; FH = familial hypercholesterolaemia; TOD = target organ damage.

A stepwise approach starts with prevention goals for all, regardless of CVD risk. This is followed by CVD risk stratification and discussion of potential benefits of treatment with the patient. If treatment is initiated, its effect must be evaluated, and subsequent treatment intensification to reach ultimate risk factor goals must be considered in all patients, taking into account additional benefit, comorbidities, and frailty, all of which converge with patient preferences in a shared decision-making process.

In the field of DM, studies have shown benefit of a stepwise approach to treatment intensification and do not support the contention of 'therapeutic nihilism' occurring in either physicians or patients. In fact, it appears that attainment of treatment goals is similar, side-effects are fewer, and patient satisfaction is significantly higher with such an approach.^{66,67} We do, however, emphasize that stopping assessment of treatment goals and/or treatment routinely after the first step is inappropriate. The evidence-based ultimate targets of treatment intensification are optimal from the perspective of CVD risk reduction and are to be considered in all patients.

3.2.3.2 Risk estimation in apparently healthy people

Apparently healthy people are those without established ASCVD, type 2 DM, or severe comorbidities. In the 2016 ESC prevention guidelines,² the Systemic Coronary Risk Estimation (SCORE) algorithm was used to estimate 10-year risk of CVD death. However, CVD morbidity (non-fatal myocardial infarction, non-fatal stroke) combined with CVD mortality better reflects the total burden of ASCVD. The updated SCORE algorithm—SCORE2—used in these guidelines (see Figure 3), estimates an individual's 10-year risk of *fatal and non-fatal* CVD events (myocardial infarction, stroke) in apparently healthy people aged 40–69 years with risk factors that are untreated or have been stable for several years.⁶⁸

Several specific considerations apply to CVD risk estimation in older people. First, the gradient of the relationship between classical risk factors, such as lipids and BP, with CVD risk attenuates with age.⁶⁹ Second, CVD-free survival dissociates from overall survival progressively with increasing age, because risk for non-CVD mortality increases ('competing risk').⁷⁰ For these reasons, traditional risk

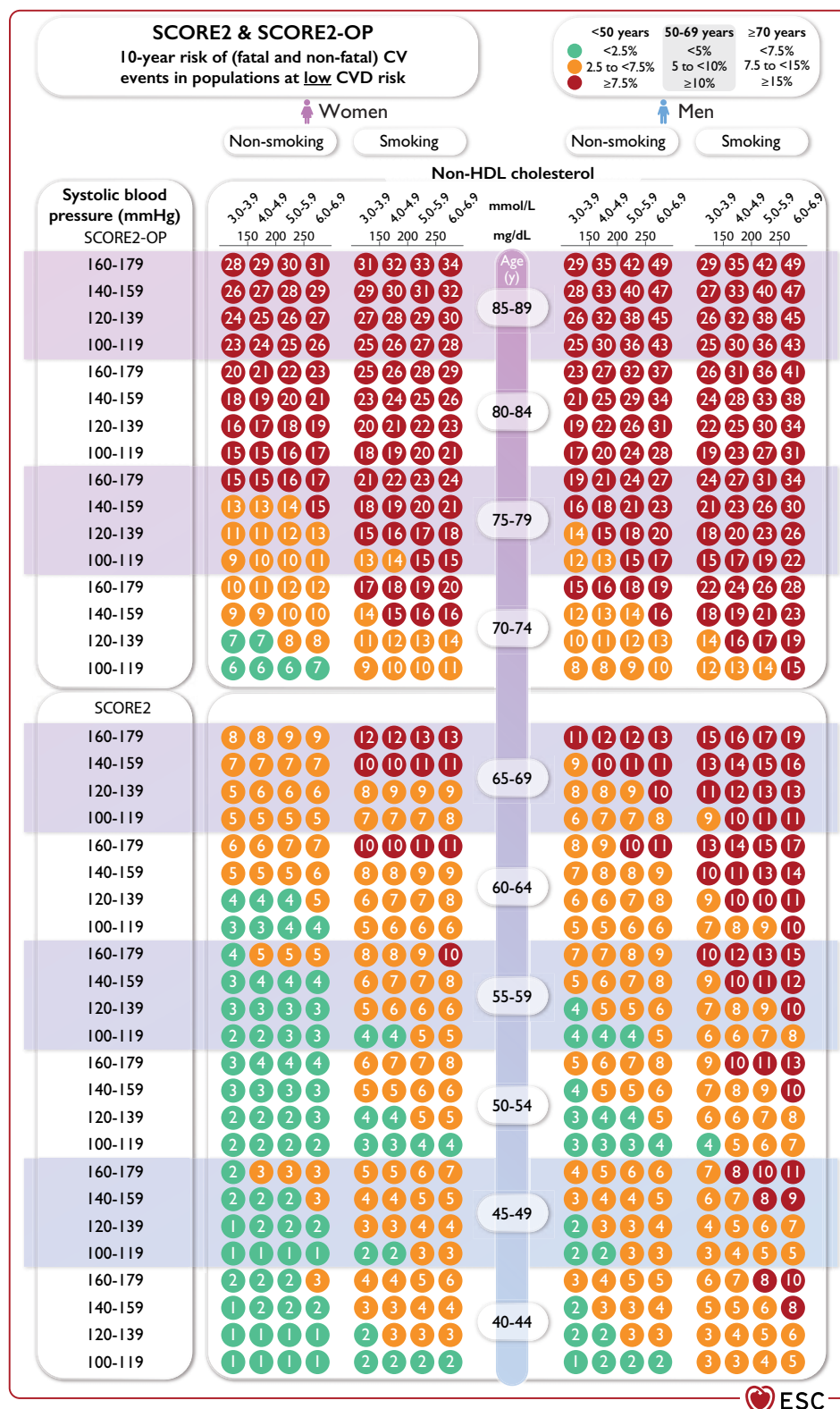


Figure 3 Systematic Coronary Risk Estimation 2 and Systematic Coronary Risk Estimation 2-Older Persons risk charts for fatal and non-fatal (myocardial infarction, stroke) cardiovascular disease.^{68,72} ASCVD = atherosclerotic cardiovascular disease; CV = cardiovascular; CVD = cardiovascular disease; SBP = systolic blood pressure; HDL-C = high-density lipoprotein cholesterol; SCORE2 = Systematic Coronary Risk Estimation 2; SCORE2-OP = Systematic Coronary Risk Estimation 2-Older Persons; TFYR = The Former Yugoslav Republic; UK = United Kingdom. For apparently healthy people aged 40–69 years, the SCORE2 algorithm⁶⁸ is used to estimate 10-year risk of fatal and non-fatal (myocardial infarction, stroke) CVD. For apparently healthy people ≥70 years of age, the SCORE2-OP is used.⁷² **Low-risk countries:** Belgium, Denmark, France, Israel, Luxembourg, Norway, Spain, Switzerland, the Netherlands, and the UK. **Moderate-risk countries:** Austria, Cyprus, Finland, Germany, Greece, Iceland, Ireland, Italy, Malta, Portugal, San Marino, Slovenia, and Sweden. **High-risk countries:** Albania, Bosnia and Herzegovina, Croatia, Czech Republic, Estonia, Hungary, Kazakhstan, Poland, Slovakia, and Turkey. **Very-high-risk countries:** Algeria, Armenia, Azerbaijan, Belarus, Bulgaria, Egypt, Georgia, Kyrgyzstan, Latvia, Lebanon, Libya, Lithuania, Montenegro, Morocco, Republic of Moldova, Romania, Russian Federation, Serbia, Syria, TFYR (Macedonia), Tunisia, Ukraine, and Uzbekistan.

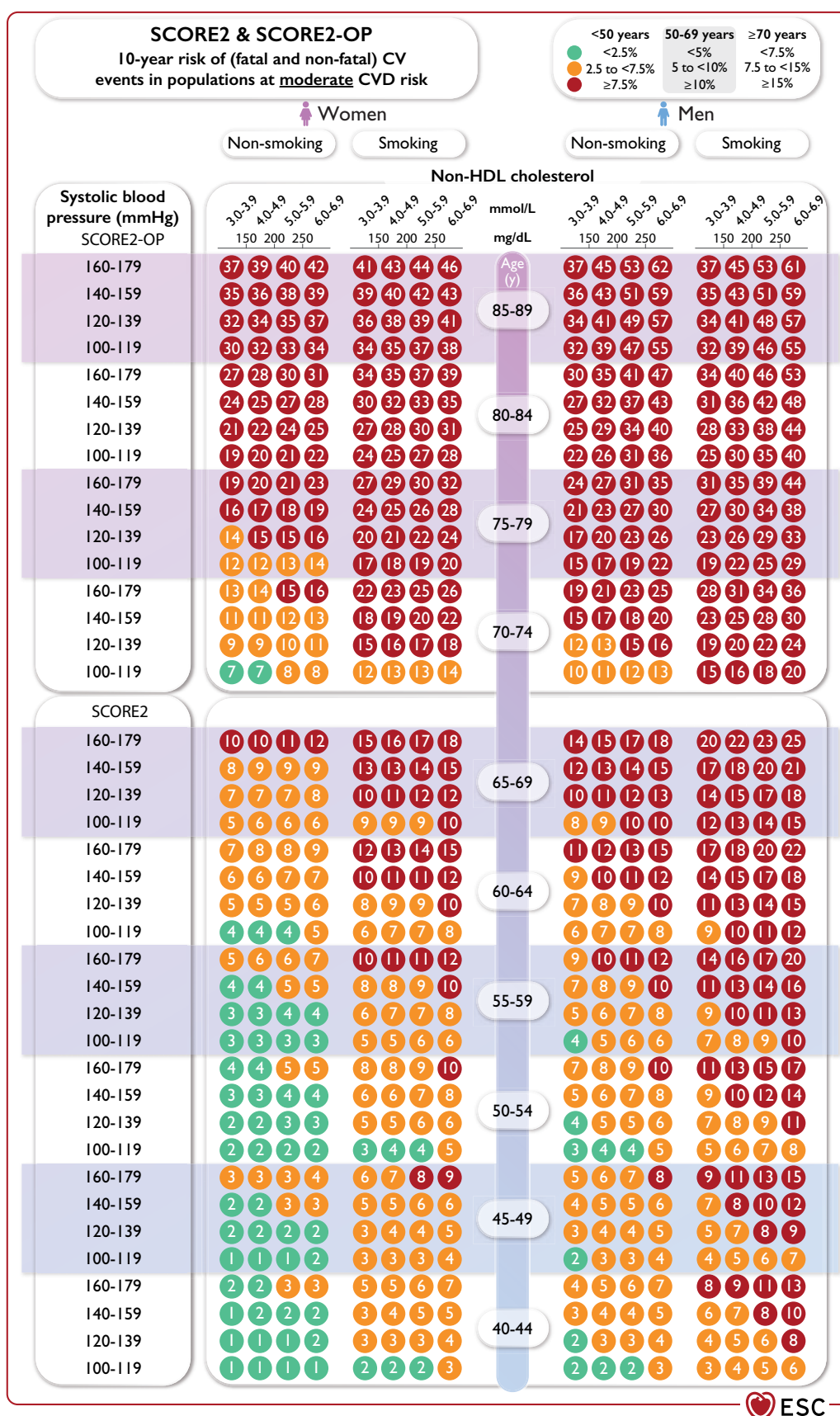


Figure 3 Continued.

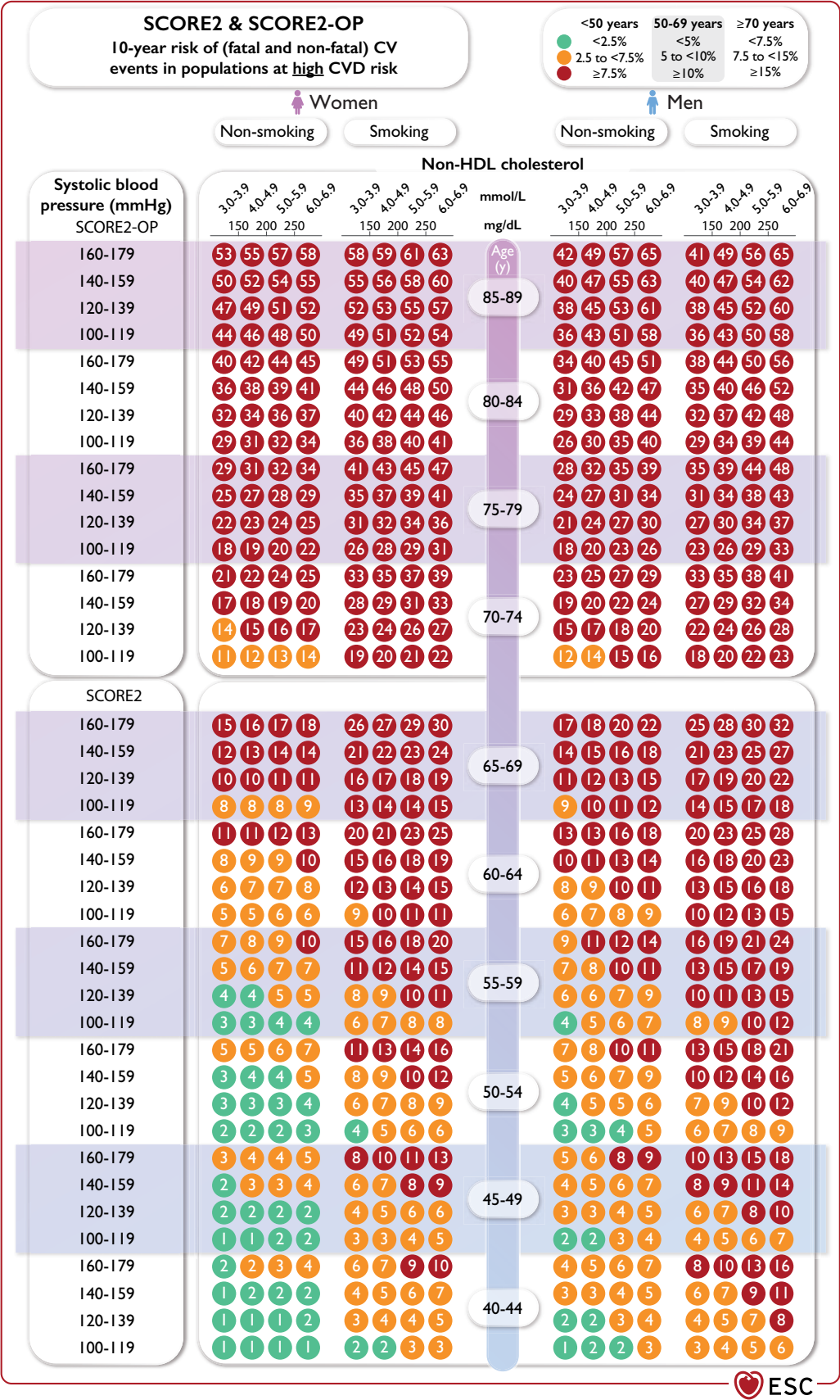


Figure 3 Continued.

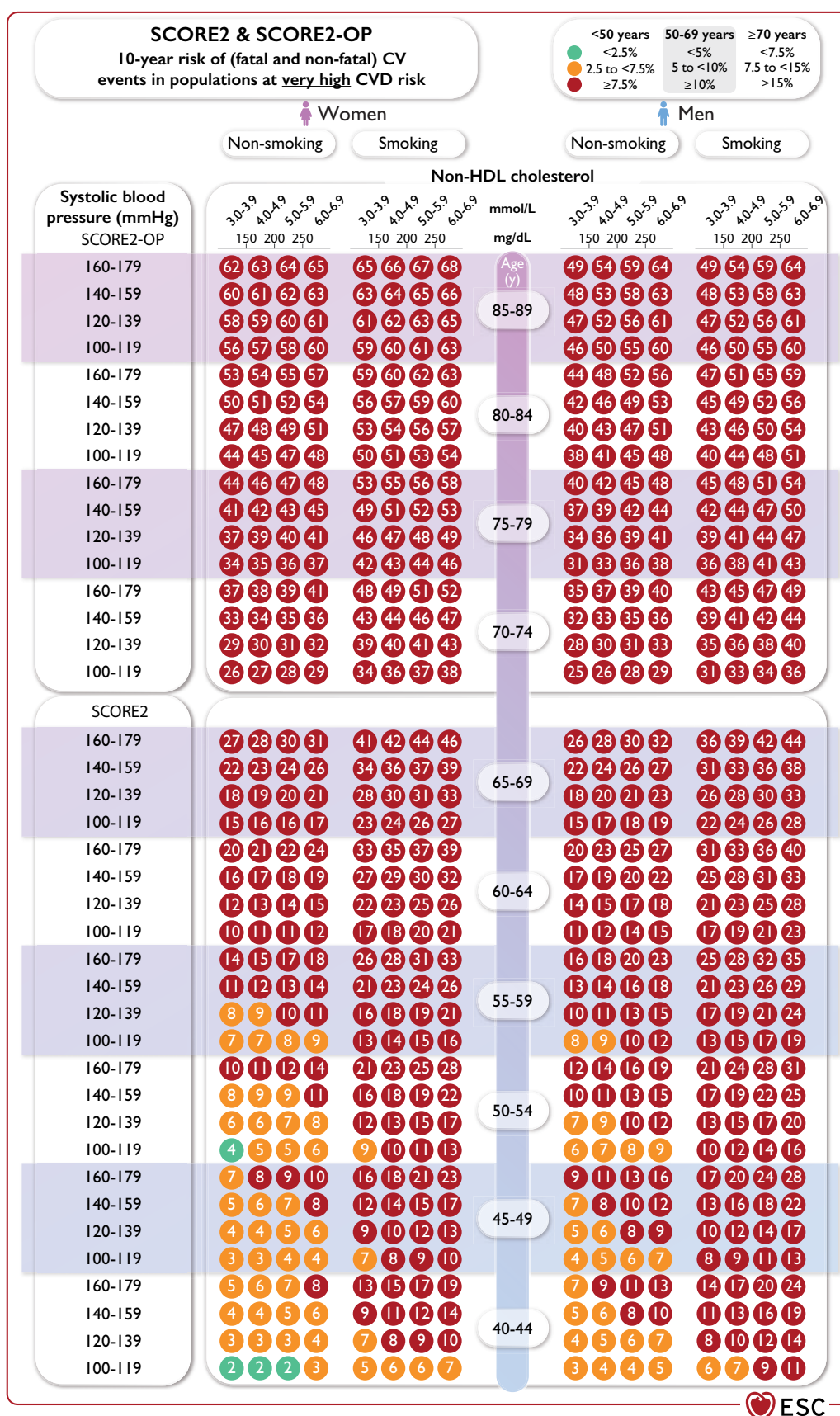


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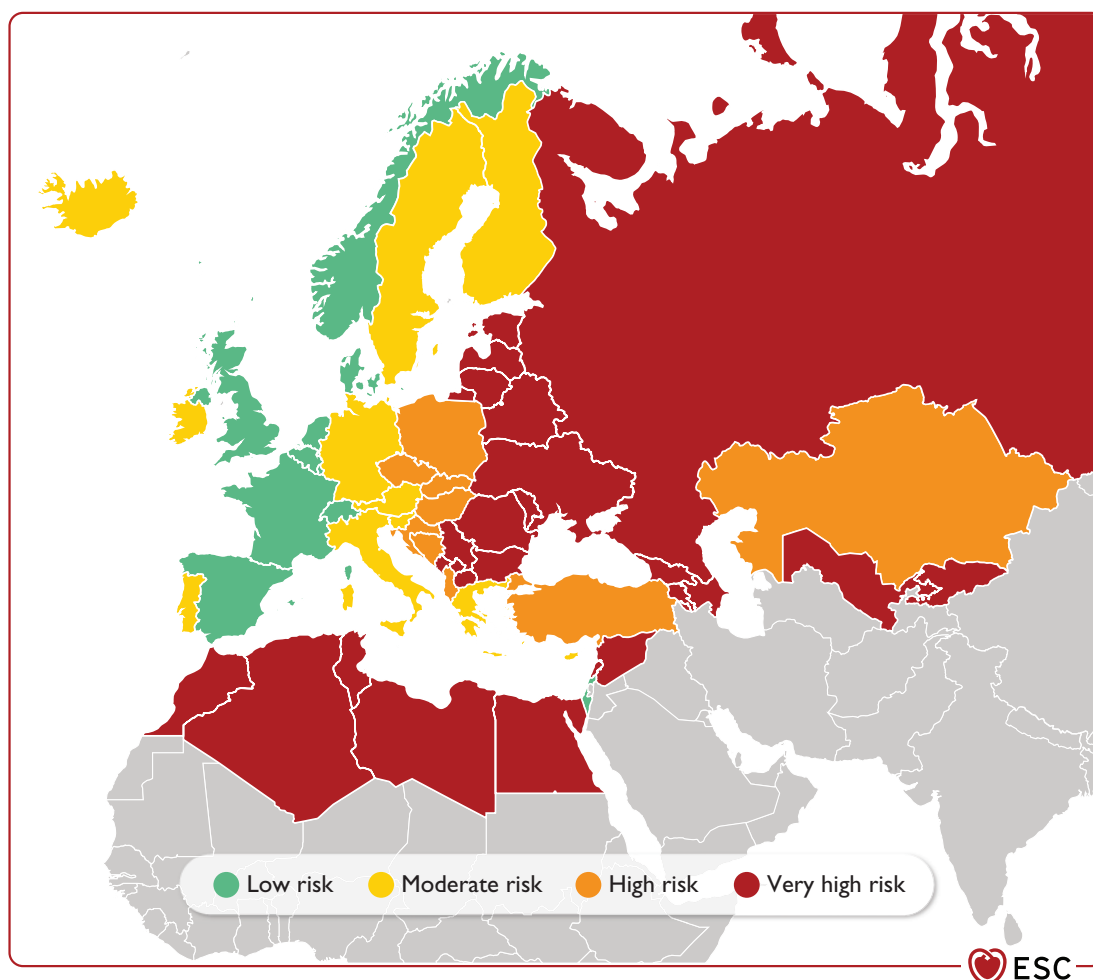


Figure 4 Risk regions based on World Health Organization cardiovascular mortality rates.^{68,72,73}

models that do not take into account the competing risk of non-CVD mortality, tend to overestimate the actual 10-year risk of CVD, and hence overestimate the potential benefit of treatment.⁷¹ The SCORE2-OP algorithm estimates 5-year and 10-year fatal and non-fatal CVD events (myocardial infarction, stroke) adjusted for competing risks in apparently healthy people aged ≥ 70 years.⁷²

SCORE2 and SCORE2-OP are calibrated to four clusters of countries (low, moderate, high, and very high CVD risk) that are grouped based on national CVD mortality rates published by the WHO (*Supplementary Table 3* and *Figure 4*).⁷³

Low-risk countries: Belgium, Denmark, France, Israel, Luxembourg, Norway, Spain, Switzerland, the Netherlands, and the United Kingdom (UK).

Moderate-risk countries: Austria, Cyprus, Finland, Germany, Greece, Iceland, Ireland, Italy, Malta, Portugal, San Marino, Slovenia, and Sweden.

High-risk countries: Albania, Bosnia and Herzegovina, Croatia, Czech Republic, Estonia, Hungary, Kazakhstan, Poland, Slovakia, and Turkey.

Very high-risk countries: Algeria, Armenia, Azerbaijan, Belarus, Bulgaria, Egypt, Georgia, Kyrgyzstan, Latvia, Lebanon, Libya, Lithuania, Montenegro, Morocco, Republic of Moldova, Romania, Russian Federation, Serbia, Syria, The Former Yugoslav Republic (Macedonia), Tunisia, Ukraine, and Uzbekistan. A multiplier approach has been used for converting CVD mortality

rates to fatal and non-fatal CVD events.⁷⁴ The SCORE2 algorithm can be accessed in the ESC CVD Risk app (freely available from app stores) and in risk charts for the four clusters of countries (*Figure 4*). The SCORE2 charts do not apply to persons with documented CVD or other high-risk conditions such as DM, FH, or other genetic or rare lipid or BP disorders, CKD, and in pregnant women.

To estimate a person's 10-year risk of total CVD events, one must first identify the correct cluster of countries and the accompanying risk table for their sex, smoking status, and (nearest) age. Within that table, one then finds the cell nearest to the person's BP and non-HDL-C. Risk estimates then need to be adjusted upwards as the person approaches the next age category.

3.2.3.3 Translating cardiovascular disease risk to treatment thresholds

While no risk threshold is universally applicable, the intensity of treatment should increase with increasing CVD risk. In individual cases, however, no lower threshold of total CVD risk precludes treatment of risk factors. Conversely, no high threshold for total CVD risk implies 'mandatory' treatment. Across the entire range of CVD risk, the decision to initiate interventions remains a matter of individual consideration and shared decision-making (see also *section 4.1*). In general, risk factor treatment recommendations are based on

Table 5 Cardiovascular disease risk categories based on SCORE2 and SCORE2-OP in apparently healthy people according to age

	<50 years	50–69 years	≥70 years ^a
Low-to-moderate CVD risk: risk factor treatment generally not recommended	<2.5%	<5%	<7.5%
High CVD risk: risk factor treatment should be considered	2.5 to <7.5%	5 to <10%	7.5 to <15%
Very high CVD risk: risk factor treatment generally recommended ^a	≥7.5%	≥10%	≥15%

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CVD = cardiovascular disease.

^aIn apparently healthy people ≥70 years old, the treatment recommendation for lipid-lowering drugs is Class IIb ('may be considered').

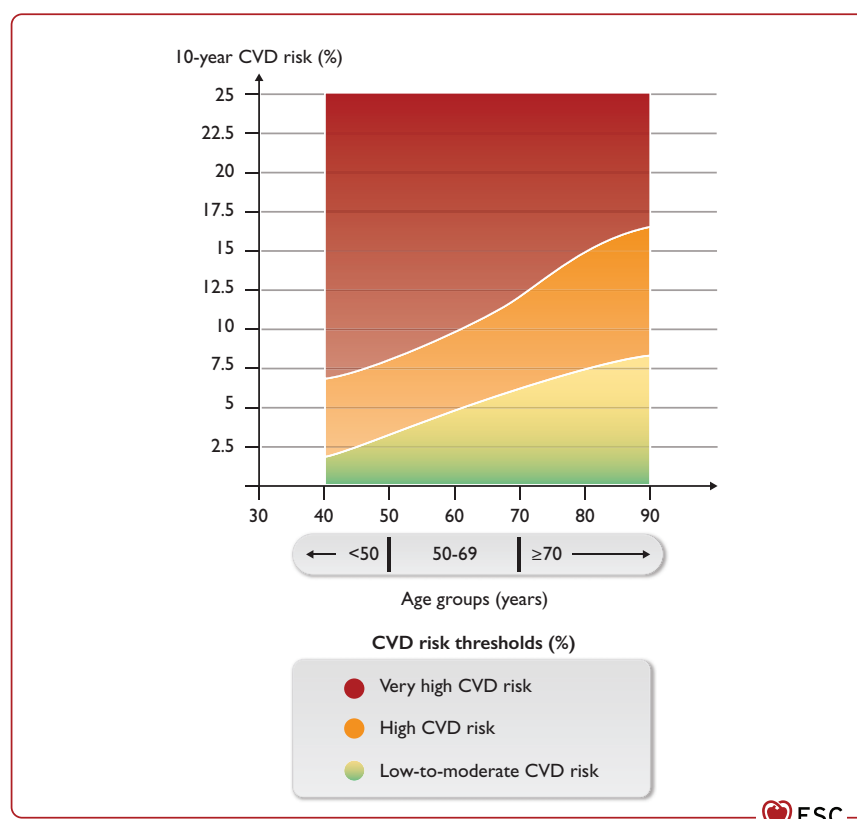
The division of the population into three distinct age groups (<50, 50–69, and ≥70 years) results in a discontinuous increase in risk thresholds for low-to-moderate, high, and very high risk. In reality, age is obviously continuous, and a sensible application of the thresholds in clinical practice would require some flexibility in handling these risk thresholds as patients move towards the next age group, or recently passed the age cut-off. Figure 5 illustrates how a continuous increase in age relates to increasing risk thresholds, and may be used as a guide for daily practice.

categories of CVD risk ('low-to-moderate', 'high', and 'very high'). The cut-off risk levels for these categories are numerically different for various age groups to avoid undertreatment in the young and to avoid overtreatment in older persons. As age is a major driver of CVD risk, but lifelong risk factor treatment benefit is higher in younger people, the risk thresholds for considering treatment are lower for younger people (Table 5).

Risk categories do not 'automatically' translate into recommendations for starting drug treatment. In all age groups, consideration of risk modifiers, lifetime CVD risk, treatment benefit, comorbidities, frailty, and patient preferences may further guide treatment decisions.

Also, note that many patients can move themselves towards a lower risk category without taking drugs just by stopping smoking. Finally, note that persons ≥70 years old may be at very high risk whilst being at target SBP, and primary prevention with lipid-lowering drugs in older persons is a Class IIb ('may consider') recommendation; see section 4.6.

In the 50–69-year age range, a 10-year CVD mortality risk threshold of 5% estimated with the previously used SCORE algorithm corresponds, on average, to a 10-year fatal and non-fatal CVD risk threshold of 10% estimated with SCORE2, as approximately the same number of people are above the risk threshold and would qualify for treatment.⁶⁸

**Figure 5** Schematic representation of increasing 10-year cardiovascular disease risk thresholds across age groups. CVD = atherosclerotic cardiovascular disease.

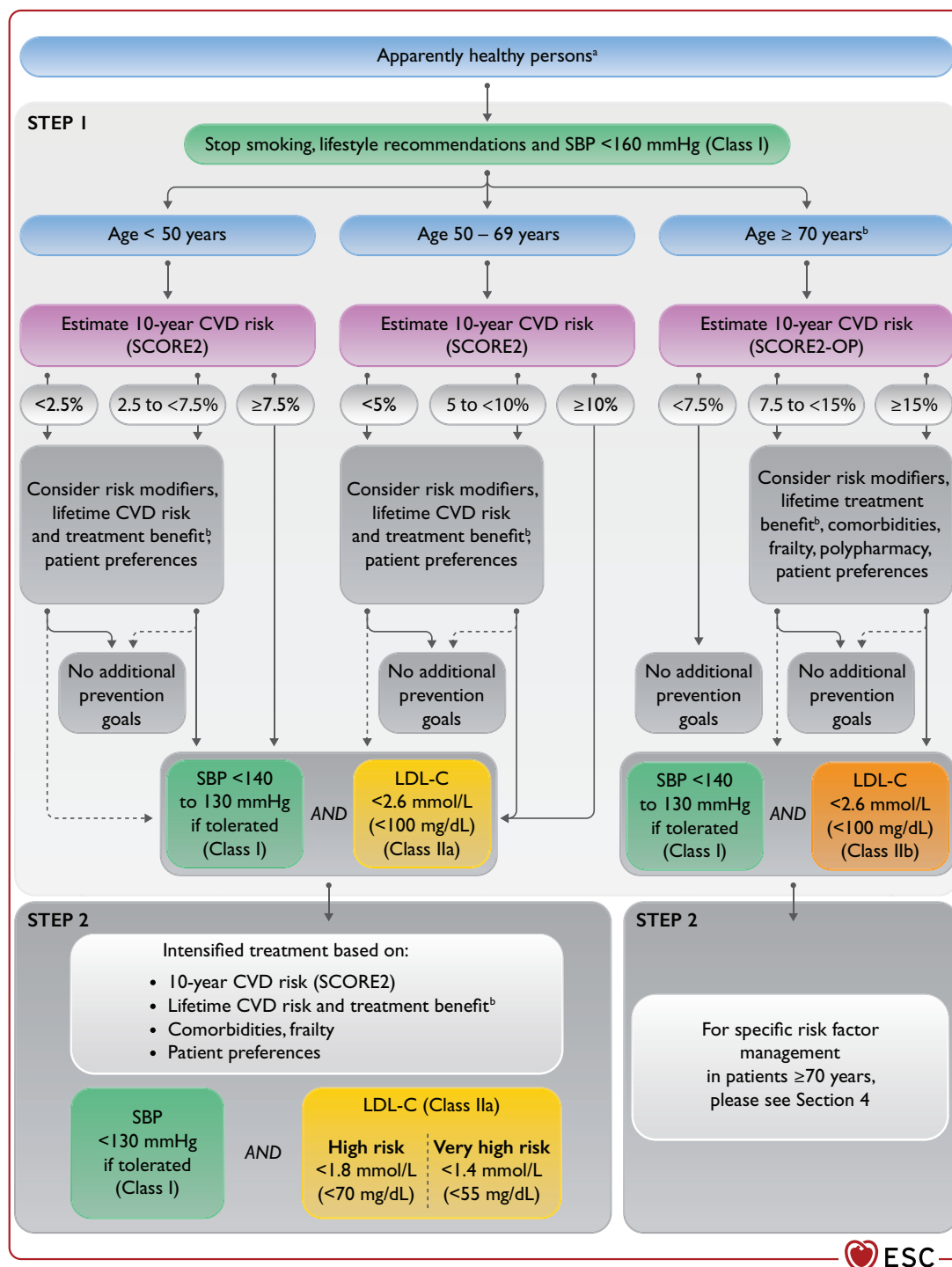


Figure 6 Flow chart of cardiovascular disease risk and risk factor treatment in apparently healthy persons. ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; CVD = cardiovascular disease; DM = diabetes mellitus; ESC = European Society of Cardiology; FH = familial hypercholesterolaemia; LDL-C = low-density lipoprotein cholesterol; LIFE-CVD = LIFETIME-perspective CardioVascular Disease; SBP = systolic blood pressure; SCORE2 = Systematic Coronary Risk Estimation 2; SCORE2-OP = Systematic Coronary Risk Estimation 2-Older Persons. Solid lines represent default options for the majority of people. Dotted lines represent alternative choices for some, depending on the patient-specific characteristics and conditions indicated in the boxes. Ultimate treatment goals for SBP (<130 mmHg) and LDL-C (according to level of risk) according to the respective ESC Guidelines are to be pursued as indicated. The stepwise approach has to be applied as a whole: after STEP 1, considering proceeding to the intensified goals of STEP 2 is mandatory. Risk scores are available in the ESC CVD Risk Calculator app for mobile devices (<https://www.escardio.org/Education/ESC-Prevention-of-CVD-Programme/Risk-assessment/esc-cvd-risk-calculation-app>) and at websites such as <https://www.u-prevent.com>. ^aDoes not include patients with CVD, DM, CKD, or FH. ^bThe LIFE-CVD model for estimating lifetime CVD risk and treatment benefit is calibrated for low- and moderate-risk regions (see Box 1).

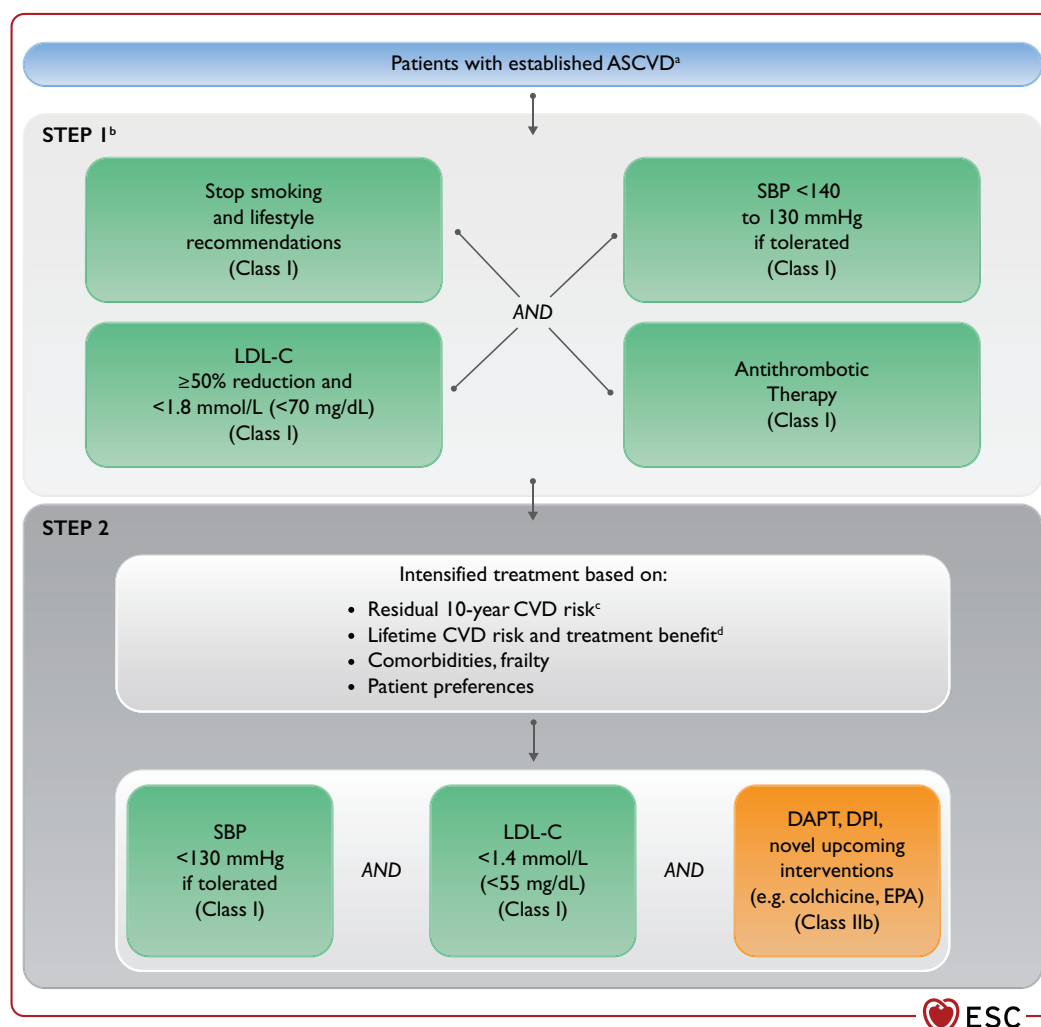


Figure 7 Flow chart of cardiovascular risk and risk factor treatment in patients with established atherosclerotic cardiovascular disease. Ultimate treatment goals for SBP (<130 mmHg) and LDL-C (according to level of risk) according to the respective ESC Guidelines^{3,4} are to be pursued as indicated. The stepwise approach has to be applied as a whole: after STEP 1, considering proceeding to the intensified goals of STEP 2 is mandatory. ACS = acute coronary syndromes; ASCVD = atherosclerotic cardiovascular disease; CR = cardiac rehabilitation; CVD = cardiovascular disease; DAPT = dual antiplatelet therapy; DM = diabetes mellitus; ESC = European Society of Cardiology; EUROASPIRE = European Action on Secondary and Primary Prevention by Intervention to Reduce Events; LDL-C = low-density lipoprotein cholesterol; SBP = systolic blood pressure; SMART = Secondary Manifestations of Arterial Disease. Risk scores are available in the ESC CVD Risk Calculator app for mobile devices (<https://www.escardio.org/Education/ESC-Prevention-of-CVD-Programme/Risk-assessment/esc-cvd-risk-calculation-app>) and at websites such as <https://www.u-prevent.com>. ^aFor patients with DM see DM flow chart (Figure 8). ^bFor patients with recent ACS, these prevention goals are part of participation in CR (Class I/A). ^cFor patients aged ≥70 years, a high 10-year risk may be associated with a lower absolute lifetime benefit from treatment due to limited life expectancy. ^dLifetime treatment benefit is expressed as extra CVD-free life gained from a certain intervention or treatment intensification.

As the 10-year CVD risk thresholds guide treatment decisions and have an impact on healthcare costs and resources, countries or regions may decide on using higher or lower treatment thresholds.

3.2.3.4 Risk estimation and risk factor treatment in apparently healthy people 50–69 years of age

Stopping smoking, lifestyle recommendations, and SBP <160 mmHg are recommended for all (Figure 6). A 10-year CVD risk (fatal and non-fatal ASCVD events) ≥10% is generally considered 'very high risk', and treatment of CVD risk factors is recommended. A 10-year CVD risk of 5 to <10% is considered 'high risk', and treatment of risk factors should

be considered, taking CVD risk modifiers, lifetime risk and treatment benefit (in low- and moderate-risk regions, Box 1), and patient preferences into account. A 10-year CVD risk <5% is considered 'low-to-moderate risk', and would generally not qualify for risk factor treatment unless one or several risk modifiers (see section 3.3) increase risk, or the estimated lifetime risk and treatment benefit is considered substantial.

3.2.3.5 Risk estimation and risk factor treatment estimation in apparently healthy people ≥70 years of age

Stop smoking, lifestyle recommendations and a SBP <160 mmHg are recommended for all (Figure 6). Age is the dominant driver of CVD

risk, and estimated 10-year CVD risk of almost all individuals ≥ 70 years exceeds conventional risk thresholds. Also, lifetime benefit of treatment in terms of time gained free of CVD is lower in older people. Therefore, the CVD risk thresholds for risk factor treatment are higher in apparently healthy people ≥ 70 years. A 10-year CVD risk $>15\%$ is generally considered 'very high risk', and treatment of ASCVD risk factors is recommended (note: the recommendation for lipid-lowering treatment in apparently healthy people ≥ 70 years is class IIb; 'may be considered'; see [section 4.6](#)). A 10-year CVD risk of 7.5 to $<15\%$ is considered 'high risk', and treatment of risk factors should be considered taking CVD risk modifiers, frailty, lifetime treatment benefit (in low and moderate risk regions, [Box 1](#)), comorbidities, polypharmacy, and patient preferences into account. Given the subjective nature of many of these factors, it is not possible to define strict criteria for these considerations. A 10-year CVD risk $<7.5\%$ is considered 'low-to-moderate risk', and would generally not qualify for risk factor treatment unless one or several risk modifiers ([section 3.3](#)) increase risk or the estimated lifetime risk and treatment benefit is considered substantial.^{75–79}

3.2.3.6 Risk estimation and risk factor treatment in apparently healthy people <50 years of age

Stopping smoking, lifestyle recommendations, and SBP <160 mmHg are recommended for all ([Figure 6](#)). The 10-year CVD risk in relatively young, apparently healthy people is on average low, even in the presence of high risk factor levels, but the lifetime CVD risk is in these circumstances very high. In apparently healthy people <50 years of age, a 10-year CVD risk $\geq 7.5\%$ is generally considered 'very high risk' as this risk relates to a high lifetime risk, and treatment of ASCVD risk factors is recommended. A 10-year CVD risk of 2.5 to $<7.5\%$ is considered 'high risk', and treatment of risk factors should be considered, taking CVD risk modifiers, lifetime risk and treatment benefit (in low- and

moderate-risk regions), and patient preferences into account. A 10-year CVD risk $<2.5\%$ is considered 'low-to-moderate risk', and would generally not qualify for risk factor treatment unless one or several risk modifiers (see [section 3.3](#)) increase risk or the estimated lifetime risk and treatment benefit is considered substantial (see [Box 1](#)) ([Figure 6](#)).^{75–78}

In risk communication with younger people, the lifetime benefit perspective may be useful, as well as discussing the potential of avoiding a devastating CVD event in the short-to-intermediate term, despite the fact that 10-year CVD risk may be very low.

CVD risk predictions, as well as predictions of lifetime benefit of risk factor treatment, are likely to be imprecise at very young age (<40 years). At that age, lipid-lowering and BP-lowering drug treatment are not usually considered, except for patients with FH or specific BP disorders. A healthy lifestyle that is maintained throughout life is more relevant for the very young. Mendelian randomization studies illustrate very nicely that relatively small differences in LDL-C or SBP maintained throughout life have large implications on CVD risk over a lifespan.⁸⁰

3.2.3.7 Risk estimation and risk factor treatment in patients with established atherosclerotic cardiovascular disease

Patients with clinically established ASCVD are, on average, at very high risk of recurrent CVD events if risk factors are not treated. Therefore, smoking cessation, adoption of a healthy lifestyle, and risk factor treatment is recommended in all patients (STEP 1). Further intensification of risk factor treatment by aiming at lower treatment goals (STEP 2) is beneficial in most patients and must be considered, taking 10-year CVD risk, comorbidities, lifetime risk and treatment benefit ([Box 1](#)), frailty, and patient preferences into account in a shared decision-making process ([Figure 7](#)).

After initial risk factor treatment and the achievement of risk factor treatment goals, the individual *residual* risk for recurrent

Box 1. Lifetime CVD risk and treatment benefit estimation

Prevention of CVD by treating risk factors is usually done with a lifetime perspective. Lifetime CVD risk can be approximated by clinical experience with clinical criteria such as age, (change in) risk factor levels, risk modifiers, etc. or estimated in apparently healthy people, patients with established ASCVD, and persons with type 2 DM with specific lifetime CVD risk scores.^{75–77} Lifetime benefit from risk factor management can be estimated by combining lifetime risk models with HRs derived from RCTs, meta-analyses of RCTs, or Mendelian randomization studies, which may provide estimates of the effects of longer-term treatment of risk factors. Online calculators (such as the ESC CVD Risk app) can be used to estimate the average lifetime benefit of smoking cessation (see also [Figure 11](#)), lipid lowering (see also [Figure 12](#)), and BP lowering (see also [Figure 15](#)) on an individual patient level expressed as extra CVD-free life-years.⁷⁸ Average lifetime benefit is easy to interpret and may improve the communication of potential therapy benefits to patients in a shared decision-making process. This may in turn increase patient engagement, self-efficacy, and motivation to adhere to lifestyle changes and drug treatment.

The lifetime risk is an estimate of the age at which there is a 50% probability that a person will either have experienced a CVD event or have died. Lifetime benefit is the numerical difference between the predicted age at which there is a 50% probability that a person will either have experienced a CVD event or have died with and without a proposed treatment. Currently there are no formal treatment thresholds for average lifetime benefit. In addition, the estimated individual lifetime benefit should be viewed in the light of the estimated duration of treatment. Duration of lifelong treatment will generally be longer in young persons compared to older people. Both treatment effect and treatment duration determine the individual 'return on investment' of risk factor treatment. In a shared decision-making process between healthcare provider and patient, the minimum desired benefit of a certain treatment needs to be established, a process in which patient preference, expected treatment harms, and costs can be taken into account.

BP = blood pressure; CVD = cardiovascular disease; DM = diabetes mellitus; ESC = European Society of Cardiology; HR = hazard ratio; RCT = randomized controlled trial.

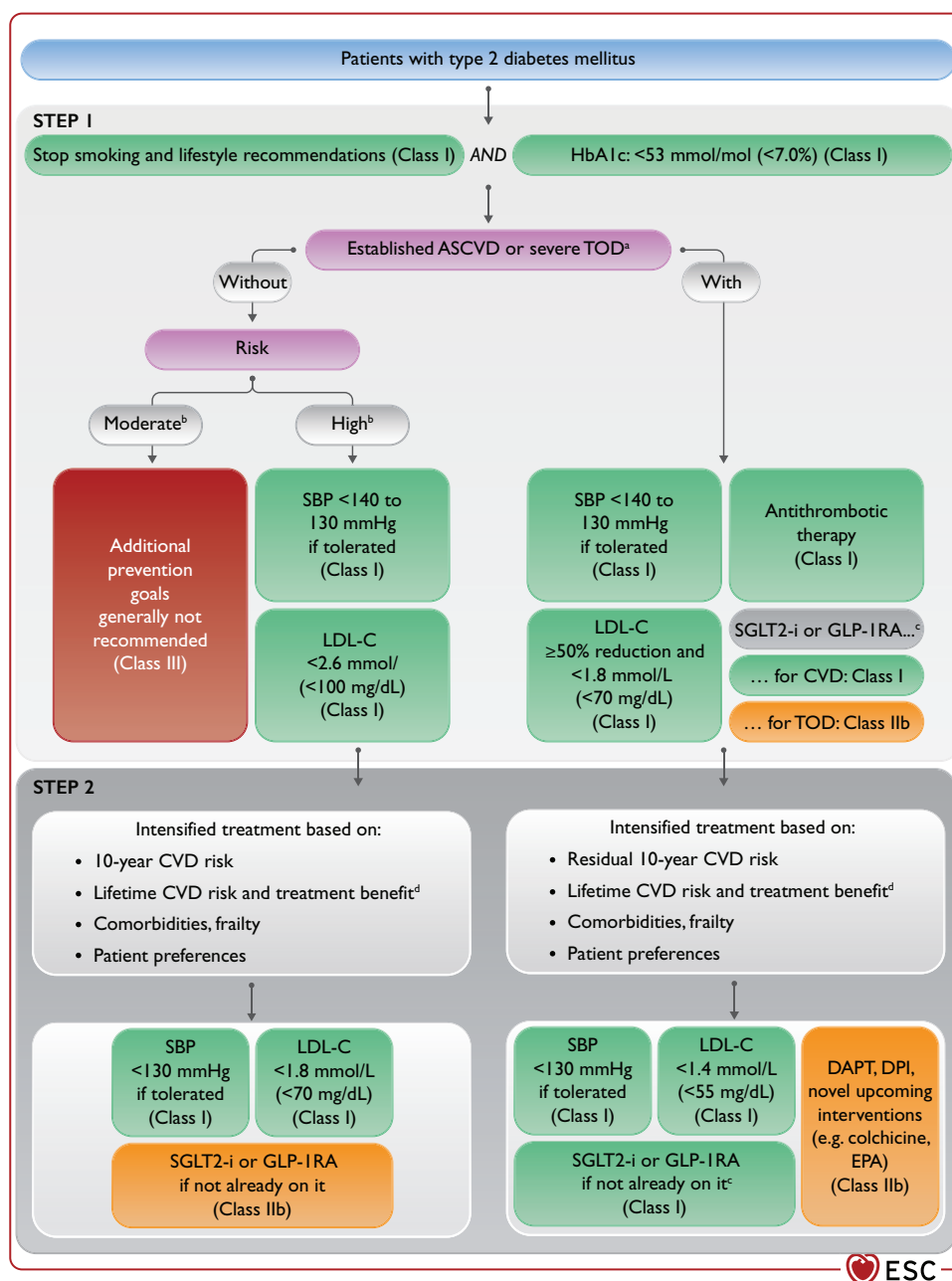


Figure 8 Flow chart of cardiovascular risk and risk factor treatment in patients with type 2 diabetes mellitus. Ultimate treatment goals for SBP (<130 mmHg) and LDL-C (according to level of risk) according to the respective ESC Guidelines^{3,4} are to be pursued as indicated. The stepwise approach has to be applied as a whole: after STEP 1, considering proceeding to the intensified goals of STEP 2 is mandatory. Risk scores are available in the ESC CVD Risk Calculator app for mobile devices (<https://www.escardio.org/Education/ESC-Prevention-of-CVD-Programme/Risk-assessment/esc-cvd-risk-calculation-app>) and at websites such as <https://www.u-prevent.com>. ACR = albumin-to-creatinine ratio; ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; CVD = cardiovascular disease; DAPT = dual antiplatelet therapy; DM = diabetes mellitus; eGFR = estimated glomerular filtration rate; ESC = European Society of Cardiology; GLP-1RA = glucagon-like peptide-1 receptor agonist; HbA1c = glycated haemoglobin; HF = heart failure; LDL-C = low-density lipoprotein cholesterol; SBP = systolic blood pressure; SGLT2 = sodium-glucose cotransporter 2; TOD = target organ damage (retinopathy, nephropathy, neuropathy). ^aSevere TOD is defined as at least one of: eGFR <45 mL/min/1.73 m² irrespective of the presence or absence of albuminuria; eGFR 46–59 mL/min/1.73 m² and microalbuminuria (ACR 30–300 mg/g or 3–30 mg/mmol); proteinuria (ACR >300 mg/g or >30 mg/mmol); presence of microvascular disease in at least three different sites (e.g. microalbuminuria plus retinopathy plus neuropathy). ^bSee Table 4 for CVD risk groups. ^cPatients with prevalent HF or CKD are recommended for SGLT2 inhibitor, and patients post stroke are recommended for GLP-1RA treatment. ^dLifetime treatment benefit is expressed as extra CVD-free life gained from a certain intervention or treatment intensification. See Box 1.

CVD varies widely and should be considered.⁸¹ It is evident that patients with a recent ACS or progressive vascular disease, and patients with DM and vascular disease, are all at exceptionally high risk for recurrent CVD events. For other patients with established ASCVD, the residual risk may be less evident and could be estimated based on clinical criteria such as age, (change in) risk factor levels, and risk modifiers, or by calculation of residual CVD risk with a calculator.

The risk of recurrent CVD is influenced mainly by classical risk factors, vascular disease site, and kidney function. Risk stratification tools for secondary prevention include the SMART (Secondary Manifestations of Arterial Disease) risk score (available in the ESC CVD Risk app) for estimating 10-year residual CVD risk in patients with stable ASCVD, defined as CAD, PAD, or cerebrovascular disease,⁸¹ and the European Action on Secondary and Primary Prevention by Intervention to Reduce Events (EUROASPIRE) risk model, which estimates 2-year risk of recurrent CVD in patients with stable CAD.⁸²

Occasionally, recurrent CVD risk is very high despite maximum (tolerated) conventional treatments. In such cases, novel but less well-established preventive treatments such as dual antithrombotic pathway inhibition,⁸³ icosapent ethyl,⁸⁴ or anti-inflammatory therapy with colchicine (see [section 4.10](#))^{85,86} may be considered.

3.2.3.8 Risk estimation and risk factor treatment in persons with type 2 diabetes mellitus

Most adults with type 2 DM are at high or very high risk for future CVD, particularly from middle age onwards. On average, type 2 DM doubles CVD risk and reduces life expectancy by 4–6 years, with absolute risks highest in those with any target organ damage (TOD). Type 2 DM also increases the risk for cardiorenal outcomes, in particular HF and CKD. Relative risks (RRs) for CVD in type 2 DM are higher at younger ages of onset and are modestly higher in women compared with men.⁸⁷ Smoking cessation and adoption of a healthy lifestyle are recommended for all people with type 2 DM, and risk factor treatment should be considered in all people with DM, at least those above the age of 40 years (see [sections 4.6](#) and [4.7](#)). Still, there is a wide range in individual risk for CVD events, especially after initial risk factor management.⁸⁸

Persons with DM with severe TOD (for definition: see [Table 4](#)) can be considered to be at very high CVD risk, similar to people with established CVD (see [Table 4](#)). Most others with DM are considered to be at high ASCVD risk.⁶⁴ However, an exception can be made for patients with well-controlled short-standing DM (e.g. <10 years), no evidence of TOD, and no additional ASCVD risk factors, who may be considered as being at moderate CVD risk.

In addition to the semi-quantitative division into three risk categories described above, DM-specific risk models may refine risk estimates and illustrate the impact of treatments. These models generally include duration of DM, glycated haemoglobin (HbA1c) level, and presence of TOD. Examples are the ADVANCE (Action in Diabetes and Vascular disease: preterAx and diamicroN-MR Controlled Evaluation) risk score, which predicts 10-year CVD risk, and the UKPDS (UK Prospective Diabetes Study) risk engine, which predicts fatal and non-fatal CVD risk and is available for use in the UK. However, we recommend cautious use of these calculators, since both are based on older cohort data^{89,90} ([Figure 8](#)).

Recommendations for CVD risk estimation

Recommendations	Class ^a	Level ^b
In apparently healthy people <70 years without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorders, estimation of 10-year fatal and non-fatal CVD risk with SCORE2 is recommended. ⁶⁸	I	B
In apparently healthy people ≥70 years without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorders, estimation of 10-year fatal and non-fatal CVD risk with SCORE2-OP is recommended. ⁷²	I	B
In apparently healthy people, after estimation of 10-year fatal and non-fatal CVD risk, lifetime risk and treatment benefit, risk modifiers, frailty, polypharmacy, and patient preferences should be considered.	IIa	C
Patients with established ASCVD and/or DM and/or moderate-to-severe renal disease and/or genetic/rarer lipid or BP disorders are to be considered at high or very high CVD risk. ^{75,77,81,88–90}	I	A
A stepwise treatment-intensification approach aiming at intensive risk factor treatment is recommended for apparently healthy people at high or very high CVD risk, as well as patients with established ASCVD and/or DM, with consideration of CVD risk, treatment benefit of risk factors, risk modifiers, comorbidities, and patient preferences. ^{66,67}	I	B
Treatment of ASCVD risk factors is recommended in apparently healthy people without DM, CKD, genetic/rarer lipid, or BP disorders who are at very high CVD risk (SCORE2 ≥7.5% for age under 50; SCORE2 ≥10% for age 50–69; SCORE2-OP ≥15% for age ≥70 years). ^{68,72}	I	C
Treatment of ASCVD risk factors should be considered in apparently healthy people without DM, CKD, genetic/rarer lipid, or BP disorders who are at high CVD risk (SCORE2 2.5 to <7.5% for age under 50; SCORE2 5 to <10% for age 50–69; SCORE2-OP 7.5 to <15% for age ≥70 years), taking CVD risk modifiers, lifetime risk and treatment benefit, and patient preferences into account.	IIa	C

ASCVD = atherosclerotic cardiovascular disease; BP = blood pressure; CKD = chronic kidney disease (see definition in [Table 4](#)); DM = diabetes mellitus; SCORE2 = Systemic Coronary Risk Estimation 2; SCORE2-OP = Systemic Coronary Risk Estimation 2-Older Persons.

^aClass of recommendation.

^bLevel of evidence.

Intensification of risk factor treatment in STEP 2 must be considered in all patients, taking into account 10-year CVD risk, comorbidities, lifetime risk and treatment benefit (Box 1), frailty, and patient preferences in a shared decision-making process.⁷⁵

3.2.3.9 Risk estimation and risk factor treatment in persons with type 1 diabetes mellitus

People with type 1 DM are at increased CVD risk, and earlier manifestation of type 1 DM relates to more life-years lost in women than men, mostly due to CVD.⁹¹ RRs of CVD are, on average, higher in type 1 vs. type 2 DM, due to an average of three to four extra decades of hyperglycaemia, and usual risk factors contribute strongly to CVD outcomes in type 1 DM.⁹² CVD risks have declined over time, commensurate with improvements in life expectancy.⁹³ Lifetime CVD risks in type 1 DM are higher with poorer glycaemic control, lower social class, and younger age of onset. The absolute risk of CVD events or CVD mortality is highest among those with any evidence of microvascular disease, particularly renal complications, and is strongly influenced by age. CVD risk stratification in persons with type 1 DM may be based on the same risk classification as for type 2 DM, summarized in Table 4, although the level of evidence for type 1 DM is weaker.

3.2.4. Communication of cardiovascular disease risk

Reducing CVD risk at the individual level begins with appropriate assessment of individual risk and effective communication of risk and anticipated risk reduction by risk factor treatment. Patient-doctor interactions are complex and communicating risk is challenging.^{94,95} There is no single ‘correct’ approach; rather, it will depend on the individual’s preferences and understanding, which may differ with education status and numeracy. Risk perception is also strongly affected by emotional factors such as fear, optimism, etc. (‘patients don’t think risk, they feel risk’).⁹⁶

It is important to explore whether patients understand their risk, the anticipated risk reduction, and the pros and cons of intervention, and to identify what is important to them. For example, one patient may focus on living free of medications, whereas another may be less able to change their lifestyle. In terms of outcomes, reducing mortality risk is crucial to some, whereas disease risk is more important to others. Short-term risk may motivate some patients, whereas lifetime benefit (see Box 1) will have more impact in others. In general, visual aids (graphs etc.) improve risk understanding, absolute risk (reduction) is better understood than RR (reduction), and the use of ‘numbers needed to treat’ is less well understood.

In apparently healthy people, the standard approach is to report absolute 10-year risk of a CVD event with SCORE2 or SCORE2-OP, which can be found at the ESC CVD Risk Calculator app (<https://www.escardio.org/Education/ESC-Prevention-of-CVD-Programme/Risk-assessment/esc-cvd-risk-calculation-app>) or at <http://www.heartscore.org> or <https://www.u-prevent.com>. In specific situations, one may opt for expressing risk in terms other than absolute 10-year risk. Examples of such situations include risks in young or very old people. In young people, lifetime risk might be more informative, as 10-year CVD risk is usually low even in the presence of risk factors. In older persons, specific risk estimation is required, taking competing non-CVD mortality into account.⁷⁸ Direct translation

Recommendation for CVD risk communication

Recommendation	Class ^a	Level ^b
An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended. ⁹⁶	I	C

CVD = cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.

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Recommendations for CVD risk modifiers

Recommendations	Class ^a	Level ^b
Stress symptoms and psychosocial stressors modify CVD risk. Assessment of these stressors should be considered. ^{100–102}	IIa	B
CAC scoring may be considered to improve risk classification around treatment decision thresholds. Plaque detection by carotid ultrasound is an alternative when CAC scoring is unavailable or not feasible. ^{103,104}	IIb	B
Multiplication of calculated risk by RR for specific ethnic subgroups should be considered. ¹⁰⁵	IIa	B
The routine collection of other potential modifiers, such as genetic risk scores, circulating or urinary biomarkers, or vascular tests or imaging methods (other than CAC scoring or carotid ultrasound for plaque determination), is not recommended.	III	B

CVD = cardiovascular disease; CAC = coronary artery calcium; RR = relative risk.
^aClass of recommendation.
^bLevel of evidence.

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of RRs to treatment decisions is not recommended, as absolute risk remains the key criterion for starting treatment.

An alternative way of expressing individual risk is to calculate a person’s ‘risk age’.⁹⁶ The risk age of a person with several ASCVD risk factors is the age of a person of the same sex with the same level of risk but with low levels of risk factors. Risk age is an intuitive and easily understood way of illustrating the likely reduction in life expectancy that a young person with a low absolute but high RR of CVD will be exposed to if preventive measures are not adopted. Risk age is also automatically calculated as part of HeartScore (<http://www.heartscore.org/>).^{97–99}

CVD risk may also be expressed with a lifetime rather than a 10-year horizon, for example, the LIFE-CVD (LIFETIME-perspective CardioVascular Disease) calculator (ESC CVD Risk Calculation app or <https://www.u-prevent.com>) (also see Box 1).⁷⁸ Lifetime CVD risk-prediction models identify high-risk individuals both in the short and long term. Such models account for predicted risk in the context of competing risks from other diseases over the remaining expected lifespan of an individual. A similar approach also employing lifetime perspective is to calculate lifetime benefit of preventive

interventions.⁷⁸ Lifetime benefit of preventive interventions can be expressed as gain in CVD-free life (years), which is easier to communicate to a patient and may support the shared decision-making process.

3.3. Potential risk modifiers

Apart from the conventional CVD risk factors included in the risk charts, additional risk factors or types of individual information can also modify calculated risk. Assessment of a potential modifier may be considered if:

- It improves measures of risk prediction, such as discrimination or reclassification (e.g. by calculation of net reclassification index)
- Public health impact is clear (e.g. number needed to screen or net benefit)
- It is feasible in daily practice
- Information is not just available on how risk increases with an unfavourable result, but also on how risk decreases if the modifier shows a favourable result
- The literature on this potential modifier is not distorted by publication bias.

Very few potential modifiers meet all of these criteria. Meta-analyses in this field are, for example, susceptible to substantial publication bias.¹⁰⁶ Also, the exact way of integrating additional information on top of regular risk calculator input parameters is mostly unknown. Finally, RCTs to determine whether the added risk information eventually leads to improved health outcomes are generally lacking.

Assessment of potential risk modifiers seems particularly relevant if the individual's risk is close to a decision threshold. In low-risk or very-high-risk situations, additional information is less likely to alter management decisions. The number of individuals in this 'grey zone' is large. Therefore, feasibility becomes a limitation as modifiers become more complex or expensive, such as some imaging techniques.

Care should be taken not to use risk modifiers solely to increase risk estimates when the modifier profile is unfavourable, but also vice versa. Although an unfavourable risk modifier may increase an individual's estimated risk, a more favourable profile than would be expected based on other patient characteristics must have the opposite effect. Finally, it is important to acknowledge that the degree to which calculated absolute risk is affected by modifiers is generally much smaller than the (independent) RRs reported for these modifiers in the literature.¹⁰⁷

Taking the above into account, we summarize the literature on several popular risk modifiers in this section.

3.3.1. Psychosocial factors

Psychosocial stress is associated, in a dose-response pattern, with the development and progression of ASCVD, independently of conventional risk factors and sex. Psychosocial stress includes stress symptoms (i.e. symptoms of mental disorders), as well as stressors such as loneliness and critical life events. The RRs of psychosocial stress are commonly between 1.2 and 2.0^{108,109} (*Supplementary Table 4*). Conversely, indicators of mental health, such as optimism and a strong sense of purpose, are associated with lower risk.¹⁰⁹ Psychosocial stress has direct biological effects, but is also highly correlated with socioeconomic and behavioural risk factors (e.g. smoking, poor adherence).^{100,109–113} Although the associations of psychosocial stress with CV health are robust, only 'vital exhaustion' has been proven to improve risk reclassification.¹⁰¹ Owing to the importance of stress symptoms among ASCVD patients, several guidelines and scientific statements recommend screening of ASCVD patients for psychological stress^{113–115} (*Box 2* and *Supplementary Table 5*). A recent prospective cohort study with a median follow-up of 8.4 years reported favourable effects of screening for depression on major ASCVD events.¹⁰²

3.3.2. Ethnicity

Europe includes many citizens whose ethnic background originates in countries such as India, China, North Africa, and Pakistan. Given the considerable variability in ASCVD risk factors between immigrant groups, no single CVD risk score performs adequately in all groups. Rather, the use of a multiplying factor would be helpful to take account of CVD risk imposed by ethnicity independent of other risk factors in the risk score. The most contemporary relevant data come from the QRISK3 findings in the UK,¹⁰⁵ although this focuses on a wider range of CVD outcomes and not simply on CVD mortality.

Immigrants from South Asia (notably India and Pakistan) present higher CVD rates independent of other risk factors, whereas adjusted CVD risks appear lower in most other ethnic groups. The reasons for such differences remain inadequately studied, as do the risks associated with other ethnic backgrounds. Based on such data, the following correction factors, based on data from the UK, could be applied when assessing CVD risk using risk calculators.¹⁰⁵ Ideally, country and risk-calculator-specific RRs should be used, as the impact of ethnicity may vary between regions and risk calculators.

- Southern Asian: multiply the risk by 1.3 for Indians and Bangladeshis, and 1.7 for Pakistanis.
- Other Asian: multiply the risk by 1.1.
- Black Caribbean: multiply the risk by 0.85.
- Black African and Chinese: multiply the risk by 0.7.

Box 2. Core topics for psychosocial assessment

Simultaneous diagnostic assessment	At least one in five patients carries a diagnosis of a mental disorder, usually presenting with bodily symptoms (e.g. chest tightness, shortness of breath). Therefore, physicians should be equally attentive to somatic as to emotional causes of symptoms.
Screening	Screening instruments assessing depression, anxiety, and insomnia are recommended (e.g. Patient Health Questionnaire, ¹¹⁶ see <i>Supplementary Table 5</i>). ^{117,118}
Stressors	There are simple questions to get into a conversation about significant stressors ¹¹² : Are you bothered by stress at work, financial problems, difficulties in the family, loneliness, or any stressful events?
Need for mental health support	Are you interested in a referral to a psychotherapist or mental health service?

3.3.3. Imaging

3.3.3.1 Coronary artery calcium

Coronary artery calcium (CAC) scoring can reclassify CVD risk upwards and downwards in addition to conventional risk factors, and may thus be considered in men and women with calculated risks around decision thresholds.^{103,104} Availability and cost-effectiveness of large-scale CAC scanning must, however, be considered in a locoregional context (see [section 2.3](#) on cost-effectiveness). If CAC is detected, its extent should be compared with what would be expected for a patient of the same sex and age. Higher-than-expected CAC increases the person's calculated risk, whereas absent or lower-than-expected CAC is associated with lower than calculated risk. CAC scoring does not provide direct information on total plaque burden or stenosis severity, and can be low or even zero in middle-aged patients with soft non-calcified plaque. Clinicians are advised to consult existing protocols for details of how to assess and interpret CAC scores.

3.3.3.2 Contrast computed tomography coronary angiography

Contrast computed tomography angiography (CCTA) allows identification of coronary stenoses and predicts cardiac events.¹¹⁹ In the SCOT-HEART (Scottish Computed Tomography of the Heart) study, 5-year rates of coronary death or myocardial infarction were reduced when CCTA was used in patients with stable chest pain.¹²⁰ The relative reduction in myocardial infarction was similar in patients with non-cardiac chest pain. Whether CCTA improves risk classification or adds prognostic value over CAC scoring is unknown.

3.3.3.3 Carotid ultrasound

Systematic use of intima-media thickness (IMT) to improve risk assessment is not recommended due to the lack of methodological standardization, and the absence of added value of IMT in predicting future CVD events, even in the intermediate-risk group.¹²¹

Plaque is defined as the presence of a focal wall thickening that is $\geq 50\%$ greater than the surrounding vessel wall, or as a focal region with an IMT measurement ≥ 1.5 mm that protrudes into the lumen.¹²² Although the evidence is less extensive than it is for CAC, carotid artery plaque assessment using ultrasonography probably also reclassifies CVD risk,^{104,122} and may be considered as a risk modifier in patients at intermediate risk when a CAC score is not feasible.

3.3.3.4 Arterial stiffness

Arterial stiffness is commonly measured using either aortic pulse wave velocity or arterial augmentation index. Studies suggest that arterial stiffness predicts future CVD risk and improves risk classification.¹²³ However, measurement difficulties and substantial publication bias¹⁰⁶ argue against widespread use.

3.3.3.5 Ankle brachial index

Estimates are that 12–27% of middle-aged individuals have an ankle brachial index (ABI) <0.9 , around 50–89% of whom do not have typical claudication.¹²⁴ An individual patient data meta-analysis concluded that the reclassification potential of ABI was limited, perhaps with the exception of women at intermediate risk.¹²⁵

3.3.3.6 Echocardiography

In view of the lack of convincing evidence that it improves CVD risk reclassification, echocardiography is not recommended to improve CV risk prediction.

3.3.4. Frailty

Frailty is a multidimensional state, independent of age and multimorbidity, that makes the individual more vulnerable to the effect of stressors. It constitutes a functional risk factor for unfavourable outcomes, including both high CV and non-CV morbidity and mortality.^{126,127}

Frailty is not the same as ageing and the two should not be confused. The incidence of frailty increases with age, but people of the same chronological age can differ significantly in terms of health status and vitality. 'Biological age' is much more important in the context of clinical status (including frailty features) and hard clinical outcomes (including CVD events).^{126,127} Similarly, although the presence of comorbidities can exacerbate frailty within an individual, frailty is not the same as multimorbidity (see [section 6.7](#)).

Frailty screening is indicated in every elderly patient, but should also be performed in every individual regardless of his/her age, when being at risk of accelerated ageing.^{126,127} Most of the tools relate to frail features, including slowness, weakness, low physical activity (PA), exhaustion, and shrinking (e.g. Fried scale, Short Physical Performance Battery, Rockwood Clinical Frailty Scale, handgrip strength, gait speed).^{126–129} Frailty assessment is important at each stage of an ASCVD trajectory. During an acute CVD event, however, frailty assessment is more difficult, and either relies on history taking or should be postponed to when patients return to a stable condition.

Frailty is a potential modifier of global CVD risk. The impact of frailty on CVD risk has been demonstrated across the spectrum of ASCVD, including people with ASCVD risk factors, patients with subclinical ASCVD, stable ASCVD, acute cerebral and coronary syndromes, and HF.^{126–130} with frailty itself rather than classical CVD risk factors predicting both all-cause and CVD mortality in the very old.^{130,131} Importantly, the ability of frailty measures to improve CVD risk prediction has not been formally assessed. Hence, we do not recommend that frailty measures are integrated into formal CVD risk assessment.

Importantly, frailty may influence treatment. Non-pharmacological interventions (e.g. balanced nutrition, micronutrient supplementation, exercise training, social activation) aiming to prevent, attenuate, or reverse frailty are of utmost importance.^{126,127,132} In terms of pharmacotherapy and device implantations, frailty assessment is not a method to determine the eligibility for any particular treatment, but rather serves to build an individualized care plan with predefined priorities. Frail individuals often have comorbidities, polypharmacy, and may be more susceptible to drug side-effects and serious complications during invasive and surgical procedures.^{126,127}

3.3.5. Family history

Family history of premature CVD is a simple indicator of CVD risk, reflecting the genetic and environment interplay.¹³³ In the few studies that simultaneously assessed the effects of family history and genetics, family history remained significantly associated with CVD after adjusting for genetic scores.^{134,135} However, family history only marginally improves the prediction of CVD risk beyond conventional ASCVD risk factors.^{136–141} Possible explanations are the varying definitions

of family history applied and that conventional ASCVD risk factors largely explain the impact of family history.

A family history of premature CVD is simple, inexpensive information that can trigger comprehensive risk assessment in individuals with a family history of premature CVD.¹³⁶

3.3.6. Genetics

The aetiology of ASCVD has a genetic component, but this information is not currently used in preventive approaches.¹⁴² Advances on polygenic risk scores for risk stratification could increase the use of genetics in prevention.^{143–145} For ASCVD, there is, however, a lack of consensus regarding which genes and corresponding single nucleotide polymorphisms should be included, and whether to use risk factor-specific or outcome-specific polygenic risk scores.¹⁴⁶ Polygenic risk scoring has shown some potential to improve ASCVD risk prediction for primary prevention,^{147–149} but the incremental prediction accuracy is relatively modest and needs further evaluation in both men and women.^{150,151} Additional evidence is also needed to evaluate the clinical utility of polygenic risk scores in other clinical settings, such as in patients with pre-existing ASCVD.¹⁵²

3.3.7. Socioeconomic determinants

Low socioeconomic status and work stress are independently associated with ASCVD development and prognosis in both sexes.^{153,154} The strongest association has been found between low income and CVD mortality, with a RR of 1.76 [95% confidence interval (CI) 1.45–2.14].¹⁵⁵ Work stress is determined by job strain (i.e. the combination of high demands and low control at work) and effort-reward imbalance. There is preliminary evidence that the detrimental impact of work stress on ASCVD health is independent of conventional risk factors and their treatment.¹⁵⁶

3.3.8. Environmental exposure

Environmental exposures with CVD risk modifying potential include air and soil pollution as well as above-threshold noise levels. Evaluating individual cumulative exposure to pollutants and noise remains challenging, but when available, might impact on individual risk assessment.

Components of outdoor air pollution include airborne particulate matter [PM; ranging in size from coarse particles 2.5–10 µm in diameter, to fine (<2.5 µm; PM_{2.5}), and ultrafine (<0.1 µm)] and gaseous pollutants (e.g. ozone, nitrogen dioxide, volatile organic compounds, carbon monoxide, sulphur dioxide), produced primarily by combustion of fossil fuels. Soil and water pollutions are also CVD risk modifiers; increased exposure to lead, arsenic, and cadmium is associated with multiple CVD outcomes including hypertension, coronary heart disease (CHD), stroke, and CVD mortality.¹⁵⁷ Ambient PM pollution recently ranked as a leading modifiable mortality risk factor and also responsible for attributable disability adjusted life-years at the global level.¹⁵⁸ A recent model estimated that loss of life expectancy due to ambient air pollution is similar to, if not exceeding, that due to tobacco smoking, and accounts for a global excess mortality estimated at 8.8 million/year.¹⁵⁹

The short-term attributable effects on mortality are linked primarily to exposure to PM, nitrogen dioxide, and ozone, with an average 1.0% increase of all-cause mortality for an increment of 10 µg/m³ in

exposure to PM_{2.5}; the long-term effects are associated mainly with PM_{2.5}. The evidence linking exposure to PM and CVD events is based on large-scale epidemiological studies and experimental studies. Associations with ASCVD mortality vary, but the majority of cohort studies link long-term air pollution with an increased risk of fatal or non-fatal CAD, and with subclinical atherosclerosis. Evidence suggests that reduction of PM_{2.5} is associated with improvements in inflammation, thrombosis, and oxidative stress, and a decrease in death from ischaemic heart disease.^{38,160,161} As sufficiently precise individual exposure estimates are hard to obtain, formal risk reclassification is difficult to quantify at present.

Recommendations for cardiovascular disease risk related to air pollution

Recommendations	Class ^a	Level ^b
Patients at (very) high risk for CVD may be encouraged to try to avoid long-term exposure to regions with high air pollution.	I IIb	C
In regions where people have long-term exposure to high levels of air pollution, (opportunistic) CVD risk screening programmes may be considered.	I IIb	C

CVD = cardiovascular disease.
^aClass of recommendation.
^bLevel of evidence.

3.3.9. Biomarkers in blood or urine

Many biomarkers have been suggested to improve risk stratification. Some may be causal [e.g. lipoprotein(a), reflecting a pathogenic lipid fraction], whereas others may reflect underlying mechanisms (e.g. C-reactive protein reflecting inflammation) or indicate early cardiac damage (e.g. natriuretic peptides or high-sensitivity cardiac troponin).

In the 2016 Guidelines,² we recommended against the routine use of biomarkers because most do not improve risk prediction, and publication bias seriously distorts the evidence.^{106,162} New studies confirm that C-reactive protein has limited additional value.¹⁰³ There is renewed interest in lipoprotein(a), but it too provides limited additional value in terms of reclassification potential.^{163,164} Cardiac biomarkers are promising,^{165,166} but further work is needed.

3.3.10. Body composition

Worldwide, BMI has increased substantially in recent decades, in children, adolescents, and adults.⁴³ In observational studies, all-cause mortality is minimal at a BMI of 20–25 kg/m², with a J- or U-shaped relation in current smokers.^{45,46} Mendelian randomization analyses suggest a linear relation between BMI and mortality in never-smokers and a J-shaped relation in ever-smokers.⁴⁴ A meta-analysis concluded that both BMI and waist circumference are similarly strongly and continuously associated with ASCVD in the elderly and the young and in men and women.⁴⁷

Among those with established ASCVD, the evidence is contradictory. Systematic reviews of patients with ACS or HF have suggested an ‘obesity paradox’ whereby obesity appears protective.^{167,168 169} However, this evidence should be interpreted with caution as reverse causality and other biases may be operating.⁴⁵

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3.3.10.1 Which index of obesity is the best predictor of cardiovascular risk?

BMI can be measured easily and is used extensively to define categories of body weight (see [Supplementary Table 6](#)). Body fat stored in visceral and other ectopic depots carries a higher risk than subcutaneous fat. Several measures of global and abdominal fat are available, of which waist circumference is the simplest to measure. The WHO thresholds for waist circumference are widely accepted in Europe. Two action levels are recommended:

- Waist circumference ≥ 94 cm in men and ≥ 80 cm in women: no further weight gain
- Waist circumference ≥ 102 cm in men and ≥ 88 cm in women: weight reduction advised.

Different cut-offs for anthropometric measurements may be required in different ethnicities.

The phenotype of 'metabolically healthy obesity', defined by the presence of obesity in the absence of metabolic risk factors, has gained interest. Long-term results support the notion that metabolically healthy obesity is a transient phase moving towards glucometabolic abnormalities rather than a specific 'state'.¹⁷⁰

3.3.10.2 Risk reclassification

The associations between BMI, waist circumference, and waist-to-hip ratio and CVD are maintained after adjustment for conventional risk factors. However, these measures did not improve CVD risk prediction as assessed by reclassification.⁴⁷

Recommendations for cardiovascular disease assessment in specific clinical conditions

Clinical condition	Recommendations	Class ^a	Level ^b
CKD	In all CKD patients, with or without DM, appropriate screening for ASCVD and kidney disease progression, including monitoring changes in albuminuria is recommended. ¹⁷²	I	C
Cancer	It is recommended to monitor cardiac dysfunction using imaging techniques and circulating biomarkers before, periodically during, and after cancer treatment. ¹⁷³	I	B
	Cardioprotection in high-risk patients (those receiving high cumulative doses or combined radiotherapy) receiving anthracycline chemotherapy may be considered for prevention of LV dysfunction. ^{174,175}	IIb	B
	Screening for ASCVD risk factors and optimization of the CVD risk profile is recommended in patients on treatment for cancer.	I	C

Continued

COPD	It is recommended that all COPD patients be investigated for ASCVD and ASCVD risk factors.	I	C
Inflammatory conditions	Assessment of total CVD risk may be considered in adults with chronic inflammatory conditions. ¹⁷⁶	IIb	B
	Multiplication of calculated total CVD risk by a factor of 1.5 should be considered in adults with rheumatoid arthritis. ^{177,178}	IIa	B
Migraine	Presence of migraine with aura should be considered in CVD risk assessment. ^{179–181}	IIa	B
	Avoidance of combined hormonal contraceptives may be considered in women with migraine with aura. ^{182,183}	IIb	B
Sleep disorders and OSA	In patients with ASCVD, obesity, and hypertension, regular screening for non-restorative sleep is indicated (e.g. by the question: 'how often have you been bothered by trouble falling or staying asleep, or sleeping too much?').	I	C
	If there are significant sleep problems, which are not responding within 4 weeks to sleep hygiene, referral to a specialist is recommended.	I	C
Mental disorders	It is recommended that mental disorders with either significant functional impairment or decreased use of healthcare systems be considered as influencing total CVD risk.	I	C
Sex-specific conditions	In women with a history of preeclampsia and/or pregnancy-induced hypertension, periodic screening for hypertension and DM should be considered. ^{184–187}	IIa	B
	In women with a history of polycystic ovary syndrome or gestational DM, periodic screening for DM should be considered. ^{188–191}	IIa	B
	In women with a history of premature or stillbirth, periodic screening for hypertension and DM may be considered. ^{192,193}	IIb	B
	Assessment of CVD risk should be considered in men with ED.	IIa	C

ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; CV = cardiovascular; CVD = cardiovascular disease; DM = diabetes mellitus; ED = erectile dysfunction; LV = left ventricular; OSA = obstructive sleep apnoea.

^aClass of recommendation.

^bLevel of evidence.

3.3.10.3 Assess risk factors and cardiovascular disease risk in persons with obesity

Comprehensive CVD risk assessment should be considered in individuals with unfavourable body composition. The main risk-related sequelae of adiposity include hypertension, dyslipidaemia, insulin resistance, systemic inflammation, a prothrombotic state, albuminuria, as well as a decline in estimated glomerular filtration rate (eGFR)¹⁷¹ and the development of type 2 DM, CVD events, as well as HF and AF.

3.4. Clinical conditions

Individual calculated risks of CVD, as evaluated by conventional risk factors in risk scores, are subject to refinement by potential risk modifiers as highlighted in [section 3.3](#). Beyond these potential modifiers, specific clinical conditions can influence CVD risk. These clinical conditions often increase the likelihood of CVD, or are associated with poorer clinical prognosis. The current section reviews some of these conditions, which are not often included in traditional risk scores but may be integrated in some national risk scores. Here we discuss how these conditions increase this risk.

Many clinical conditions share common CVD and ASCVD risk factors and therefore treating these allows a synergistic reduction in the overall burden of disease.

3.4.1. Chronic kidney disease

Worldwide, the total number of individuals with chronic kidney disease (CKD) who are not treated with kidney replacement therapy was approximately 850 million in 2017.¹⁹⁴ This number accounts to a prevalence of 10–12% among men and women. CKD is the third fastest growing cause of death globally.¹⁹⁵

CKD is defined as abnormalities of kidney structure or function, present for >3 months, with health implications. Criteria and markers of kidney damage, especially kidney disease due to DM, are albuminuria [albumin-to-creatinine ratio (ACR) >30 mg/g in spot urine specimens] and glomerular filtration rate (GFR) <60 mL/min/1.73 m². GFR can be estimated (eGFR) from calibrated serum creatinine and estimating equations using the CKD-EPI (Chronic Kidney Disease Epidemiology) Collaboration formula. Kidney disease severity is differentiated into stages (categories) according to the level of GFR and albuminuria; a patient with an eGFR <60 mL/min/1.73 m² is classified as having CKD stage 3a, which represents an advanced kidney function impairment.¹⁷²

Among persons with CKD, CVD is the leading cause of morbidity and death.¹⁹⁶ Even after adjustment for known CAD risk factors, including DM and hypertension, mortality risk progressively increases with worsening CKD.¹⁹⁷ As GFR declines below approximately 60–75 mL/min/1.73 m², the probability of developing CAD increases linearly,¹⁹⁸ with up to triple the CVD mortality risk when reaching an eGFR of 15 mL/min/1.73 m². Kidney disease is associated with a very high CVD risk. Among persons with CKD, there is a high prevalence of traditional CAD risk factors, such as DM and hypertension. The use of CAC score to risk stratify patients with CKD might be a promising tool.^{199–203} Furthermore, persons with CKD are also exposed to other non-traditional ASCVD risk factors such as uraemia-related ones, including inflammation, oxidative stress, and promoters of vascular calcification. CKD and kidney failure not only increase the risk of CAD, they also modify its clinical presentation and cardinal symptoms.²⁰⁴

3.4.2. Atrial fibrillation

Atrial fibrillation (AF) appears to be associated with an increased risk of death and of CVD and kidney disease.²⁰⁵ Furthermore, AF appears to be a stronger risk factor for CVD in women than in men.²⁰⁶

The prevalence of AF ranges between 2% and 4%, and a 2.3-fold rise is expected, owing in part to ageing of the population and intensified searching for undiagnosed AF, as well as lower CV death.²⁰⁷ The age-adjusted incidence, prevalence, and lifetime risk of AF are lower in women vs. men and in non-white vs. white cohorts.^{208,209} The lifetime AF risk estimate is now 1 in 3 individuals of European ancestry at an index age of 55 years.²¹⁰ ASCVD risk factor burden and comorbidities, including lifestyle factors, and age significantly affect the lifetime risk for AF development.^{211–213} The observed effect of clinical ASCVD risk factor burden and multiple comorbidities on the lifetime risk of AF (significantly increasing from 23.4% among individuals with an optimal clinical risk factor profile to 33.4% and 38.4% in those with borderline and elevated clinical risk factors, respectively²¹⁴) suggests that early intervention and control of modifiable ASCVD risk factors could reduce incident AF. The continuum of unhealthy lifestyle, risk factor(s), and CVDs can contribute to atrial remodelling/cardiomyopathy and development of AF that commonly results from a combined effect of multiple interacting factors (*Figure 9*).²¹⁵ Risk factor and CVD management reduces AF burden. Targeted therapy of underlying conditions may significantly improve maintenance of sinus rhythm in patients with persistent AF and HF.²¹⁶ However, studies addressing isolated management of specific conditions alone (e.g. hypertension) yielded inconsistent findings.²¹⁷

The overall annual risk of ischaemic stroke in patients with AF is 5%, but varies considerably according to comorbidities.²¹⁵ Cardioembolic strokes associated with AF are usually more severe, and often recurrent.²¹⁸ Furthermore, AF appears to be a stronger predictor of stroke in women than in men.²¹⁵ AF is also associated with impaired cognitive function, ranging from mild cognitive impairment to dementia.²¹⁹ AF is independently associated with a two-fold increased risk of all-cause mortality in women and a 1.5-fold increased risk in men.²²⁰ In one population, the most common causes of death were HF (14.5%), malignancy (23.1%), and infection/sepsis (17.3%), while stroke-related mortality was only 6.5%.²²¹ These data indicate that, in addition to anticoagulation and HF treatment, comorbid conditions need to be actively treated to reduce AF-related mortality and morbidity.

Regarding PA, both sedentary lifestyles and very high levels of PA are associated with development of AF (U-shaped association), through different mechanisms. Furthermore, when AF develops in athletes it is not associated with the same increased risk of stroke.

3.4.3. Heart failure

Heart failure (HF) of ischaemic origin constitutes a severe clinical manifestation of ASCVD. Conversely, HF itself (predominantly of ischaemic aetiology) increases the risk of CVD events (myocardial infarction, arrhythmias, ischaemic stroke, CV death).

Asymptomatic LV dysfunction (systolic or/and diastolic dysfunction) as well as overt symptomatic HF [across the spectrum of LVEF, i.e. HF with reduced ejection fraction (HFrEF), HF with mid-range ejection fraction,²²² and HF with preserved ejection fraction (HFpEF)] increases the risk of urgent CV hospitalizations (including hospitalizations due to HF worsening) and CV and all-cause deaths.

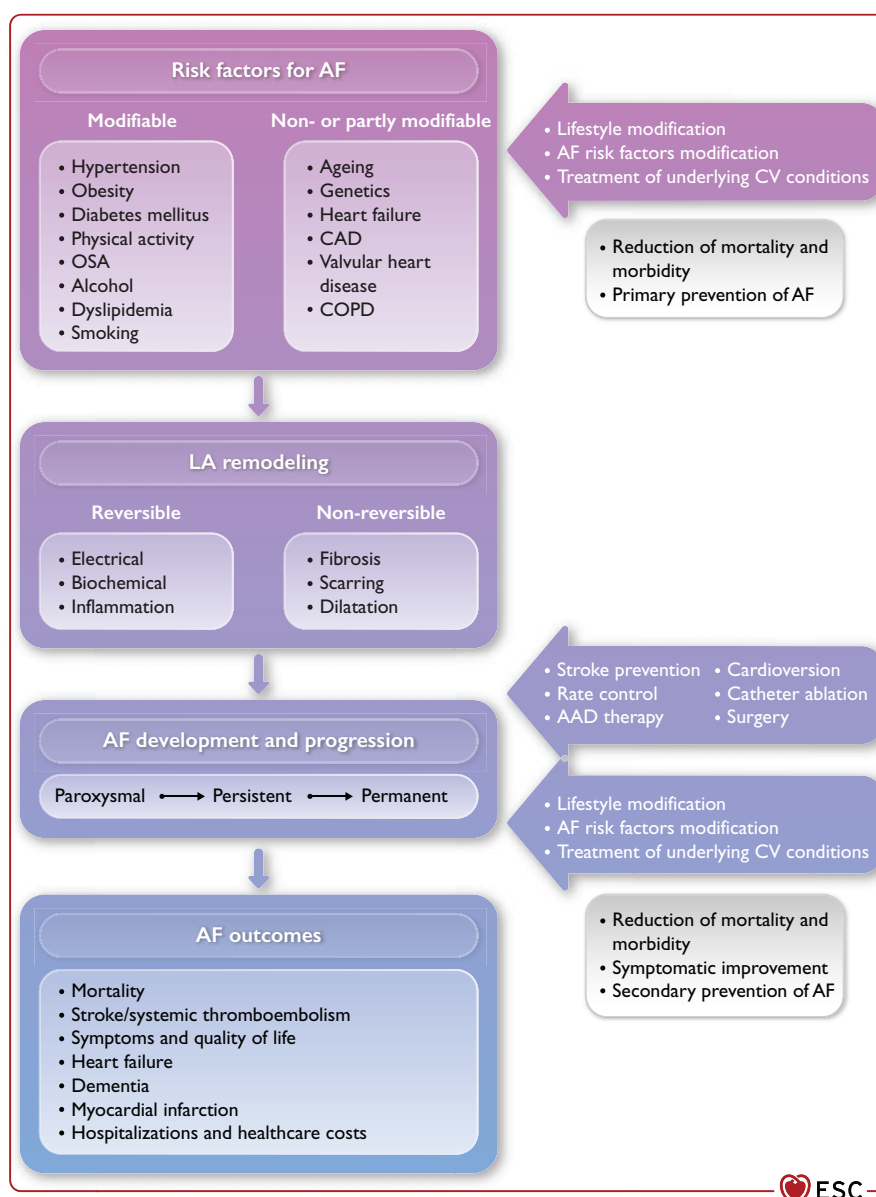


Figure 9 The role of risk factors and comorbidities in atrial fibrillation.²¹⁵ AF = atrial fibrillation; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; CV = cardiovascular; DM = diabetes mellitus; HF = heart failure; OSA = obstructive sleep apnoea.

These unfavourable effects on clinical outcomes have been demonstrated in asymptomatic subjects without overt CVD, in patients with acute and previous myocardial infarction, in patients with acute and previous stroke, and in patients with other clinical manifestations of CVD.²²³

The diagnosis of ischaemic HF positions individuals at very high CV risk, and justifies recommendations as for secondary prevention therapeutic strategies. Additionally, for patients with symptomatic HFrEF, several drugs are recommended to reduce the risk of CV morbidity and mortality (see [section 6.2](#)).

3.4.4. Cancer

In patients with cancer, there is an overlap between cancer and ASCVD risk factors, with shared biological mechanisms and genetic predispositions. Prevention and treatment of these is therefore

beneficial in reducing both CVD as well as cancer risk. Moreover, the rates of the extent of CVD risk depend on both the CVD toxicity of treatments and patient-related factors. Owing to recent improvements in clinical outcomes for many patients with cancer, CVD mortality may ultimately exceed those from most forms of cancer recurrence.^{224,225}

The rapidly expanding variety of novel anticancer drugs/adjuvant therapies has demonstrated a wide range of both early and late CVD side-effects, including cardiomyopathy, LV dysfunction, HF, hypertension, CAD, arrhythmias, and other injuries. Therefore, effective strategies for the prediction and prevention of CVD toxicities are critically important. The latency and severity of radiotherapy cardiotoxicity, as well as accelerated atherosclerosis and cerebral vascular disease, is related to multiple factors, including the dose (total per fraction), the volume of the heart irradiated, concomitant

administration of other cardiotoxic drugs, and patient factors (which include, amongst other factors, younger age, traditional risk factors, and history of heart disease).^{226,227} Furthermore, radio- and chemotherapy may exert direct vascular effects and increase atherosclerosis-related CVD outcomes.^{227,228}

3.4.4.1 Diagnosis and screening

Signs or symptoms of cardiac dysfunction should be monitored before and periodically during and after cancer treatment for early detection of abnormalities in patients receiving potentially cardiotoxic chemotherapy. Detection of subclinical abnormalities using imaging and measurement of circulating biomarkers (such as cardiac troponins and natriuretic peptides) is currently recommended.^{173,229} Measures of myocardial strain, particularly systolic global longitudinal strain, may precede a significant decline in LVEF.^{230–233}

3.4.4.2 Prevention of cardiotoxicity and cardiovascular risk factors

RCTs of preventive therapy with renin-angiotensin-aldosterone system (RAAS) inhibitors and/or beta-blockers after trastuzumab or anthracyclines have reported contradictory results.^{230,234,235} The main benefits are less marked LV remodelling or a reduced decline in LVEF observed with cardiac magnetic resonance, but translation into better outcomes remains speculative.

Exercise should be strongly advised. In particular, aerobic exercise is considered a promising non-pharmacological strategy to prevent and/or treat chemotherapy toxicity.²³⁶ A study showed a significantly higher risk of CVD in survivors of childhood cancer than in non-cancer adult controls, and particularly in survivors of adult-onset cancer with underlying ASCVD risk factors.²³⁷ Therefore, aggressive management of ASCVD risk factors in this population is recommended.

3.4.5. Chronic obstructive pulmonary disease

Chronic obstructive pulmonary disease (COPD) is a complex, progressive respiratory disorder and currently the fourth leading cause of death worldwide. It is characterized by chronic airflow limitation with respiratory symptoms and is associated with an increased inflammatory response and abnormalities of the airways caused by significant exposure to noxious particles or gases (mainly smoking). Although COPD is recognized and thoroughly investigated as a CVD comorbidity, its role as an ASCVD risk factor is not well established. Nevertheless, COPD patients have a two- to three- fold increased risk of CVD compared with age-matched controls when adjusted for tobacco smoking. Patients with mild-to-moderate COPD are 8–10 times more likely to die from ASCVD than respiratory failure, having higher rates of hospitalization and death due to CVD, stroke, and HF.^{238,239} CVD also runs undiagnosed; less than one-third of COPD patients with electrocardiographic (ECG) evidence of myocardial infarction are diagnosed with CVD.²⁴⁰ CVD mortality increases by 28%, and the frequency of non-fatal coronary events by 20%, for every 10% decrease in the forced expiratory volume in 1 second (FEV1).²⁴¹ Acute COPD exacerbations, mainly due to infections, are frequent and are responsible for a four-fold increase of CVD events.²⁴² The risk of both myocardial infarction and ischaemic stroke is increased during the 3 months after an acute exacerbation.²⁴³

The high prevalence of CVD in COPD patients may be explained by the fact that both diseases share common risk factors, such as smoking, ageing, hypertension, and dyslipidaemia.²⁴⁴ Metabolic

syndrome and reduced PA is present in 34% of COPD patients, with its most prevalent components being hypertension (56%), abdominal obesity (39%), and hyperglycaemia (44%).²⁴⁵ CVD may be caused by hypoxia during exercise due to lung hyperinflation, high resting heart rates, impaired vasodilatory capacity, and peripheral, cardiac, and neurohumoral sympathetic stress. Atherosclerosis and coronary artery calcification may be the result of oxidative stress, and reductions in antiaging molecules causing both lung and vascular ageing.²⁴⁶ Systemic inflammation is prominent in COPD, with circulating biomarkers in high concentrations and associated with increased mortality.²⁴⁷ Troponin is elevated during an acute exacerbation of COPD, and 10% of hospitalized patients meet the definition of acute myocardial infarction (AMI).²⁴⁸ B-natriuretic peptide level, if elevated, increases the mortality risk.²⁴⁹

Systemic inflammation and oxidative stress caused by COPD promote vascular remodelling, stiffness, and atherosclerosis, and induce a 'procoagulant' state that affects all vasculature types.²⁵⁰ Cognitive impairment and dementia due to cerebral microvascular damage is correlated with COPD severity; patients have a 20% increased risk for both ischaemic and haemorrhagic stroke, which may be up to seven-fold higher following an acute exacerbation.²⁵¹ PAD is present in about 9% of COPD patients,²⁵² who have an almost doubled risk of developing PAD,²⁵³ as well as an increased prevalence of carotid plaques related to the disease severity.²⁵⁴ Finally, COPD is positively associated with abdominal aortic aneurysm, regardless of smoking status.²⁵⁵

Cardiac arrhythmias are common and may be due to the haemodynamic effects (pulmonary hypertension, diastolic dysfunction, atrial structural, and electrical remodelling) caused by the disease in combination with autonomic imbalance and abnormal ventricular repolarization.²⁵⁶ AF is frequent, directly associated with FEV1, usually triggered by acute exacerbations of COPD, and an independent predictor of in-hospital COPD mortality.^{257,258} COPD is also a risk factor for ventricular tachycardia independent of LVEF,²⁵⁹ and for sudden cardiac death independent of CVD risk profile.²⁶⁰

Unrecognized ventricular dysfunction is common in COPD,²⁶¹ although HF is 3.8 times more common in COPD patients than in controls.²⁶² Patients with frequent acute exacerbations have a high frequency of diastolic dysfunction; HFpEF risk is higher because of a high prevalence of hypertension and DM.²⁶³

Considering these facts, it seems of utmost importance to screen COPD patients for ASCVD and ASCVD risk factors, bearing in mind that COPD affects the accuracy of CVD diagnostic tests. Achieving adequate exercise is difficult, vasodilators for myocardial perfusion scanning may be contraindicated because of the risk of bronchospasm, and stress or transthoracic echocardiography is often disturbed by poor ultrasound windows. Computed tomography coronary angiography or magnetic resonance imaging may be alternatives, but remain expensive, time consuming, and not always available.

The use of COPD medications (i.e. long-acting muscarinic antagonists and long-acting beta agonists) is not associated with overall CV adverse events in patients with stable COPD. Olodaterol may reduce the risk of overall CV adverse events and formoterol may decrease the risk of cardiac ischaemia. Long-acting beta agonists may reduce the incidence of hypertension, but may also increase the risk of HF, so should be used with caution in HF patients.²⁶⁴

3.4.6. Inflammatory conditions

Inflammatory conditions increase CVD risk both acutely and over time. The best evidence for chronic inflammation increasing CVD risk is available for rheumatoid arthritis, which increases CVD risk by approximately 50% beyond established risk factors.¹⁷⁶ Hence, a low threshold for assessment of total CVD risk is appropriate in adults with rheumatoid arthritis, and one should consider increasing the risk estimate based on the level of disease activity.¹⁷⁶ There is also evidence for an approximately 20% increased CVD risk in patients with active inflammatory bowel disease.²⁶⁵

In other chronic inflammatory conditions, such as psoriasis¹⁷⁷ and ankylosing spondylitis,¹⁷⁸ CVD risk may also be increased. However, the strength of the evidence is less strong, as is the independence of such increased risks from the classical ASCVD risk factors. Nonetheless, it seems prudent to at least consider CVD risk assessment in patients with any chronic inflammatory condition, and to take into account the presence of such conditions when there is doubt regarding initiation of preventive interventions. The cumulative disease burden and recent degree of inflammation are important determinants of the risk-enhancing effect.

Apart from optimal anti-inflammatory treatment, CVD risk in inflammatory conditions should be treated with similar interventions as in the general high-risk population, as there is evidence that traditional methods to lessen risk (e.g. lipid-lowering treatment) are just as beneficial in preventing ASCVD.

3.4.7. Infections (human immunodeficiency virus, influenza, periodontitis)

Infection with human immunodeficiency virus (HIV) is associated with a 19% increased risk of LEAD and CAD beyond that explained by traditional atherosclerotic risk factors.^{266,267} However, for those with sustained CD4 cell counts <200 cells/mm³, the risk of incident LEAD events is nearly two-fold higher, whereas for those with sustained CD4 cell counts ≥500 cells/mm³, there is no excess risk of incident LEAD events compared with uninfected people.²⁶⁸

CVD and influenza have long been associated, due to an overlap in the peak incidence of each disease during winter months. Epidemiological studies have noted an increase in CV deaths during influenza epidemics, indicating that CV complications of influenza infection, including acute ischaemic heart disease and, less often, stroke, are important contributors to morbidity and mortality during influenza infection.

The risk of AMI or stroke is more than four times higher after a respiratory tract infection, with the highest risk in the first 3 days after diagnosis.²⁶⁹ Preventing influenza, particularly by means of vaccination, could prevent influenza-triggered AMI.²⁷⁰

Studies have linked periodontal disease to both atherosclerosis and CVD,^{271–273} and serological studies have linked elevated antibody titres of periodontal bacteria to atherosclerotic disease.²⁷⁴ Nevertheless, if active treatment or prevention of periodontitis improves, clinical prognosis requires further studies despite preliminary evidence.^{275–277}

3.4.8. Migraine

Migraine is a highly prevalent condition affecting around 15% of the general population.²⁷⁸ There are two main types of migraine—migraine without aura, which is the most common subtype, and

migraine with aura, which accounts for about one-third of all migraines; in many patients the two forms coexist.

Available data indicate that migraine overall is associated with a two-fold increased risk of ischaemic stroke and a 1.5-fold increase in the risk of cardiac ischaemic disease.^{179–181,279,280} The associations are more evident for migraine with aura.^{179,180,280} Given the young mean age of the population affected by migraine, the absolute increase in risk is small at the individual level, but high at the population level because of the high migraine prevalence.²⁸¹

Several lines of evidence also indicate that the vascular risk of subjects with migraine may be magnified by cigarette smoking¹⁸² and by the use of combined hormonal contraceptives.^{183,281–283} Contraception using combined hormonal contraceptives should therefore be avoided in women with migraine.^{282,283} However, further information is needed as good-quality studies assessing risk of stroke associated with low-dose oestrogen use in women with migraine are lacking.

3.4.9. Sleep disorders and obstructive sleep apnoea

Sleep disturbances or abnormal sleep durations are associated with increased CVD risk.^{284–286} Regarding sleep duration, 7 h seems to be optimal for CV health.²⁸⁷

In the general population, the prevalence of general sleep disturbances is around 32.1%: 8.2% for insomnia, 6.1% for parasomnia, 5.9% for hypersomnolence, 12.5% for restless legs disorder and limb movements during sleep, and 7.1% for sleep-related breathing disorder [e.g. obstructive sleep apnoea (OSA)].²⁸⁸ All sleep disturbances are strongly associated with mental disorders and share hyperarousal as an underlying mechanism.^{289,290}

The most important sleep-related breathing disorder is OSA, which is characterized by repetitive episodes of apnoea, each exceeding 10 seconds. Despite the strong associations of OSA with CVD, including hypertension, stroke, HF, CAD, and AF, treatment of OSA by positive airway pressure (PAP) has failed to improve hard CV outcomes in patients with established CVD.^{291–293} Therefore, interventions that include behaviour change (reduction of obesity, alcohol abstinence), sleep hygiene, and stress reduction in addition to PAP are needed.^{290,294} Regarding hypertension and OSA, there are modest effects of PAP on BP levels, but only in patients with ABPM-confirmed resistant hypertension who use PAP for more than 5.8 h/night.²⁹⁵

3.4.10. Mental disorders

The 12-month prevalence of mental disorders or mental health disorders in the general European population is between 27% and 38% depending on sources and definitions.²⁹⁶ All mental disorders (e.g. anxiety disorders, somatoform disorders, substance disorders, personality disorders, mood disorders, and psychotic disorders) are associated with the development of CVD and reduced life expectancy in both sexes.^{297–300} The risk increases with the severity of the mental disturbance and vigilance for (often non-specific) symptoms is crucial.³⁰¹ The onset of CVD is associated with an approximately 2–3-fold increased risk of mental disorders compared to a healthy population.^{115,302} In this context, screening should be performed at every consultation (or 2–4 times/year). The 12-month prevalence of mental disorders in CVD patients is around 40%, leading to significantly worse prognosis.^{100,108,303,304} The onset of CVD increases the risk of committing suicide.³⁰⁵ In this context, awareness of anxiety and depression symptoms should be increased.

The precise mechanism by which mental disorders increase CVD remains uncertain. The detrimental effects are potentially caused by unhealthy lifestyle, increased exposure to socioeconomic stressors, and cardiometabolic side-effects of some medications,¹¹³ but also by direct effects of the amygdala-based fear-defence system and other direct pathophysiological pathways.³⁰³ Abuse of psychostimulants (e.g. cocaine) is a powerful trigger of myocardial ischaemia.³⁰⁶ Further, the capacity of these patients to adaptively use the health-care systems is impaired due to their mental condition (e.g. not being able to trust other people and seek help, impaired capacity to be adherent).¹⁰⁰ Barriers on the part of healthcare providers are stigmatizing attitudes, insufficient mental health literacy, and lack of confidence in mental healthcare.^{307–309} Although patients with mental disorders have an increased CVD risk, they receive a lower rate of recognition and treatment of traditional ASCVD risk factors.³¹⁰ Preliminary evidence suggests that taking mental disorders into account improves classical CVD risk models.^{311,312}

Certain categories of patients with learning difficulties and associated disorders (such as Down's syndrome) are at increased risk of CVD disease, but perhaps not specifically ASCVD. However, health inequalities and the prevalence of CV risk factors may be greater in these populations, although epidemiology research is scarce.

3.4.11. Non-alcoholic fatty liver disease

Non-alcoholic fatty liver disease (NAFLD) has been associated with an increased risk of myocardial infarction and stroke. NAFLD represents accumulation of ectopic fat; persons with NAFLD are often overweight or obese, and not uncommonly have abnormal BP, glucose, and lipid levels. A recent study investigating whether NAFLD increases CV risk beyond traditional risk factors³¹³ shows that after adjusting for established risk factors, the associations did not persist. Nevertheless, patients with NAFLD should have their CVD risk calculated, be screened for DM, and be recommended a healthy lifestyle with a reduction of alcohol intake.

3.4.12. Sex-specific conditions

3.4.12.1 Obstetric conditions

Pre-eclampsia (defined as pregnancy-related hypertension accompanied by proteinuria) occurs in 1–2% of all pregnancies and is associated with an increase in CVD risk by a factor of 1.5–2.7 compared with all women,^{185,186,314} while the RR of developing hypertension is 3¹⁸⁷ and DM is 2.^{184,185} It has not been established whether the increased CVD risk after preeclampsia occurs independently of CV risk factors. The rationale for screening these women for the occurrence of hypertension and DM is, however, quite strong. At present, no separate risk model for women with a history of hypertensive disorders of pregnancy seems necessary, despite their higher baseline risk.³¹⁵

Pregnancy-related hypertension affects 10–15% of all pregnancies. The associated risk of later CVD is lower than for preeclampsia but is still elevated (RR 1.7–2.5).^{193,314,316,317} Also, the risk for sustained or future hypertension is elevated (RRs vary, from 2.0 to 7.2 or even higher).^{187,318} Again, however, there was incomplete adjustment for conventional risk factors. The risk of developing DM is also elevated in these women (RR 1.6–2.0).^{314,319} Both preterm (RR 1.6) and still-birth (RR 1.5) have been associated with a moderate increase in risk of CVD.³¹⁶

Finally, gestational DM confers a sharply elevated risk of future DM, with up to 50% of affected women developing DM within 5 years after pregnancy, and an up to two-fold increased risk of CVD in the future.^{188,320} Screening by fasting glucose or HbA1c may be preferable to oral glucose tolerance testing.^{191,321}

3.4.12.2 Non-obstetric conditions

Polycystic ovary syndrome affects 5% of all women in their fertile years.^{322,323} It has been associated with an increased risk of CVD.³¹⁴ The risk of developing hypertension is probably increased, but data are conflicting.³²⁴ Polycystic ovary syndrome is associated with a higher risk of developing DM (RR 2–4),^{189,190} suggesting that periodic screening for DM is appropriate.

Premature menopause occurs in roughly 1% of women ≤ 40 years of age. Up to 10% of women experience an early menopause, defined as that occurring by 45 years of age.^{314,325} Early menopause is associated with an increased risk of CVD (RR 1.5).^{326–328} A linear inverse relationship between earlier menopause and CHD risk has been found, whereby each 1-year decrease in age at menopause portended a 2% increased risk of CHD.³²⁹

3.4.12.3 Erectile dysfunction

Erectile dysfunction (ED), defined as the consistent inability to reach and maintain an erection satisfactory for sexual activity, has a multifactorial cause. It affects almost 40% and more than 50% of men over 40 years and 60 years of age, respectively.^{330,331} Men with ED have an increased risk of all-cause mortality [odds ratio (OR) 1.26, 95% CI 1.01–1.57] and CVD mortality (OR 1.43, 95% CI 1.00–2.05). ED and CVD share common risk factors (hypercholesterolaemia, hypertension, insulin resistance and DM, smoking, obesity, metabolic syndrome, sedentary lifestyle, and depression) and a common pathophysiological basis of aetiology and progression.^{332,333}

Medication used to prevent CVD, such as aldosterone receptor antagonists, some beta-blockers, and thiazide diuretics, can cause ED.^{330,332–335} ED is associated with subclinical vascular disease,³³⁶ and precedes CAD, stroke, and PAD by a period that usually ranges from 2 to 5 years (average 3 years). Men with ED have a 44–59% higher risk for total CV events, 62% for AMI, 39% for stroke, and 24–33% for all-cause mortality, with a higher risk in those with severe ED.^{337–341}

There is strong evidence that CVD risk assessment is needed in men presenting with ED.^{336,342} In men with ED and low-to-intermediate CVD risk, detailed risk profiling by, for example, CAC score is suggested, but so far not supported by evidence.^{338,341} Assessment of ED severity and physical examination should be part of the first-line CVD risk assessment in men.^{333,341} Lifestyle changes are effective in improving sexual function in men: these include vigorous physical exercise,^{334,343} improved nutrition, weight control, and smoking cessation.^{343–345}

4. Risk factors and interventions at the individual level

4.1. Treatment recommendations: classes, grades, and freedom of choice

Clear communication about risks and benefits is crucial before any treatment is initiated. Risk communication is discussed in [section 3.2.4](#),

Table 6 Treatment goals for different patient categories

Patient category	Prevention goals (STEP 1)	Intensified/additional prevention goals ^a (STEP 2)
Apparently healthy persons	For BP and lipids: initiation of drug treatment based on CVD risk assessment (Table 5) or SBP >160 mmHg	
<50 years	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL)	SBP <130 mmHg if tolerated ^b LDL-C <1.8 mmol/L (70 mg/dL) and ≥50% reduction in high-risk patients LDL-C <1.4 mmol/L (55 mg/dL) and ≥50% reduction in very-high-risk patients
50 - 69 years	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL)	SBP <130 mmHg if tolerated ^b LDL-C <1.8 mmol/L (70 mg/dL) and ≥50% reduction in high-risk patients LDL-C <1.4 mmol/L (55 mg/dL) and ≥50% reduction in very-high-risk patients
≥70 years	Stop smoking and lifestyle optimization SBP <140 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL)	For specific risk factor management in patients ≥70 years old, please see relevant sections in section 4.
Patients with CKD	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL) and ≥50% LDL-C reduction Otherwise according to ASCVD and DM history	LDL-C <1.8 mmol/L (70 mg/dL) in high-risk patients and <1.4 mmol/L (55 mg/dL) in very-high risk patients (see Table 4)
Patients with FH	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL) and ≥50% LDL-C reduction Otherwise according to ASCVD and DM history	LDL-C <1.8 mmol/L (70 mg/dL) in high-risk patients and <1.4 mmol/L (55 mg/dL) in very-high risk patients (see Table 4)
People with type 2 DM		
Well-controlled short-standing DM (e.g. <10 years), no evidence of TOD and no additional ASCVD risk factors	Stop smoking and lifestyle optimization	
Without established ASCVD or severe TOD (see Table 4 for definitions)	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b LDL-C <2.6 mmol/L (100 mg/dL) HbA1c <53 mmol/mol (7.0%)	SBP <130 mmHg if tolerated ^b LDL-C <1.8 mmol/L (70 mg/dL) and ≥50% reduction SGLT2 inhibitor or GLP-1RA
With established ASCVD and/or severe TOD (see Table 4 for definitions)	Stop smoking and lifestyle optimisation SBP <140 down to 130 mmHg if tolerated ^b LDL-C <1.8 mmol/L (70 mg/dL) HbA1c <64 mmol/mol (8.0%) SGLT2 inhibitor or GLP1-RA CVD: antiplatelet therapy	SBP <130 mmHg if tolerated ^b LDL-C <1.4 mmol/L (55 mg/dL) and ≥50% reduction SGLT2 inhibitor or GLP-1RA if not already on <i>May additionally consider novel upcoming treatments: DAPT, dual pathway inhibition, a colchicine, icosapent ethyl</i>
Patients with established ASCVD	Stop smoking and lifestyle optimization SBP <140 down to 130 mmHg if tolerated ^b Intensive oral lipid-lowering therapy aiming at ≥50% LDL-C reduction and LDL-C <1.8 mmol/L (70 mg/dL) Antiplatelet therapy	SBP <130 mmHg if tolerated ^b LDL-C <1.4 mmol/L (55 mg/dL) <i>May additionally consider novel upcoming treatments: DAPT, dual pathway inhibition, colchicine, icosapent ethyl, etc.</i>

ASCVD = atherosclerotic cardiovascular disease; BP = blood pressure; CKD = chronic kidney disease; CVD = cardiovascular disease; DAPT = dual antiplatelet therapy; DBP = diastolic blood pressure; DM = diabetes mellitus; EAS = European Atherosclerosis Society; ESC = European Society of Cardiology; FH = familial hypercholesterolaemia; GLP-1RA = glucagon-like peptide-1 receptor agonist; HbA1c = glycated haemoglobin; LDL-C = low-density lipoprotein cholesterol; SBP = systolic blood pressure (office); SGLT2 = sodium-glucose cotransporter 2; TOD = target organ damage.

^aDepending on 10-year (residual) risk and/or estimated lifetime benefit (see Table 4 for details), comorbidities, and patient preference. Levels of evidence of intensified goals vary, see recommendation tables in sections 4.6 and 4.7. For CKD and FH, LDL-C targets are taken from the 2019 ESC/EAS Guidelines for the treatment of dyslipidaemias.³

^bOffice DBP treatment target range <80 mmHg.

and benefits of individual treatment are the topic of this section. In all scenarios where recommendations for individual interventions to reduce risk are ‘strong’ (class I or IIa), it is important to realize that many patients who have received appropriate risk information often (in up to 50% of cases, some studies suggest) consciously opt to forego the proposed intervention. This applies not only to lifestyle measures, but also to drug interventions. Apparently, what professionals feel is sufficient risk reduction for a reasonable effort or initiation of a drug with few side-effects does not always correspond to patients’ views. The reverse is also true: not only may some patients at (very) high risk forego interventions, some patients with low-to-moderate risk may be highly motivated to decrease their risk even further. Hence, treatment recommendations are never ‘imperative’ for (very) high risk patients, nor are interventions ever ‘prohibited’ for patients at low-to-moderate risk. There is evidence that a higher proportion of women, compared to men, have a low awareness of their CVD risk and the need for therapeutic interventions. This warrants efforts to improve awareness, risk assessment, and treatment in women.^{52,346–351}

4.2. Optimizing cardiovascular risk management

4.2.1. Goals of clinicianpatient communication

Clinicians should provide a personalized presentation of guidelines to improve understanding, encourage lifestyle changes, and support adherence to drug therapy. Applying this in daily practice faces different barriers.³⁵² Patients’ ability to adopt a healthy lifestyle depends on cognitive and emotional factors, the impact of a diagnosis or symptoms, socioeconomic factors, educational level, and mental health. Perceived susceptibility to illness and the anticipated severity of the consequences are also prominent components of patients’ motivation.³⁵³

4.2.2. How to improve motivation?

Communication strategies such as motivational interviewing are useful.³⁵⁴ Consultation sessions may include a family member or friend, especially for elderly patients. Connection is paramount: focus before greeting; listen intently; agree on what matters most; connect with the person’s story; and explore emotions.³⁵⁵ The OARS (Open-ended questions, Affirmation, Reflective listening, and Summarizing) principle helps patients to present their perceptions, and clinicians to summarize. The SMART (Specific, Measurable, Achievable, Realistic, Timely) principle may help with setting goals for behavioural change.^{353,356} Healthcare professionals must consider capability, opportunity (physical, social, or environmental) and motivation for behavioural change.³⁵⁷ Multidisciplinary behavioural approaches that combine the knowledge and skills of different caregivers are recommended.³⁵⁸

4.2.3. Optimizing drug adherence

Medication adherence ranges from 50% for primary ASCVD prevention to 66% for secondary prevention.³⁵⁹ Physicians should consider non-adherence in every patient and inquire non-judgmentally about it.³⁶⁰ Approximately 9% of cases of ASCVD in Europe can be attributed to poor medication adherence.³⁶¹ Contributors to non-adherence include polypharmacy, complexity of drug/dose regimes, poor doctor-patient relationship, lack of disease acceptance, beliefs

about consequences and side-effects, intellectual/cognitive abilities, mental disorders, physical limitations, financial aspects, and living alone.^{360,362–364} Importantly, only substantial risk reduction motivates patients for preventive drug treatment, which obviates the need for appropriate risk communication.^{365,366} Depression is another important factor, and adequate treatment thereof improves adherence.^{367,368}

Mobile phone applications may improve adherence to both medication and behavioural changes.³⁶⁹ Their use is easy and probably cost-effective.³⁷⁰

4.2.4. Treatment goals

In the subsequent sections, different domains of individual treatment are discussed. Table 6 summarizes the treatment goals and some key interventions for different categories of patients. For additional information on risk categories and the principle of a stepwise approach to treatment targets, please refer to section 3.2.3.1. For details on treatment goals, how to achieve them, strengths of recommendations and levels of supporting evidence, please go to the relevant subsections.

4.3. Optimizing lifestyle

4.3.1. Physical activity and exercise

Recommendations for physical activity

Recommendations	Class ^a	Level ^b
It is recommended for adults of all ages to strive for at least 150 - 300 min a week of moderate-intensity or 75 - 150 min a week of vigorous-intensity aerobic PA, or an equivalent combination thereof, to reduce all-cause mortality, CV mortality, and morbidity. ^{371,372}	I	A
It is recommended that adults who cannot perform 150 min of moderate-intensity PA a week should stay as active as their abilities and health condition allow. ^{373,374}	I	B
It is recommended to reduce sedentary time to engage in at least light activity throughout the day to reduce all-cause and CV mortality and morbidity. ^{375–377}	I	B
Performing resistance exercise, in addition to aerobic activity, is recommended on 2 or more days per week to reduce all-cause mortality. ^{378,379}	I	B
Lifestyle interventions, such as group or individual education, behaviour-change techniques, telephone counselling, and use of consumer-based wearable activity trackers, should be considered to increase PA participation. ^{380–382}	IIa	B

CV = cardiovascular; PA = physical activity.

^aClass of recommendation.

^bLevel of evidence.

PA reduces the risk of many adverse health outcomes and risk factors in all ages and both sexes. There is an inverse relationship between moderate-to-vigorous PA and all-cause mortality, CV morbidity and

Table 7 Classification of physical activity intensity and examples of absolute and relative intensity levels.

Absolute intensity			Relative intensity		
Intensity	MET ^a	Examples	%HR _{max}	RPE (Borg scale score)	Talk test
Light	1.1–2.9	Walking <4.7 km/h, light household work	57–63	10–11	
Moderate	3–5.9	Walking at moderate or brisk pace (4.1–6.5 km/h), slow cycling (15 km/h), painting/decorating, vacuuming, gardening (mowing lawn), golf (pulling clubs in trolley), tennis (doubles), ballroom dancing, water aerobics	64–76	12–13	Breathing is faster but compatible with speaking full sentences
Vigorous	≥6	Race-walking, jogging, or running, cycling >15 km/h, heavy gardening (continuous digging or hoeing), swimming laps, tennis (singles)	77–95	14–17	Breathing very hard, incompatible with carrying on a conversation comfortably

%HR_{max} = percentage of measured or estimated maximum heart rate (220–age); MET = metabolic equivalent of task; RPE = rating of perceived exertion (Borg-scale 6–20); VO₂ = oxygen consumption.

^aMET is estimated as the energy cost of a given activity divided by resting energy expenditure: 1 MET = 3.5 mL oxygen kg⁻¹ min⁻¹ VO₂.

Modified from ³⁹²

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mortality, as well as incidence of type 2 DM.^{371–373,383–387} The reduction in risk continues across the full range of PA volumes, and the slope of risk decline is steepest for the least active individuals.^{371–374,386,387} More information on PA prescription can be found in a recent ESC Guideline.³⁸⁸

4.3.1.1 Physical activity prescription

PA should be individually assessed and prescribed in terms of frequency, intensity, time (duration), type, and progression.³⁸⁹ Recommendations regarding pre-participation screening can be found in previous ESC Guidelines.³⁸⁸ Interventions shown to increase PA level or reduce sedentary behaviour include behaviour theory-based interventions, such as goal-setting, re-evaluation of goals, self-monitoring, and feedback.^{372,380,381} Using a wearable activity tracker may help increase PA.³⁸² Most important is to encourage activity that people enjoy and/or can include in their daily routines, as such activities are more likely to be sustainable.

4.3.1.2 Aerobic physical activity

Examples of aerobic PA include walking, jogging, cycling, etc.³⁸⁹ Adults are recommended to perform at least 150–300 min a week of moderate-intensity PA, or 75–150 min of vigorous-intensity PA, or an equivalent combination of both, spread throughout the week.^{371,372} Additional benefits are gained with even more PA. Practising PA should still be encouraged in individuals unable to meet the minimum. In sedentary individuals, a gradual increase in activity level is recommended. When older adults or individuals with chronic conditions cannot achieve 150 min of moderate-intensity PA a week, they should be as active as their abilities and conditions allow.^{371–375,384,385} PA accumulated in bouts of even <10 min is associated with favourable outcomes, including mortality.^{371,390}

PA can be expressed in absolute or relative terms.³⁸⁹ Absolute intensity is the amount of energy expended per minute of activity, assessed by oxygen uptake per unit of time (mL/min or L/min) or by metabolic equivalent of task (MET). A compendium of the energy cost in MET values for various activities is available.³⁹¹ An absolute measure does not consider individual factors such as body weight, sex, and fitness level.³⁸⁹

Relative intensity is determined based on an individual's maximum (peak) effort, e.g. percentage of cardiorespiratory fitness (%VO₂ max), percentage of maximum (peak) heart rate (%HR_{max}) or using rating of perceived exertion according to the Borg scale. Less fit individuals generally require a higher level of effort than fitter people to perform the same activity. A relative intensity measure is necessary to provide an individualized PA prescription.³⁸⁹

Classification for both absolute and relative intensity and examples are presented in Table 7.

4.3.1.3 Resistance exercise

Resistance exercise in addition to aerobic PA is associated with lower risks of total CV events and all-cause mortality.^{378,379,393–395} The suggested prescription is one to three sets of 8–12 repetitions at the intensity of 60–80% of the individual's 1 repetition maximum at a frequency of at least 2 days a week in a variety of 8–10 different exercises involving each major muscle group. For older adults or deconditioned individuals, it is suggested to start with one set of 10–15 repetitions at 40–50% of 1 repetition maximum.³⁸⁹ In addition, older adults are recommended to perform multicomponent PA that combines aerobic, muscle-strengthening, and balance exercises to prevent falls.³⁷²

4.3.1.4 Sedentary behaviour

Sedentary time is associated with greater risk for several major chronic diseases and mortality.^{371,372,375–377,396–399} For physically inactive adults, light-intensity PA, even as little as 15 minutes a day, is likely to produce benefits. There is mixed evidence to suggest how activity bouts that interrupt sedentary behaviour are associated with health outcomes.^{375,398,400}

4.3.2. Nutrition and alcohol

Recommendations for nutrition and alcohol

Recommendations	Class ^a	Level ^b
A healthy diet is recommended as a cornerstone of CVD prevention in all individuals. ^{401,402}	I	A

Continued

It is recommended to adopt a Mediterranean or similar diet to lower risk of CVD. ^{403,404}	I	A
It is recommended to replace saturated with unsaturated fats to lower the risk of CVD. ^{405–409}	I	A
It is recommended to reduce salt intake to lower BP and risk of CVD. ⁴¹⁰	I	A
It is recommended to choose a more plant-based food pattern, rich in fibre, that includes whole grains, fruits, vegetables, pulses, and nuts. ^{411,412}	I	B
It is recommended to restrict alcohol consumption to a maximum of 100 g per week. ^{413–415}	I	B
It is recommended to eat fish, preferably fatty, at least once a week and restrict (processed) meat. ^{406,416–418}	I	B
It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake. ^{419,420}	I	B

CVD = cardiovascular disease; BP = blood pressure.
^aClass of recommendation.
^bLevel of evidence.

Dietary habits influence CV risk, mainly through risk factors such as lipids, BP, body weight, and DM.^{401,402} Table 8 summarizes the characteristics of a healthy diet. Although recommendations about nutrients and foods remain important for CV health, there is a growing concern about environmental sustainability, supporting a shift from an animal- to a more plant-based food pattern.^{411,412}

4.3.2.1 Fatty acids

Risk of CHD is reduced when dietary saturated fats are replaced appropriately (Figure 10). This is also the case when replacing meat and dairy foods.^{406,407} Polyunsaturated fats (-25%), monounsaturated fats (-15%), and to a lesser extent carbohydrates from whole grains (-9%), were all associated with reduced CHD risk when isocalorically substituted for dietary saturated fat.^{408,409}

Reducing saturated fatty acid intake to less than 10% of energy may have additional benefits.⁴⁰⁵ However, the LDL-C-lowering effect of substituting polyunsaturated fatty acids (PUFAs) for saturated fatty acids may be less in obese (5.3%) than in normal-weight persons (9.7%).⁴²¹

Trans fatty acids, formed during industrial processing of fats, have unfavourable effects on total cholesterol (increase) and HDL-C (decrease). On average, a 2% increase in energy intake from trans fatty acids is associated with a 23% higher CHD risk.⁴²² A regulation of the European Union (EU) Commission has set the upper limit to 2 g per 100 g of fat (April 2019) (https://ec.europa.eu/food/safety/labelling_nutrition/trans-fat-food_en).

When guidelines to lower saturated fat intake are followed, reductions in dietary cholesterol intake follow.

4.3.2.2 Minerals and vitamins

A reduction in sodium intake may reduce SBP by, on average, 5.8 mmHg in hypertensive, and 1.9 mmHg in normotensive patients.⁴¹⁰

Table 8 Healthy diet characteristics

Adopt a more plant- and less animal-based food pattern
Saturated fatty acids should account for <10% of total energy intake, through replacement by PUFAs, MUFAs, and carbohydrates from whole grains
Trans unsaturated fatty acids should be minimized as far as possible, with none from processed foods
<5 g total salt intake per day
30–45 g of fibre of per day, preferably from wholegrains
≥200 g of fruit per day (≥2–3 servings)
≥200 g of vegetables per day (≥2–3 servings)
Red meat should be reduced to a maximum of 350–500 g a week, in particular processed meat should be minimized
Fish is recommended 1–2 times per week, in particular fatty fish
30 g unsalted nuts per day
Consumption of alcohol should be limited to a maximum of 100 g per week
Sugar-sweetened beverages, such as soft drinks and fruit juices, must be discouraged

MUFA = monounsaturated fatty acid; PUFA = polyunsaturated fatty acid.

The DASH (Dietary Approaches to Stop Hypertension) trial showed a dose–response relation between sodium reduction and BP reduction.⁴²³ In a meta-analysis, salt reduction of 2.5 g/day resulted in a 20% reduction of ASCVD events (RR 0.80).⁴¹⁰ A U- or J-shaped relation between a low salt intake and ASCVD is debated.⁴²⁴ Underlying illness and malnutrition may explain both low food and salt intakes as well as increased ASCVD.^{410,425,426} The totality of evidence warrants salt reduction to prevent CHD and stroke.

In most Western countries, salt intake is high (≈9–10 g/day), whereas the recommended maximum intake is 5 g/day. Optimal intake might be as low as ≈3 g/day. Salt reduction can be achieved by dietary choices (fewer processed foods) and the reformulation of foods by lowering their salt content (see section 5.2.2).

Potassium (e.g. in fruits and vegetables) has favourable effects on BP and risk of stroke (RR 0.76).⁴²⁷

As for vitamins, observational studies have found inverse associations between vitamins A and E and risk of ASCVD. However, intervention trials have failed to confirm these findings. Also, trials of supplementation with B vitamins (B6, folic acid, and B12), and vitamins C and D have not shown beneficial effects.^{428,429}

4.3.2.3 Fibre

Each 7 g/day higher intake of total fibre is associated with a 9% lower risk of CAD (RR 0.91).⁴³⁰ A 10 g/day higher fibre intake was associated with a 16% lower risk of stroke (RR 0.84) and a 6% lower risk of type 2 DM (RR 0.94).^{431,432} A high fibre intake may reduce postprandial glucose responses after carbohydrate-rich meals and also lower triglyceride levels.⁴³³

4.3.2.4 Specific foods and food groups

4.3.2.4.1. Fruits, vegetables, and pulses. A meta-analysis reported a 4% lower risk in CV mortality for each additional serving of fruits

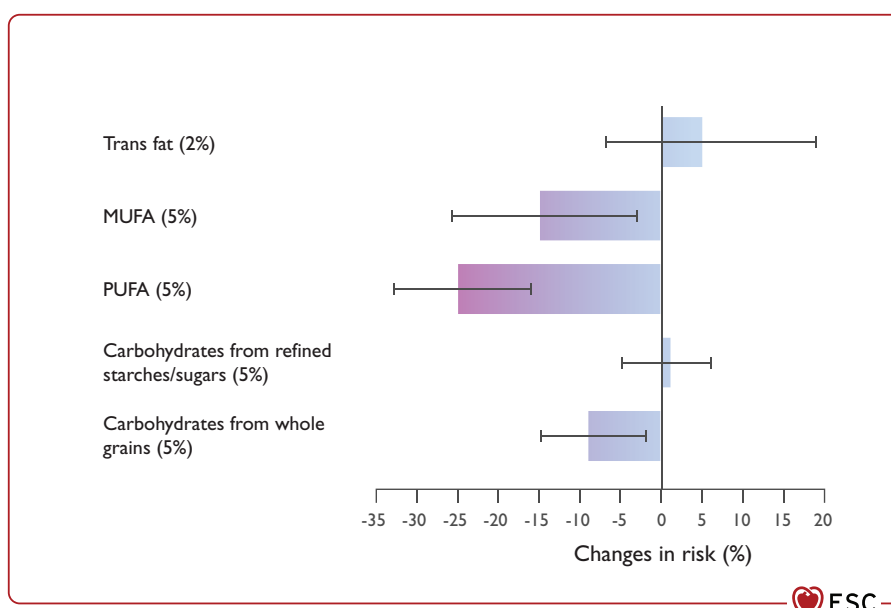


Figure 10 Estimated percentage change in risk of coronary heart disease associated with isocaloric substitutions of saturated fat for other types of fat or carbohydrates. Reproduced from Sacks *et al.*⁴⁰⁹ MUFA = monounsaturated fatty acid; PUFA = polyunsaturated fatty acid.

(equivalent to 77 g) and vegetables (equivalent to 80 g) per day, while all-cause mortality was not reduced further with intakes of more than five servings.⁴³⁴ A meta-analysis reported an 11% lower risk for stroke associated with three to five daily servings of fruits and vegetables and of 26% with five servings a day compared with fewer than three servings.^{435,436} A single portion of pulses (legumes) a day lowers LDL-C by 0.2 mmol/L and is associated with a lower risk of CHD.^{437,438}

4.3.2.4.2. Nuts. A meta-analysis of prospective cohort studies suggested that daily consumption of 30 g of (mixed) nuts was associated with a ≈30% lower risk of ASCVD.⁴³⁷ Both pulses and nuts contain fibre and other bioactive components.⁴³⁸

4.3.2.4.3. Meat. From both a health and an environmental point of view, a lower consumption of meat, especially processed meat, is recommended.⁴¹¹ A restriction of red meat may have little or no effect on major cardiometabolic outcomes.⁴¹⁶ However, substituting red meat with high-quality plant foods (i.e. nuts, soy, and legumes) does improve LDL-C concentrations.⁴⁰⁶ A recent analysis showed that higher intake of processed meat and unprocessed red meat is associated with a 7% and 3%, respectively, increased risk of ASCVD.⁴¹⁷

By reducing processed meats, salt intake will also be reduced. The World Cancer Research Fund recommends limiting red meat consumption to 350–500 g per week.⁴³⁹

4.3.2.4.4. Fish and fish oil supplements. Studies indicate that eating fish, particularly fish rich in n-3 PUFA, at least once a week, is associated with a 16% lower risk of CAD,⁴¹⁸ and eating fish two to four times a week is associated with a 6% lower risk of stroke.⁴⁴⁰ The highest risk was observed in the range of no or very low intakes.

Several meta-analyses and a recent Cochrane review showed no benefits of fish oils on CV outcomes and/or mortality,^{441–443}

although a 7% lower risk of CHD events was observed. A meta-analysis of 13 RCTs included the results of VITAL (Vitamin D and Omega-3 Trial), ASCEND (A Study of Cardiovascular Events in Diabetes), and REDUCE-IT (Reduction of Cardiovascular Events with Icosapent Ethyl—Intervention Trial).⁴⁴⁴ In the analysis excluding REDUCE-IT, fish oil reduced total ASCVD (RR 0.97) and CHD death (RR 0.92).⁴⁴⁴ Including REDUCE-IT (a study done in participants with high triglycerides, comparing very high icosapent ethyl doses vs. mineral oil placebo) strengthened the results.⁴⁴⁴ However, this is the only study that tested a high icosapent ethyl dose and questions have been raised regarding the choice of placebo. Very recently, STRENGTH (Long-Term Outcomes Study to Assess Statin Residual Risk with Epanova in High Cardiovascular Risk Patients with Hypertriglyceridemia) failed to demonstrate benefit of a combined eicosapentaenoic acid and docosahexaenoic acid preparation.⁴⁴⁵

4.3.2.4.5. Alcoholic beverages. The upper safe limit of drinking alcoholic beverages is about 100 g of pure alcohol per week. How this translates into number of drinks depends on portion size, the standards of which differ per country, mostly between 8 and 14 g per drink. This limit is similar for men and women.⁴¹³ Drinking above this limit lowers life expectancy.

Results from epidemiological studies have suggested that, whereas higher alcohol consumption is roughly linearly associated with a higher risk of all stroke subtypes, coronary disease, HF, and several less common CVD subtypes, it appeared approximately log-linearly associated with a lower risk of myocardial infarction.⁴¹³ Moreover, Mendelian randomization studies do not support the apparently protective effects of moderate amounts vs. no alcohol against ASCVD, suggesting that the lowest risks for CVD outcomes are in abstainers and that any amount of alcohol uniformly increases BP and BMI.^{414,415} These data challenge the concept that moderate alcohol consumption is universally associated with lower CVD risk.

4.3.2.4.6. Soft drinks and sugar. Regular consumption of sugar-sweetened beverages (i.e. two servings per day compared with one serving per month) was associated with a 35% higher risk of CAD in women in the Nurses' Health Study, whereas artificially sweetened beverages were not associated with CAD. In the EPIC (European Prospective Investigation into Cancer and Nutrition) cohort, both artificially and sugar-sweetened soft drinks were associated with all-cause mortality, while only the former was associated with circulatory diseases.⁴¹⁹ The WHO guideline recommends a maximum intake of 10% of energy from free sugars (mono- and disaccharides), which includes added sugars as well as sugars present in fruit juices.⁴²⁰

4.3.2.4.7. Coffee. Non-filtered coffee contains LDL-C-raising cafestol and kahweol, and may be associated with an up to 25% increased risk of ASCVD mortality by consumption of nine or more drinks a day.⁴⁴⁶ Non-filtered coffee includes boiled, Greek, and Turkish coffee and some espresso coffees. Moderate coffee consumption (3–4 cups per day) is probably not harmful, perhaps even moderately beneficial.⁴⁴⁷

4.3.2.4.8. Functional foods. Functional foods containing phytosterols (plant sterols and stanols) are effective in lowering LDL-C levels by an average of 10% when consumed in amounts of 2 g/day.⁴⁴⁸ The effect is in addition to that obtained with a low-fat diet or use of statins. No studies with clinical endpoints have been performed yet.

Red yeast rice supplements are not recommended and may even cause side-effects.⁴⁴⁹

4.3.2.4.9. Dietary patterns. Studying the impact of a total dietary pattern shows the full preventive potential of diet. The Mediterranean diet includes high intakes of fruits, vegetables, pulses, wholegrain products, fish, and olive oil, moderate consumption of alcohol, and low consumption of (red) meat, dairy products, and saturated fatty acids. Greater adherence to a Mediterranean diet is associated with a 10% reduction in CV incidence or mortality and an 8% reduction in all-cause mortality.⁴⁰³ Following a Mediterranean diet enriched with nuts over a 5-year period, compared with a control diet, lowered the risk of ASCVD by 28% and by 31% with a diet enriched with extra-virgin olive oil.⁴⁰⁴

Also, a shift from a more animal-based to a plant-based food pattern may reduce ASCVD.⁴¹¹

4.3.3. Body weight and composition

Recommendations for body weight

Recommendations	Class ^a	Level ^b
It is recommended that overweight and obese people aim for a reduction in weight to reduce BP, dyslipidaemia, and risk of type 2 DM, and thus improve their CVD risk profile. ^{450,451}	I	A
While a range of diets are effective for weight loss, it is recommended that a healthy diet in regard to CVD risk is maintained over time. ^{452–454}	I	A

Continued

Bariatric surgery for obese high-risk individuals should be considered when lifestyle change does not result in maintained weight loss. ⁴⁵⁵	IIa	B
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CVD = cardiovascular disease; BP = blood pressure; DM = diabetes mellitus.
^aClass of recommendation.
^bLevel of evidence.

4.3.3.1 Treatment goals and modalities

Although diet, exercise, and behaviour modification are the main therapies for overweight and obesity, they are often unsuccessful in the long term. Yet, maintaining even a moderate weight loss of 5–10% from baseline has salutary effects on risk factors including BP, lipids, and glycaemic control,^{450,451} as well as on premature all-cause mortality.⁴⁵⁶ Weight loss is associated with lower morbidity but higher mortality in (biologically) older adults (the 'obesity paradox'). In this group, emphasis should be less on weight loss and more on maintaining muscle mass and good nutrition.

4.3.3.2 Diets for weight loss

Energy restriction is the cornerstone of management. PA is essential to maintain weight loss and prevent rebound weight gain, but is not reviewed here. Hypocaloric diets may be categorized as:

1. Diets that aim to reduce ASCVD, including plant-based^{457,458} and hypocaloric Mediterranean diets,^{458,459} with modifications to suit local food availability and preferences.
2. Changes to the fat and carbohydrate macronutrient composition of the diet, including low or very low carbohydrate diets (with 50–130 g and 20–49 g carbohydrates/day, respectively), moderate carbohydrate diets (>130–225 g carbohydrates/day), and low-fat diets (<30% of energy from fat).
3. High-protein diets to preserve lean muscle mass and enhance satiety.
4. Diets focusing on specific food groups (e.g. increasing fruit and vegetables or avoiding refined sugars).
5. Diets that restrict energy intake for specified time periods, for example on 2 days a week or alternate days (intermittent fasting) or during certain hours of the day (time-restricted eating).

These diets give broadly similar short-term weight loss.^{452–454} By 12 months, the effects tend to diminish.⁴⁵³ Benefits of the Mediterranean diet, however, tend to persist. The quality of nutrients in a diet, for example substituting unsaturated for saturated fats (see section 4.3.2.1) and including fibre-rich carbohydrates⁴⁶⁰ determines whether a diet is healthy in the long term.

Low or very low carbohydrate diets may have advantages regarding appetite control, lowering triglycerides, and reducing medications for type 2 DM.⁴⁶¹ Such diets may be ketogenic and need medical or at least dietetic supervision. Studies beyond 2 years are scarce. Extreme carbohydrate intakes should be avoided in the long term and plant substitutions of fat and protein for carbohydrates are advantageous over animal ones.⁴⁶²

Intermittent fasting diets produce equivalent weight loss to continuous energy restriction when matched for energy intake.⁴⁶³

Medications approved in Europe as aids to weight loss (orlistat, naltrexone/bupropion, high-dose liraglutide) may supplement lifestyle change to achieve weight loss and maintenance, although

sometimes at the expense of side-effects. Meta-analysis of medication-assisted weight loss found favourable effects on BP, glycaemic control, and ASCVD mortality.⁴⁶⁴

A very effective treatment option for extreme obesity or obesity with comorbidities is bariatric surgery. A meta-analysis indicated that patients undergoing bariatric surgery had over 50% lower risks of total, ASCVD, and cancer mortality compared with people of similar weight who did not have surgery.⁴⁵⁵

4.4. Mental healthcare and psychosocial interventions

Recommendations for mental healthcare and psychosocial interventions at the individual level

Recommendations	Class ^a	Level ^b
Patients with mental disorders need intensified attention and support to improve adherence to lifestyle changes and drug treatment. ^{3,465}	I	C
In ASCVD patients with mental disorders, evidence-based mental healthcare and interdisciplinary cooperation are recommended. ^{100,113,466}	I	B
ASCVD patients with stress should be considered for referral to psychotherapeutic stress management to improve CV outcomes and reduce stress symptoms. ^{467–469}	IIa	B
Patients with CHD and moderate-to-severe major depression should be considered for antidepressive treatment with an SSRI. ^{470,471}	IIa	B
In patients with HF and major depression, SSRIs, SNRIs, and tricyclic antidepressants are not recommended. ^{472,473 c}	III	B

ASCVD = atherosclerotic cardiovascular disease; CHD = coronary heart disease; CV = cardiovascular; HF = heart failure; SNRI = serotonin-noradrenaline reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor.

^aClass of recommendation.

^bLevel of evidence.

^cDetails explaining this recommendation are provided in the [supplementary material section 2.1](#).

Treatment of an unhealthy lifestyle will reduce CVD risk as well as improve mental health. Smoking cessation, for instance, has a positive effect on depression outcomes,^{474,475} as do exercise therapy^{113,476} and healthy dietary practices.⁴⁷⁷ Evidence-based interventions for smoking cessation, and improving PA and diet, are considered useful and applicable for persons with mental disorders.^{465,478–480}

Mental disorders are associated with an increased risk of CVD and a worse prognosis in patients with ASCVD, due to CVD events or other death causes, including suicide.^{100,113,305} Mental-health treatments effectively reduce stress symptoms and improve quality of life. Several observational studies indicate that treatment or remission of depression reduces CVD risk.^{113,481–484} Psychological interventions in patients with CHD may reduce cardiac mortality (RR 0.79) and alleviate psychological symptoms.⁴⁶⁶ Psychotherapy focusing on stress management in ASCVD patients improves CVD outcomes. In SUPRIM (Secondary Prevention in Uppsala Primary Health Care project), patients in the intervention group had a 41% lower rate of

fatal and non-fatal first recurrent ASCVD events [hazard ratio (HR 0.59)] and fewer recurrent AMIs (HR 0.55).⁴⁶⁷ In SWITCHD (Stockholm Women's Intervention Trial for Coronary Heart Disease), the intervention yielded a substantial reduction in all-cause mortality (OR 0.33).⁴⁶⁸ A recent RCT reported that cardiac rehabilitation (CR) enhanced by stress management produced significant reductions in ASCVD events compared with standard CR alone (HR 0.49).⁴⁶⁹ Concerning psychopharmacotherapy of patients with CHD and depression, selective serotonin reuptake inhibitor (SSRI) treatment lowers rates of CHD readmission (risk ratio 0.63) and all-cause mortality (risk ratio 0.56).⁴⁷⁰ A recent RCT reported that, in patients with ACS and depression, treatment with the SSRI, escitalopram, resulted in a lower rate of the composite endpoint of all-cause mortality, myocardial infarction, or percutaneous coronary intervention (PCI) (HR 0.69).⁴⁷¹ Collaborative care for patients with CHD and depression has small beneficial effects on depression, but significantly reduces short-term major cardiac events.⁴⁸⁵

Concerning side-effects of psychopharmacological treatments, many psychiatric drugs are associated with an increased risk of sudden cardiac death.⁴⁸⁶ In patients with HF, antidepressants are associated with increased risk of cardiac and all-cause mortality (HR 1.27; for details see [supplementary material for section 4.4](#)).⁴⁷² Therefore, ASCVD patients with complex mental disorders, and particularly those needing psychiatric drug treatment, require interdisciplinary cooperation.

4.5. Smoking intervention

Recommendations for smoking intervention strategies

Recommendations	Class ^a	Level ^b
All smoking of tobacco should be stopped, as tobacco use is strongly and independently causal of ASCVD. ^{487,488}	I	A
In smokers, offering follow-up support, nicotine replacement therapy, varenicline, and bupropion individually or in combination should be considered. ^{489–494}	IIa	A
Smoking cessation is recommended regardless of weight gain, as weight gain does not lessen the ASCVD benefits of cessation. ⁴⁹⁵	I	B

ASCVD = atherosclerotic cardiovascular disease.

^aClass of recommendation.

^bLevel of evidence.

4.5.1. Smoking cessation

Stopping smoking is potentially the most effective of all preventive measures, with substantial reductions in (repeat) myocardial infarctions or death.^{487,488} Lifetime gains in CVD-free years are substantial at all ages, and benefits are obviously even more substantial if other complications from smoking would be accounted for. From age 45 years, gains of 3–5 years persist in men to age 65 and in women to age 75 years ([Figure 11](#)). Even in heavy smokers (≥ 20 cigarettes/day), cessation lowers CVD risk within 5 years, although it remains elevated beyond 5 years. Total health benefits will be even larger because of gain in non-CVD health.

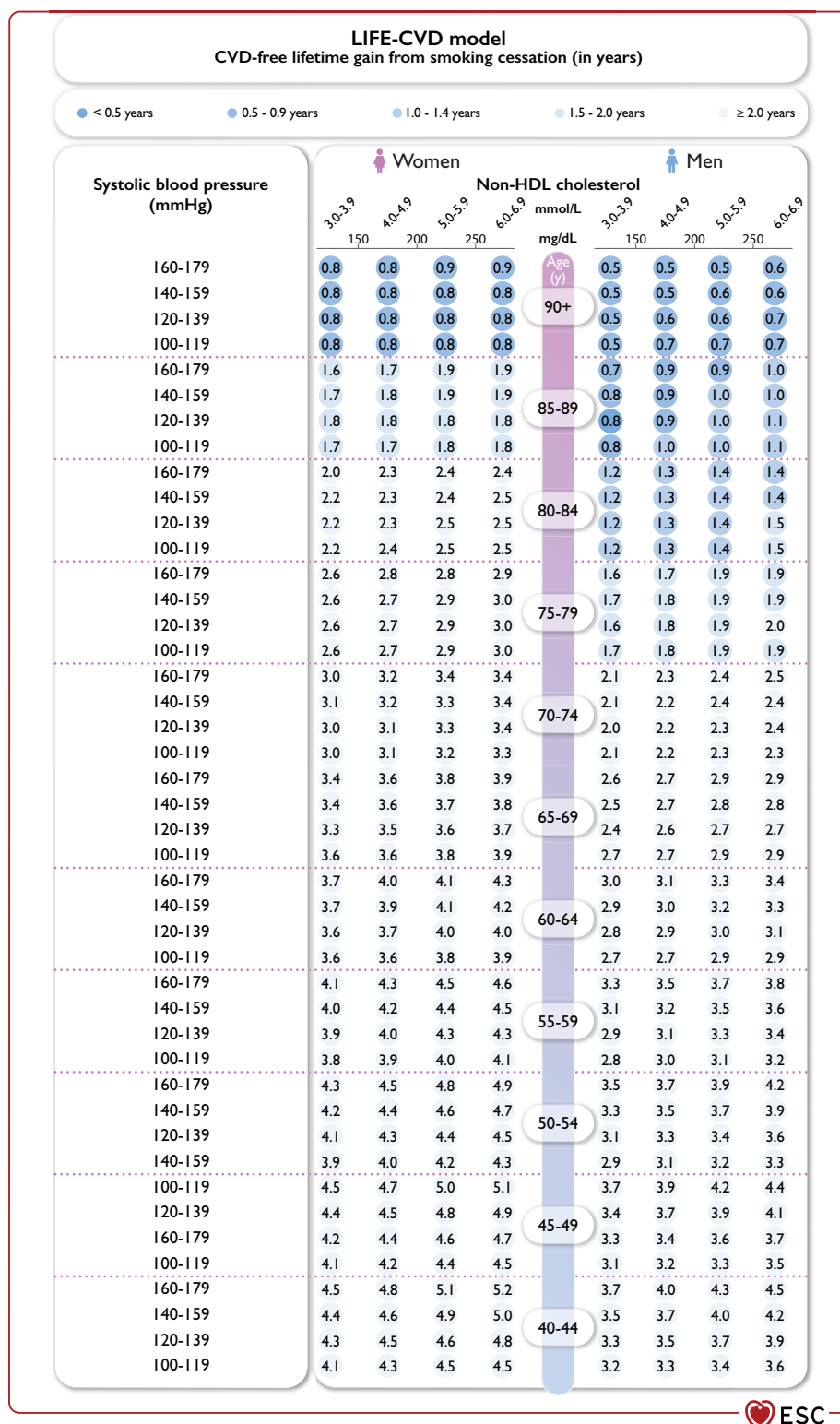


Figure 11 Lifetime atherosclerotic cardiovascular disease benefit from smoking cessation for apparently healthy persons, based on the following risk factors: age, sex, systolic blood pressure, and non-high-density lipoprotein-cholesterol. The model is currently validated for low- and moderate-risk countries. CVD = cardiovascular disease; ESC = European Society of Cardiology; HDL-C = high-density lipoprotein cholesterol; HR = hazard ratio; LIFE-CVD = LIFETIME-perspective CardioVascular Disease; SBP = systolic blood pressure. The lifetime benefit is expressed as 'years of median life expectancy free from myocardial infarction or stroke' gained from smoking cessation. The lifetime benefit is calculated by estimating lifetime CVD risk with the LIFE-CVD model⁷⁶ multiplied by the HR compared to sustained smoking (0.60) from a meta-analysis of studies on the CVD risk of smoking⁴⁹⁶ and multiplied by the HR (0.73) for non-CVD competing mortality.⁴⁹⁷ For individualized estimations of lifetime benefit, this table can be used or the electronic version of LIFE-CVD, assessable via the ESC CVD risk app or <https://u-prevent.com/>.

Table 9 'Very brief advice' for smoking cessation

'Very brief advice' on smoking is a proven 30-second clinical intervention, developed in the UK, which identifies smokers, advises them on the best method of quitting, and supports subsequent quit attempts. There are three elements to very brief advice:

- ASK - establishing and recording smoking status
- ADVISE - advising on the best ways of stopping
- ACT - offering help

UK = United Kingdom.

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Quitting must be encouraged in all smokers, and passive smoking should be avoided as much as possible. Very brief advice may be advantageous when time is limited (Table 9). A major impetus for cessation occurs at the time of diagnosis or treatment of CVD. Prompting a person to try to quit, brief reiteration of CV and other benefits of quitting, and agreeing on a specific plan with a follow-up arrangement are evidence-based interventions.

Smokers who quit may expect an average weight gain of 5 kg, but the health benefits of tobacco cessation outweigh risks from weight gain.⁴⁹⁵ Persistent or reuptake of smoking is common in patients with CHD, in particular in those with severe depression and environmental exposures.⁴⁹⁸ Mood-management therapies may improve outcomes in patients with current or past depression.⁴⁹⁹

4.5.2. Evidence-based drug interventions

Drug support for stopping smoking should be considered in all smokers who are ready to undertake this action. Evidence-based drug interventions include nicotine-replacement therapy (NRT), bupropion, varenicline, and cytosine (not widely available).^{489–491} All forms of NRT (chewing gum, transdermal nicotine patches, nasal spray, inhaler, sublingual tablets) are effective. Combination vs. single-form NRT and 4 mg vs. 2 mg gum can increase success.⁴⁹² NRT shows no adverse effects in patients with ASCVD,⁴⁹³ but evidence of efficacy in this group is inconclusive.⁴⁹⁴ In patients with ASCVD, varenicline (RR 2.6), bupropion (RR 1.4), telephone therapy (RR 1.5), and individual counselling (RR 1.6) all increase success rates.⁴⁹⁴ The antidepressant, bupropion, aids long-term smoking cessation with similar efficacy to NRT.⁴⁹⁰

Varenicline 1 mg *b.i.d.* (twice a day) increases quitting rates more than two-fold compared with placebo.⁴⁹¹ The RR for abstinence vs. NRT was 1.25 and vs. bupropion, 1.4. Lower or variable doses are also effective and reduce side-effects. Varenicline beyond the 12-week standard regimen is well tolerated. Varenicline initiated in hospital following ACS is efficacious and safe.⁵⁰⁰

The main side-effect of varenicline is nausea, but this usually subsides. A causal link between varenicline and neuropsychiatric adverse events is unlikely.⁵⁰¹ Varenicline, bupropion, and NRT do not increase serious CV adverse event risks during or after treatment.⁵⁰²

Cytosine is effective for smoking cessation, but evidence to date is limited.⁴⁹¹

4.5.2.1 Electronic cigarettes

Electronic cigarettes (e-cigarettes) simulate combustible cigarettes by heating nicotine and other chemicals into a vapour. E-cigarettes

deliver nicotine without most of the tobacco chemicals, and are probably less harmful than tobacco.

Recent evidence suggests that e-cigarettes are probably more effective than NRT in terms of smoking cessation.^{503–505} The long-term effects of e-cigarettes on CV and pulmonary health, however, require more research.⁵⁰⁶ Dual use with cigarettes should be avoided. Furthermore, as e-cigarettes are addictive, their use should be subject to similar marketing controls as standard cigarettes, especially the flavoured varieties that appeal to children.⁵⁰⁷ Despite being lower in toxicants than regular cigarettes, 'heat-not-burn' cigarettes do contain tobacco and should be discouraged.

4.6. Lipids

This section covers recommendations for the diagnosis and treatment of unfavourable blood lipid levels. More detail and guidance for complex cases/tertiary care, including genetic lipid disorders, are available in the 2019 ESC/European Atherosclerosis Society (EAS) Guidelines for the management of dyslipidaemias.³

Recent evidence has confirmed that the key initiating event in atherogenesis is the retention of LDL and other cholesterol-rich lipoproteins within the arterial wall. The causal role of LDL-C, and other apo-B-containing lipoproteins, in the development of ASCVD is demonstrated beyond any doubt by genetic, observational, and interventional studies.²⁰ Meta-analysis of clinical trials has indicated that the relative reduction in CVD risk is proportional to the absolute reduction of LDL-C, irrespective of the drug(s) used to achieve such change, with no evidence of a lower limit for LDL-C values or 'J-curve' effect.²¹ The absolute benefit of lowering LDL-C depends on the absolute risk of ASCVD and the absolute reduction in LDL-C, so even a small absolute reduction in LDL-C may translate to significant absolute risk reduction in a high- or very-high-risk patient.²² A recent LDL-C target-driven RCT in patients after ischaemic stroke or transient ischaemic attack (TIA) demonstrated a target LDL-C level of <1.8 mmol/L (70 mg/dL) with the use of statin and, if required, ezetimibe, was associated with a lower CVD risk than those who had a target range of 2.3–2.8 mmol/L (90–110 mg/dL).⁵⁰⁸ Studies on the clinical safety of (very) low achieved LDL-C values have not caused particular concerns, although monitoring for longer periods is required.

4.6.1. Measurement of lipids and lipoproteins

4.6.1.1 Fasting vs. non-fasting measurements

Non-fasting sampling of lipid parameters is recommended for general risk screening, since it has the same prognostic value as fasting samples.^{509,510} In patients with metabolic syndrome, DM, or hypertriglyceridaemia, calculated LDL-C from non-fasting samples should be interpreted with care.

Table 10 Corresponding non-high-density lipoprotein cholesterol and apolipoprotein B levels for commonly used low-density lipoprotein cholesterol goals

LDL-C	Non-HDL-C	Apolipoprotein B
2.6 mmol/L (100 mg/dL)	3.4 mmol/L (131 mg/dL)	100 mg/dL
1.8 mmol/L (70 mg/dL)	2.6 mmol/L (100 mg/dL)	80 mg/dL
1.4 mmol/L (55 mg/dL)	2.2 mmol/L (85 mg/dL)	65 mg/dL

HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol.

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4.6.1.2 Low-density lipoprotein cholesterol measurement

LDL-C can be measured directly, but in most studies and many laboratories, LDL-C is calculated using the Friedewald formula:

- In mmol/L: LDL-C = total cholesterol – HDL-C – (0.45 × triglycerides)
- In mg/dL: LDL-C = total cholesterol – HDL-C – (0.2 × triglycerides)

The calculation is only valid when the concentration of triglycerides is <4.5 mmol/L (~400 mg/dL), and not precise when LDL-C is very low [<1.3 mmol/L (50 mg/dL)]. In patients with low LDL-C levels and/or hypertriglyceridaemia (≤800 mg/dL), alternative formulae are available^{511,512} or LDL-C can be measured directly.

4.6.1.3 Non-high-density lipoprotein cholesterol

The non-HDL-C value is calculated by subtracting HDL-C from total cholesterol. Non-HDL-C, unlike LDL-C, does not require the triglyceride concentration to be <4.5 mmol/L (400 mg/dL). It also has an advantage in that it is accurate in a non-fasting setting, and may be more accurate in patients with DM. There is evidence for a role of non-HDL-C as a treatment target as it captures the information regarding all apolipoprotein-B-containing lipoproteins.⁵¹³ We suggest it as a reasonable alternative treatment goal for all patients, particularly for those with hypertriglyceridaemia or DM. How non-HDL-C levels correspond to commonly used LDL-C goals is shown in Table 10.

4.6.1.4 Apolipoprotein B

Apolipoprotein B provides a direct estimate of the total concentration of atherogenic lipid particles, particularly in patients with elevated triglycerides. However, on average, the information conferred by apolipoprotein B is similar to that of calculated LDL-C.⁵¹⁴ How apolipoprotein B levels correspond to commonly used LDL-C goals is shown in Table 10.

4.6.2. Defining lipid goals

4.6.2.1 Low-density lipoprotein cholesterol goals

Recommendation on low-density lipoprotein cholesterol goals^a

Recommendation	Class ^b	Level ^c
A stepwise treatment-intensification approach is recommended for apparently healthy people at high or very high CVD risk, as well as patients with established ASCVD and/or DM with consideration of CVD risk, treatment benefit, risk modifiers, comorbidities, and patient preferences.	I	C

ASCVD = atherosclerotic cardiovascular disease; DM = diabetes mellitus.

^aRecommendation from section 3.2.

^bClass of recommendation.

^cLevel of evidence.

LDL-C goals are summarized in the recommendations below. As not all drugs are tolerated or available/affordable, treatment should focus on achieving LDL-C levels as close as possible to the given goals. Treatment should be a shared decision-making process between physicians and the patient.

As explained earlier in these guidelines (section 3.2.3.1), we propose a stepwise approach to treatment goals, also for LDL-C (Figures 6–8). This approach may seem novel but, in reality, resembles clinical practice, where treatment intensification is considered based on anticipated benefit, side-effects, and—importantly—patient preferences. The ultimate lipid goals are the same as in the 2019 ESC/EAS dyslipidaemia Guidelines.³ Evidence from glucose-lowering treatment studies indicates that stepwise treatment does not compromise goal attainment, and is associated with fewer side-effects and higher patient satisfaction.^{66,67} In specific cases (at very high risk), the physician may opt to merge both steps and proceed directly to the low LDL-C target level of STEP 2. In apparently healthy people, lifetime treatment benefit of LDL-C reduction may play a role in shared decision-making, together with risk modifiers, comorbidities, patient preference, and frailty. Figure 12 may support decision-making, as it shows the estimated lifetime benefits in years-free-of-CVD in relation to the total CVD risk profile, calibrated in low-to-moderate CVD risk countries.

After STEP 1, treatment intensification with STEP 2 must be considered in all patients. Given that *lower is better*, we encourage liberal intensification of treatment, particularly if submaximal doses of (low-cost) generic statins are used and side-effects are not apparent.

The treatment goal of LDL-C <1.4 mmol/L (55 mg/dL) in STEP 2, in patients with established ASCVD or without ASCVD but at very high risk, is lower than the lowest LDL-C goal of 1.8 mmol/L (70 mg/dL) in the 2016 ESC prevention Guidelines.² This low goal was established based on data from recent Mendelian randomization studies,⁸⁰ meta-analyses from the Cholesterol Treatment Trialists' Collaboration,²¹ RCTs such as IMPROVE-IT (Improved Reduction of Outcomes: Vytorin Efficacy International Trial),⁵¹⁵ and—more recently—proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor clinical outcome studies.^{516–518} The class and level of evidence supporting this LDL-C target of <1.4 mmol/L (55 mg/dL) for patients with ASCVD is identical to that in the recent ESC/EAS dyslipidaemia guidelines.³ For primary prevention in very-high-risk patients, however, the class of recommendation is lower (Class I in the dyslipidaemia guidelines, Class IIa in the current guidelines), because the Task Force was less unanimous with regards to this low LDL-C target in the primary prevention context.

For patients with ASCVD who experience a second vascular event within 2 years (not necessarily of the same type as the first) while taking maximum tolerated statin-based therapy, an even lower LDL-C goal of <1.0 mmol/L (40 mg/dL) may be considered. Importantly, there are no differences in the RR reductions between men and women and between younger and older patients (at least up to age 75 years), or between those with and without DM.³

4.6.2.2 Triglyceride-rich lipoproteins and their remnants

There are no treatment goals for triglycerides, but <1.7 mmol/L (150 mg/dL) is considered to indicate lower risk, whereas higher levels indicate a need to look for other risk factors.

4.6.2.3 High-density lipoprotein cholesterol

To date, no specific goals for HDL-C levels have been determined in clinical trials, although low HDL-C is associated with (residual) risk in ASCVD patients. PA and other lifestyle factors, rather than drug treatment, remain important means of increasing HDL-C levels.

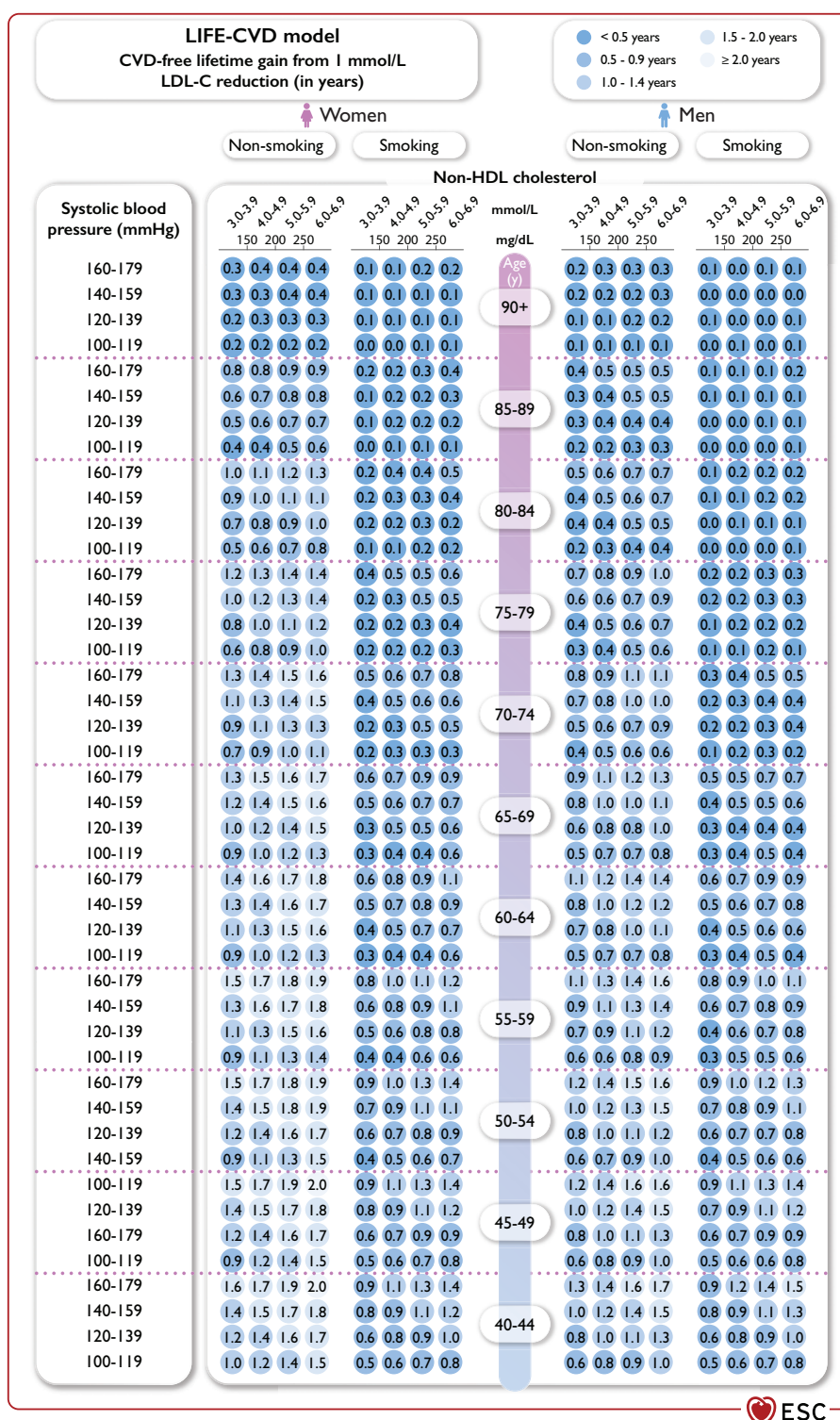


Figure 12 Average years-free-of-cardiovascular disease gained per 1 mmol/L (40 mg/dL) low-density lipoprotein cholesterol reduction in apparently healthy persons. The model is currently validated for low- and moderate-risk countries. Lifetime benefit of 1 mmol/L LDL-C lowering for apparently healthy persons, based on the following risk factors: age, sex, current smoking, SBP, and non-HDL-C. The lifetime benefit is expressed as 'years of median life expectancy free from myocardial infarction or stroke' gained from 1 mmol/L LDL-C lowering. For 2 mmol/L LDL-C lowering, the average effect is almost twice as large, and so on. The lifetime benefit is calculated by estimating lifetime CVD risk with the LIFE-CVD model⁷⁶ multiplied by the HR (0.78) from a meta-analysis of the effect of lipid lowering.²² For individualized estimations of lifetime benefit, this table can be used or the electronic version of LIFE-CVD, assessable via the ESC CVD risk app or <https://u-prevent.com/>. CVD = cardiovascular disease; ESC = European Society of Cardiology; HDL-C = high-density lipoprotein cholesterol; HR = hazard ratio; LDL-C = low-density lipoprotein cholesterol; LIFE-CVD = LIFEtime-perspective CardioVascular Disease; SBP = systolic blood pressure.

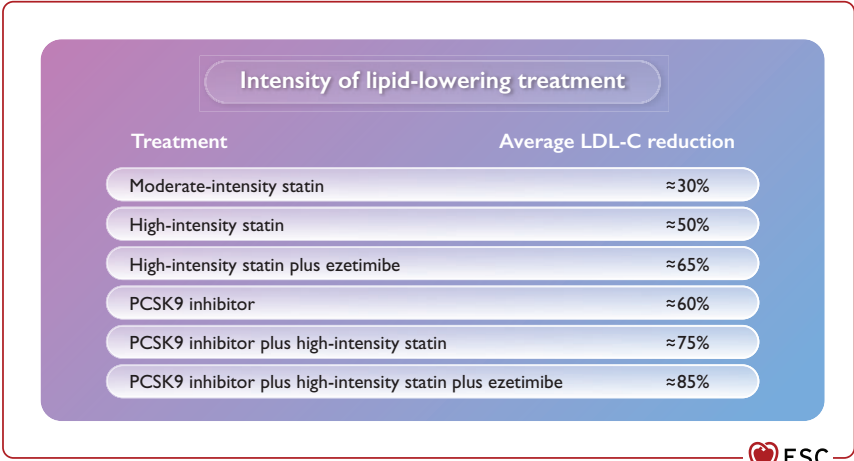


Figure 13 Expected low-density lipoprotein cholesterol reductions for combination therapies. LDL-C = low-density lipoprotein cholesterol; PCSK9 = proprotein convertase subtilisin/kexin type 9. Adapted from Mach et al.³

4.6.3. Strategies to control dyslipidaemias

The presence of dyslipidaemias secondary to other conditions must be excluded before beginning treatment, as treatment of underlying disease may improve hyperlipidaemia without requiring lipid-lowering therapy. This is particularly true for hypothyroidism. Secondary dyslipidaemias can also be caused by alcohol abuse, DM, Cushing’s syndrome, diseases of the liver and kidneys, as well as by drugs (e.g. corticosteroids). In addition, lifestyle optimization is crucial in all patients with higher than optimal lipid levels.

4.6.3.1 Strategies to control low-density lipoprotein cholesterol

4.6.3.1.1. Diet and lifestyle modifications. Dietary factors influence the development of ASCVD, either directly or through their action on traditional risk factors, such as plasma lipids, BP, or glucose levels. Consistent evidence from epidemiological studies indicates that higher consumption of fruit, non-starchy vegetables, nuts, legumes, fish, vegetable oils, yoghurt, and wholegrains, along with a lower intake of red and processed meats, foods higher in refined carbohydrates, and salt, is associated with a lower incidence of CV events.⁵¹⁹ Moreover, the replacement of animal fats, including dairy fat, with vegetable sources of fats and PUFAs may decrease the risk of ASCVD.⁴⁰⁷ More detail on lifestyle recommendations can be found earlier in this section.

4.6.3.1.2. Drugs for treatment of dyslipidaemias. The currently available lipid-lowering drugs include inhibitors of 3-hydroxy-3-methylglutaryl-coenzyme A reductase (statins), fibrates, bile acid sequestrants, selective cholesterol absorption inhibitors (e.g. ezetimibe), and—more recently—PCSK9 inhibitors. Bempedoic acid, an oral cholesterol synthesis inhibitor, has recently been approved in several countries. Usage is mainly intended in combination with ezetimibe in patients with statin intolerance. ASCVD outcome trials are not expected before the end of 2022. Additionally, inclisiran, a new small interfering ribonucleic acid, has shown to reduce LDL-C by 50–55% when applied subcutaneously twice a year. These results were obtained either on top of statin or without other lipid-lowering therapies, and with almost no side-effects. Inclisiran has been

approved in several European countries. Results from the ASCVD outcomes trial are expected for 2023.

The expected LDL-C reductions in response to therapy are shown in Figure 13, and may vary widely among individuals. Therefore, monitoring the effect on LDL-C levels is recommended, with assessment of LDL-C levels 4–6 weeks after any treatment strategy initiation or change.

Recommendations for pharmacological low-density lipoprotein cholesterol lowering for those <70 years of age (for recommendations for persons aged ≥70 years, see respective recommendations tables).

Recommendations	Class ^a	Level ^b
It is recommended that a high-intensity statin is prescribed up to the highest tolerated dose to reach the LDL-C goals set for the specific risk group. ^{21,520,521}	I	A
An ultimate ^c LDL-C goal of <1.4 mmol/L (55 mg/dL) and LDL-C reduction of ≥50% from baseline should be considered in apparently healthy persons <70 years at very high risk. ^{21,22,522}	IIa	C
An ultimate ^c LDL-C goal of <1.8 mmol/L (70 mg/dL) and LDL-C reduction of ≥50% from baseline should be considered in apparently healthy persons <70 years at high risk. ^{21,22,522}	IIa	C
In patients with established ASCVD, lipid-lowering treatment with an ultimate ^c LDL-C goal of <1.4 mmol/L (55 mg/dL) and a ≥50% reduction in LDL-C vs. baseline is recommended. ^{21,508,515–517,522}	I	A
If the goals are not achieved with the maximum tolerated dose of a statin, combination with ezetimibe is recommended. ⁵¹⁵	I	B
For primary prevention patients at very high risk, but without FH, if the LDL-C goal is not achieved on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor may be considered.	IIb	C

Continued

For secondary prevention patients not achieving their goals on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor is recommended. ^{516,517}	I	A
For very-high-risk FH patients (that is, with ASCVD or with another major risk factor) who do not achieve their goals on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor is recommended.	I	C
If a statin-based regimen is not tolerated at any dosage (even after rechallenge), ezetimibe should be considered. ^{515,523–525}	IIa	B
If a statin-based regimen is not tolerated at any dosage (even after rechallenge), a PCSK9 inhibitor added to ezetimibe may be considered. ^{523,524,526}	IIb	C
If the goal is not achieved, statin combination with a bile acid sequestrant may be considered.	IIb	C
Statin therapy is not recommended in premenopausal female patients who are considering pregnancy or are not using adequate contraception.	III	C

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ASCVD = atherosclerotic cardiovascular disease; FH = familial hypercholesterolaemia; LDL-C = low-density lipoprotein cholesterol; PCSK9 = proprotein convertase subtilisin/kexin type 9.

^aClass of recommendation.

^bLevel of evidence.

^cA stepwise approach to LDL-C targets is recommended; see [section 3.2.3.1](#) and [Figures 6 and 7](#).

Adapted from ³

4.6.3.1.3. Statins. Statins decrease LDL-C, thereby reducing ASCVD morbidity and mortality as well as the need for coronary artery interventions. Statins also lower triglycerides, and may reduce pancreatitis risk. Therefore, they are the drug of first choice in patients at increased risk of ASCVD.³

4.6.3.1.3.1. Adverse effects, interactions, and adherence to statin therapy

The most frequent adverse effect of statin therapy is myopathy, but this is rare. A meta-analysis ruled out any contribution to an increase in non-CV mortality.⁵²² Increased blood sugar and HbA1c levels (i.e. increased risk of type 2 DM) can occur after treatment initiation and are dose dependent, in part linked to slight weight gain, but the benefits of statins outweigh the risks for the majority of patients.⁵²⁷ Adhering to lifestyle changes when prescribed a statin should lessen the risk of DM. Increased levels of liver enzymes may occur during statin therapy, and are usually reversible. Routine monitoring of liver enzyme values is not indicated.

Although 5–10% of patients receiving statins complain of myalgia, in most cases it is not attributable to statins.³ The risk of myopathy (severe muscular symptoms) can be minimized by identifying vulnerable patients and/or by avoiding statin interactions with specific drugs. Rhabdomyolysis is extremely rare. As statins are prescribed on a long-term basis, possible interactions with other drugs deserve particular and continuous attention, as many patients will receive

pharmacological therapy for concomitant conditions. In practice, management of a patient with myalgia but without a major increase in creatine kinase is based on trial and error, and usually involves switching to a different statin or use of a very low dosage several days a week, with a gradual increase in frequency and dosage. A management algorithm may help to manage these patients.³

4.6.3.1.4. Cholesterol absorption inhibitors (ezetimibe). The combination of statin with ezetimibe brings a benefit that is in line with meta-analyses showing that LDL-C reduction has benefits independent of the approach used.^{3,21} The beneficial effect of ezetimibe is also supported by genetic studies.⁵²⁸ Together, these data support the position that ezetimibe should be considered as second-line therapy, either on top of statins when the therapeutic goal is not achieved, or when a statin cannot be prescribed.

4.6.3.1.5. Proprotein convertase subtilisin/kexin type 9 inhibitors. PCSK9 inhibitors (monoclonal antibodies to PCSK9) decrease LDL-C by up to 60%, either as monotherapy or in addition to the maximum tolerated dose of statin and/or other lipid-lowering therapies, such as ezetimibe. Their efficacy appears to be largely independent of background therapy. In combination with high-intensity or maximum tolerated statins, alirocumab and evolocumab reduced LDL-C by 46–73% more than placebo, and by 30% more than ezetimibe.^{516,517} Among patients in whom statins cannot be prescribed, PCSK9 inhibition reduced LDL-C levels when administered in combination with ezetimibe.⁵²⁹ Both alirocumab and evolocumab effectively lower LDL-C levels in patients who are at high or very high CVD risk, including those with DM, with a large reduction in future ASCVD events.^{516,517} PCSK9 inhibitors also lower triglycerides, raise HDL-C and apolipoprotein A-I, and lower lipoprotein(a), although the relative contributions of these lipid modifications remain unknown. PCSK9 inhibitors are costly, and their cost-effectiveness, long-term safety, and effect in primary prevention are as yet unknown. We recommend considering cost-effectiveness in a loco-regional context before implementing recommendations that involve their use. Recommendations for the use of PCSK9 inhibitors are described in the Recommendations for pharmacological LDL-C lowering. Inclisiran is a long-acting hepatic PCSK9 synthesis inhibitor that also lowers LDL-C levels considerably.⁵³⁰ Its effect on clinical outcomes remains to be established.

4.6.3.2 Strategies to control plasma triglycerides

Although CVD risk is increased when fasting triglycerides are >1.7 mmol/L (150 mg/dL),⁵³¹ the use of drugs to lower triglyceride levels may only be considered in high-risk patients when triglycerides are >2.3 mmol/L (200 mg/dL) and triglycerides cannot be lowered by lifestyle measures. The available pharmacological interventions include statins, fibrates, PCSK9 inhibitors, and n-3 PUFAs (in particular icosapent ethyl in doses of 2–4 g/day; see [section 4.3.2.4.4](#)).

Recommendations for the treatment of hypertriglyceridaemia are shown in the Recommendations below.

4.6.3.2.1. Fibrates. Fibrates are used primarily for triglyceride lowering and, occasionally, for increasing HDL-C. Evidence supporting the use of these drugs for CVD event reduction is limited, and given the strong evidence favouring statins, routine use of these drugs in CVD prevention is not recommended.³ To prevent pancreatitis, when

triglycerides are >10 mmol/L (900 mg/dL), they must be reduced not only by drugs, but also by restriction of alcohol, treatment of DM, withdrawal of oestrogen therapy, etc. In patients with severe primary hypertriglyceridaemia, referral to a specialist must be considered.

An evidence-based approach to the use of lipid-lowering nutraceuticals could improve the quality of the treatment, including therapy adherence, and achievement of the LDL-C goal in clinical practice. However, it has to be clearly stressed that there are still no outcome studies proving that nutraceuticals can prevent CVD morbidity or mortality.⁵³²

4.6.4. Important groups

Recommendations for drug treatments of patients with hypertriglyceridaemia.

Recommendations	Class ^a	Level ^b
Statin treatment is recommended as the first drug of choice for reducing CVD risk in high-risk individuals with hypertriglyceridaemia [triglycerides >2.3 mmol/L (200 mg/dL)]. ⁵³³	I	A
In patients taking statins who are at LDL-C goal with triglycerides >2.3 mmol/L (200 mg/dL), fenofibrate or bezafibrate may be considered. ^{534–536}	IIb	B
In high-risk (or above) patients with triglycerides >1.5 mmol/L (135 mg/dL) despite statin treatment and lifestyle measures, n-3 PUFAs (icosapent ethyl 2 × 2 g/day) may be considered in combination with a statin. ⁸⁴	IIb	B

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CVD = cardiovascular disease; LDL-C = low-density lipoprotein cholesterol; PUFA = polyunsaturated fatty acid.

^aClass of recommendation.

^bLevel of evidence.

Adapted from ³

4.6.4.1 Women

The proportional reductions per mmol/L reduction in LDL-C in major vascular events, major coronary events, coronary revascularization, and stroke are similar in women and men. In addition, the relative effects of non-statin drugs that lower LDL-C (ezetimibe and PCSK9 inhibitors, on top of high-intensity statin therapy) are also similar in both women and men.³

4.6.4.2 Older patients (≥70 years)

Compared to the 2019 ESC/EAS dyslipidaemia guidelines,³ we provide a single cut-off for identifying 'older persons' as those ≥70 years of age, as opposed to 75 years, for reasons of consistency with other parts of the current guidelines. As a result, class and level of evidence have been modified in some age groups, in particular the category of patients between 70 and 75 years. Although a single age cut-off is now used, it is important to stress that all such age cut-offs are relatively arbitrary, and biological age influences this threshold in clinical practice. For example, a very fit 75-year-old person may qualify for a treatment normally reserved for those <70 and, conversely, a very frail 65-year-old person should sometimes be considered 'older'. General recommendations for lipid-lowering treatment in older patients are summarized below.

Recent evidence has strengthened the role of LDL-C as an ASCVD risk factor in older patients.⁵³⁷ Evidence from trials indicates that statins and other lipid-lowering drugs produce significant reductions in major vascular events irrespective of age.^{538,539} However, there is less direct evidence of statin benefit in those without evidence of ASCVD. Under the age of 70 years, statins are recommended for primary prevention depending on the level of risk. Above that age, initiation of statin treatment for primary prevention may be considered when at (very) high risk, but we explicitly recommend also taking other arguments into account, such as risk modifiers, frailty, estimated life-time benefit, comorbidities, and patient preferences (see [section 3.2.3.3](#) and [Figure 12](#)). In case of renal function impairment or risk for drug interactions, the statin dose should be up-titrated carefully. In terms of LDL-C targets, there is insufficient evidence to support targets for primary prevention in older patients. Although the conventional LDL-C target of <2.6 mmol/L (100 mg/dL) may seem reasonable, the results of ongoing primary prevention trials in older patients must be awaited [STAREE (STAtin Therapy for Reducing Events in the Elderly) trial; [clinicaltrials.gov](#) registration: NCT02099123]. Frailty, polypharmacy, and muscle symptoms remain relevant factors to consider in older patients.

Recommendations for the treatment of dyslipidaemias in older people (≥70 years).

Recommendations	Class ^a	Level ^b
Treatment with statins is recommended for older people with ASCVD in the same way as for younger patients. ^{538,539}	I	A
Initiation of statin treatment for primary prevention in older people aged ≥70 may be considered, if at high risk or above. ^{538,539}	IIb	B
It is recommended that the statin is started at a low dose if there is significant renal impairment and/or the potential for drug interactions.	I	C

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ASCVD = atherosclerotic cardiovascular disease.

^aClass of recommendation.

^bLevel of evidence.

Adapted from ³

4.6.4.3 Diabetes mellitus

Lowering of LDL-C in patients with DM is consistently associated with lower CVD risk. Similar to prevention in apparently healthy individuals, we propose a stepwise approach to lipid control, dependent on risk, estimated lifetime benefit, comorbidities, and patient preferences ([Figure 8](#)). PCSK9 inhibitors can also be used in patients with DM not reaching their LDL-C targets with statins and/or ezetimibe.

Recommendations for the treatment of dyslipidaemias in diabetes mellitus.

Recommendations	Class ^a	Level ^b
In patients with type 2 DM at very high risk (e.g. with established ASCVD and/or severe TOD ^c), intensive lipid-lowering therapy, ultimately ^d aiming at ≥50% LDL-C reduction and an LDL-C of <1.4 mmol/L (55 mg/dL) is recommended. ^{21,22,522,540,541}	I	A

Continued

In patients with type 2 DM >40 years at high risk, lipid-lowering treatment with an ultimate LDL-C goal of $\geq 50\%$ LDL-C reduction and an LDL-C of <1.8 mmol/L (70 mg/dL) is recommended. ^{540,541}	I	A
Statin therapy may be considered in persons aged ≤ 40 years with type 1 or type 2 DM with evidence of TOD and/or an LDL-C level >2.6 mmol/L (100 mg/dL), as long as pregnancy is not being planned.	IIb	C
If the LDL-C goal is not reached, statin combination with ezetimibe should be considered. ^{515,542}	IIa	B

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ASCVD = atherosclerotic cardiovascular disease; DM = diabetes mellitus; eGFR = estimated glomerular filtration rate; LDL-C = low-density lipoprotein cholesterol; TOD = target organ damage.

^aClass of recommendation.

^bLevel of evidence.

^cSevere TOD in this specific context includes eGFR <45 mL/min/1.73 m²; eGFR 46–79 mL/min/1.73 m² plus microalbuminuria; proteinuria; presence of microvascular disease in at least three different sites (e.g. albuminuria plus retinopathy plus neuropathy). See Table 4 for details.

^dA stepwise approach to LDL-C targets is recommended; see section 3.2.3.1 and Figure 8.

Adapted from ³

4.6.4.4 Chronic kidney disease

Patients with CKD are at high or very high risk of ASCVD, and have a characteristic dyslipidaemia (high triglycerides, normal LDL-C, and low HDL-C). Statin therapy or statin therapy in combination with ezetimibe (which allows larger LDL-C reductions without increasing the statin dose) has a beneficial effect on ASCVD outcomes in CKD.⁵⁴³ For patients with end-stage renal disease, however, we recommend that hypolipidaemic therapy should not be initiated (see Recommendations below). If patients with CKD already on a hypolipidaemic therapy enter end-stage renal disease, the therapy may be maintained.

Recommendations for lipid management in patients with moderate-to-severe chronic kidney disease (Kidney Disease Outcomes Quality Initiative stages 3–5).

Recommendations	Class ^a	Level ^b
The use of statins or statin/ezetimibe combination is recommended in patients with non-dialysis-dependent, stage 3–5 CKD. ^{525,544,545}	I	A
In patients already on statins, ezetimibe, or a statin/ezetimibe combination at the time of dialysis initiation, continuation of these drugs should be considered, particularly in patients with ASCVD.	IIa	C
In patients with dialysis-dependent CKD who are free of ASCVD, commencing statin therapy is not recommended. ^{546,547}	III	A

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ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease.

^aClass of recommendation.

^bLevel of evidence.

Adapted from ³

Table 11 Dutch Lipid Clinic Network diagnostic criteria for familial hypercholesterolaemia

Criteria (choose only one score per group, the highest applicable; diagnosis is based on the total number of points obtained)	Points
1) Family history	
First-degree relative with known premature (men aged <55 years; women <60 years) coronary or vascular disease, or first-degree relative with known LDL-C above the 95 th percentile	1
First-degree relative with tendinous xanthomata and/or arcus cornealis, or children aged <18 years with LDL-C above the 95 th percentile	2
2) Clinical history	
Patient with premature (men aged <55 years; women <60 years) CAD	2
Patient with premature (men aged <55 years; women <60 years) cerebral or peripheral vascular disease	1
3) Physical examination	
Tendinous xanthomata	6
Arcus cornealis before age 45 years	4
4) LDL-C levels (without treatment)	
LDL-C ≥ 8.5 mmol/L (326 mg/dL)	8
LDL-C 6.5–8.4 mmol/L (251–325 mg/dL)	5
LDL-C 5.0–6.4 mmol/L (191–250 mg/dL)	3
LDL-C 4.0–4.9 mmol/L (155–190 mg/dL)	1
5) DNA analysis	
Functional mutation in the <i>LDLR</i> , <i>apolipoprotein B</i> , or <i>PCSK9</i> genes	8
A 'definite' FH diagnosis requires >8 points	
A 'probable' FH diagnosis requires 6–8 points	
A 'possible' FH diagnosis requires 3–5 points	

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CAD = coronary artery disease; DNA = deoxyribonucleic acid; FH = familial hypercholesterolaemia; LDL-C = low-density lipoprotein cholesterol; LDLR = low-density lipoprotein receptor; PCSK9 = proprotein convertase subtilisin/kexin type 9.

4.6.4.5 Familial Hypercholesterolaemia

Patients who could have genetic dyslipidaemias, such as heterozygous FH, can be identified by extreme lipid abnormalities and/or family history (Table 11). An LDL-C >4.9 mmol/L (190 mg/dL) in therapy-naïve patients requires careful evaluation for possible FH. However, in the presence of premature ASCVD or family history, possible FH should be considered at lower LDL-C levels. Besides genetic testing (not always affordable), use of the Dutch Clinical Lipid Network criteria (Table 11) is recommended to identify possible FH. Homozygous FH is rare and should always be placed under the care of lipid experts.

Treatment guidelines for people with FH can be found in the 2019 ESC/EAS dyslipidaemia Guidelines.³

4.7. Blood pressure

Hypertension is one of the most important preventable causes of premature morbidity and mortality. It affects more than 150 million

Summary of recommendations for the clinical management of hypertension

Recommendations	Class ^a	Level ^b
Classification of BP		
It is recommended that BP should be classified as optimal, normal, high-normal, or grades 1 - 3 hypertension, according to office BP.	I	C
Diagnosis of hypertension		
It is recommended to base the diagnosis of hypertension on:	I	C
• Repeated office BP measurements, on more than one visit, except when hypertension is severe (e.g. grade 3 and especially in high-risk patients)		
or	I	C
• Out-of-office BP measurement with ABPM and/ or HBPM when feasible.		
Assessment of HMOD		
To evaluate for the presence of HMOD, measurement of serum creatinine, eGFR, electrolytes, and ACR is recommended for all patients. A 12-lead ECG is recommended for all patients, and echocardiography is recommended for those with ECG abnormalities or signs/symptoms of LV dysfunction. Fundoscopy or retinal imaging is recommended for patients with grades 2 or 3 hypertension and all hypertensive patients with DM. ^{548–551}	I	B
Thresholds for initiation of drug treatment of hypertension		
For grade 1 hypertension, treatment initiation based on absolute CVD risk, estimated lifetime benefit, and the presence of HMOD is recommended. ^{552,553}	I	C
For patients with grade 2 hypertension or higher, drug treatment is recommended. ^{4,552}	I	A
Office BP treatment targets		
It is recommended that the first objective of treatment is to lower BP to <140/90 mmHg in all patients, and that subsequent BP targets are tailored to age and specific comorbidities. ^{552,554}	I	A
In treated patients aged 18–69 years, it is recommended that SBP should ultimately be lowered to a target range of 120 - 130 mmHg in most patients. ^{552,554–556}	I	A
In treated patients aged ≥70 years, it is recommended that SBP should generally be targeted to <140 and down to 130 mmHg if tolerated. ^{552,554,557}	I	A
In all treated patients, DBP is recommended to be lowered to <80 mmHg. ^{555,558,559}	I	A
Treatment of hypertension: lifestyle interventions		
Lifestyle interventions are recommended for people with high-normal BP or higher. ^c	I	A

Continued

Treatment of hypertension: drug treatment

It is recommended to initiate antihypertensive treatment with a two-drug combination in most patients, preferably as a single-pill combination. Exceptions are frail older patients and those with low-risk, grade 1 hypertension (particularly if SBP <150 mmHg). ^{560–565}	I	B
It is recommended that the preferred combinations include a RAS blocker (i.e. an ACE inhibitor or ARB) with a CCB or diuretic, but other combinations of the five major classes can be used (ACE inhibitor, ARB, beta-blocker, CCB, thiazide/thiazide-like diuretic). ^{566–569}	I	A
It is recommended, if BP remains uncontrolled with a two-drug combination, that treatment be increased to a three-drug combination, usually a RAS blocker with a CCB and a diuretic, preferably as a single-pill combination. ^{563,570,571}	I	A
It is recommended, if BP is not controlled by a three-drug combination, that treatment should be increased by the addition of spironolactone, or if not tolerated, other diuretics such as amiloride or higher doses of other diuretics, an alpha-blocker or beta-blocker, or clonidine. ^{555,572–574}	I	B
The combination of two RAS blockers is not recommended. ^{575,576}	III	A
Management of CVD risk in hypertensive patients		
Statin therapy is recommended for many patients with hypertension. ^d	Section 4.6	
Antiplatelet therapy is indicated for secondary prevention in patients with hypertension. ^e	Section 4.9	

ABPM = ambulatory blood pressure monitoring; ACE = angiotensin-converting enzyme; ACR = albumin-to-creatinine ratio; ARB = angiotensin receptor blocker; ASCVD = atherosclerotic cardiovascular disease; BP = blood pressure; CCB = calcium channel blocker; DBP = diastolic blood pressure; DM = diabetes mellitus; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; HBPM = home blood pressure monitoring; HMOD = hypertension-mediated organ damage; LV = left ventricular; RAS = renin–angiotensin system; SBP = systolic blood pressure.

^aClass of recommendation.

^bLevel of evidence.

^cSee section 4.3 for details.

^dSee section 4.6 for details.

^eSee section 4.9 for details.

people across Europe, over 1 billion globally, with a prevalence of ~30–45% in adults, increasing with age to more than 60% in people aged >60 years, and accounting for ~10 million deaths globally per annum.⁵⁷⁷ Despite extensive evidence for the effectiveness of BP-lowering treatments at reducing CVD risk and death, the detection, treatment, and control of BP in Europe and globally remains suboptimal.⁵⁷⁸

This section covers recommendations for the diagnosis and treatment of hypertension to be applied in routine primary and secondary care. More detail and guidance for complex cases/tertiary care are available in the 2018 ESC/European Society of Hypertension (ESH) Guidelines for the management of arterial hypertension.⁴

4.7.1. Definition and classification of hypertension

BP is classified according to seated office BP (Table 12), with approximately corresponding values according to ABPM or home BP average values in Table 13.

4.7.2. Blood pressure measurement

4.7.2.1 Office blood pressure measurement

Office BP should be measured in standardized conditions using validated auscultatory or (semi)automatic devices, as described in Table 14.

4.7.2.2 Unattended automated office blood pressure measurement

Repeated automated office BP readings may improve the reproducibility of BP measurement. If the patient is seated alone and unobserved, unattended automated office BP measurement may reduce or eliminate the 'white-coat' effect, and unattended automated office BP measurements are usually lower than conventional office BP measurements, and more similar to ambulatory daytime BP or home BP values. There is limited information on the prognostic value of unattended automated office BP measurements.⁴

4.7.2.3 Ambulatory blood pressure monitoring

ABPM is the average of repeated automated measurements of BP during the daytime, night-time, and over 24 h. ABPM is a better pre-

dictor of hypertension-mediated organ damage (HMOD) and clinical outcomes than office BP, and identifies 'white-coat' hypertension and masked hypertension (see below). Diagnostic thresholds for hypertension are lower with ABPM than office BP (Table 12).⁴

4.7.2.4 Home blood pressure monitoring

Home BP is the average of all BP readings performed with a validated semiautomatic monitor, for at least 3 consecutive days (ideally 6–7 days), with readings in the morning and evening, taken seated in a quiet room after 5 min of rest. Home BP monitoring (HBPM) thresholds for the diagnosis of hypertension are lower than those for office BP (Table 12). Patient self-monitoring may have a beneficial effect on medication adherence and BP control.⁴

Clinical indications for ambulatory or home monitoring are shown in Table 15.

4.7.3 Screening and diagnosis of hypertension

Ideally, all adults should be screened for the presence of hypertension,^{578,579} but most countries lack the required resources and infrastructure. Formally, these guidelines recommend opportunistic screening at least in susceptible individuals, such as those who are overweight or have a family history of hypertension (see section 3.1).

Table 12 Categories for conventionally measured seated office blood pressure^a

Category	SBP (mmHg)		DBP (mmHg)
Optimal	<120	and	<80
Normal	120–129	and/or	80–84
High-normal	130–139	and/or	85–89
Grade 1 hypertension	140–159	and/or	90–99
Grade 2 hypertension	160–179	and/or	100–109
Grade 3 hypertension	≥180	and/or	≥110
Isolated systolic hypertension ^b	≥140	and	<90

BP = blood pressure; DBP = diastolic blood pressure; SBP = systolic blood pressure.

^aBP category is defined according to seated clinic BP and by the highest level of BP, whether systolic or diastolic.

^bIsolated systolic hypertension is graded 1, 2, or 3 according to SBP values in the ranges indicated.

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Table 13 Definitions of hypertension according to office, ambulatory, and home blood pressure

Category	SBP (mmHg)		DBP (mmHg)
Office BP ^a	≥140	and/or	≥90
Ambulatory BP			
Daytime (or awake) mean	≥135	and/or	≥85
Night-time (or asleep) mean	≥120	and/or	≥70
24-h mean	≥130	and/or	≥80
Home BP mean	≥135	and/or	≥85

BP = blood pressure; DBP = diastolic blood pressure; SBP = systolic blood pressure.

^aRefers to conventional office BP rather than unattended office BP.

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Table 14 Considerations in blood pressure measurement

Patients should be seated comfortably in a quiet environment for 5 min before BP measurements.

Three BP measurements should be recorded, 1–2 min apart, and additional measurements if the first two readings differ by >10 mmHg. BP is recorded as the average of the last two BP readings.

Additional measurements may have to be performed in patients with unstable BP values due to arrhythmias, such as in patients with AF, in whom manual auscultatory methods should be used as most automated devices have not been validated for BP measurement in AF.

Use a standard bladder cuff (12–13 cm wide and 35 cm long) for most patients, but use larger and smaller cuffs for larger (arm circumference >32 cm) and smaller (arm circumference <26 cm) arms, respectively.

The cuff should be positioned at the level of the heart with the back and arm supported, to avoid muscle contraction and isometric-exercise-dependant increases in BP.

When using auscultatory methods, use phase I and V (sudden reduction/disappearance) Korotkoff sounds to identify SBP and DBP, respectively.

Measure BP in both arms at the first visit to detect possible between-arm differences. Use the arm with the higher value as the reference.

Measure BP 1 min and 3 min after standing from the seated position in all patients at the first measurement to exclude orthostatic hypotension. Lying and standing BP measurements should also be considered in subsequent visits in older people, in people with DM, and in other conditions in which orthostatic hypotension may frequently occur. Initial orthostatic hypotension may occur <1 min after standing and may be difficult to detect with conventional measurement techniques.

Record heart rate and use pulse palpation to exclude arrhythmia.

AF = atrial fibrillation; BP = blood pressure; DBP = diastolic blood pressure; DM = diabetes mellitus; SBP = systolic blood pressure.

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Table 15 Indications for home blood pressure monitoring or ambulatory blood pressure monitoring

Conditions in which white-coat hypertension is more common, for example:
● Grade 1 hypertension on office BP measurement
● Marked office BP elevation without HMOD
Conditions in which masked hypertension is more common, for example:
● High-normal office BP
● Normal office BP in individuals with HMOD or at high total CV risk
Postural and post-prandial hypotension in untreated and treated patients
Evaluation of resistant hypertension
Evaluation of BP control, especially in treated higher-risk patients
Exaggerated BP response to exercise
When there is considerable variability in the office BP
Evaluating symptoms consistent with hypotension during treatment
Specific indications for ABPM rather than HBPM:
● Assessment of nocturnal BP values and dipping status (e.g. suspicion of nocturnal hypertension, such as in sleep apnoea, CKD, DM, endocrine hypertension, or autonomic dysfunction)

ABPM = ambulatory blood pressure monitoring; BP = blood pressure; CKD = chronic kidney disease; CV = cardiovascular; DM = diabetes mellitus; HBPM = home blood pressure monitoring; HMOD = hypertension-mediated organ damage.

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When hypertension is suspected, the diagnosis of hypertension should be confirmed, either by repeated office BP measurements over a number of visits, or by 24-h ABPM or HBPM (Figure 14).

4.7.3.1 White-coat and masked hypertension

White-coat hypertension refers to BP that is elevated in the office but is normal when measured by ABPM or HBPM. It occurs in up to 30–40% of patients. The risk associated with white-coat hypertension is lower than sustained hypertension but may be higher than normotension. People with white-coat hypertension should receive lifestyle advice to reduce their CV risk and be offered BP measurement at least every 2 years by ABPM or HBPM because of high rates of transition to sustained hypertension. Routine drug treatment for white-coat hypertension is not indicated.

Masked hypertension refers to patients with a normal office BP but an elevated BP on ABPM or HBPM. These patients often have HMOD and are at a CV risk level at least equivalent to sustained hypertension. It is more common in younger people and in those with high-normal office BP. In masked hypertension, lifestyle changes are recommended, and drug treatment should be considered to control ‘out-of-office’ BP, with periodic monitoring of BP, usually with HBPM.

4.7.4. Clinical evaluation and risk stratification in hypertensive patients

The routine work-up for hypertensive patients is shown in Table 16. Alongside clinical examination, this is designed to:

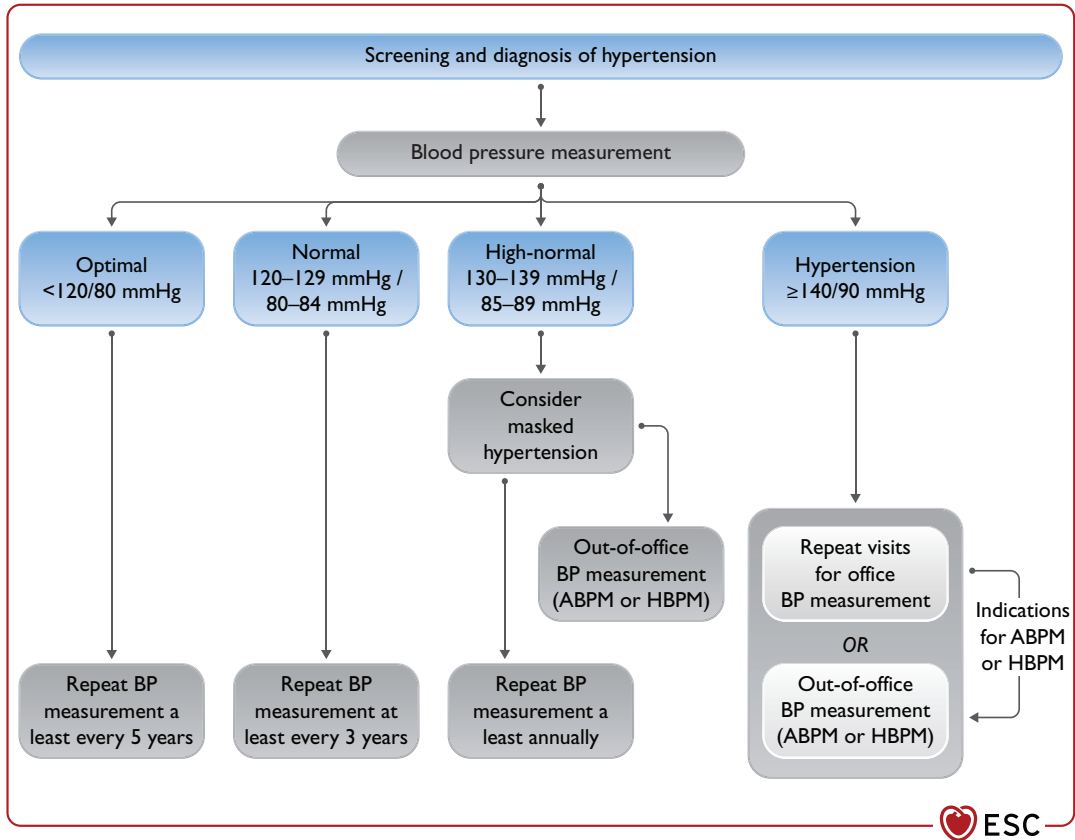


Figure 14 Screening and diagnosis of hypertension. ABPM = ambulatory blood pressure monitoring; BP = blood pressure; HBPM = home blood pressure monitoring.

Table 16 Routine tests for patients with hypertension

Routine tests
Haemoglobin and/or haematocrit
Fasting blood glucose and/or HbA1c
Blood lipids: total cholesterol, LDL-C, HDL-C, triglycerides
Blood potassium and sodium
Blood uric acid
Blood creatinine and eGFR
Blood liver function tests
Urine analysis: microscopic; urinary protein by dipstick or, ideally, ACR
12-lead ECG

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ACR = albumin-to-creatinine ratio; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; HbA1c = glycated haemoglobin; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol.

- Assess risk factors for ASCVD (see [section 3.2](#)), or the presence of cardiac, vascular, or renal disease
- Detect evidence of HMOD, e.g. LV hypertrophy, renal disease, or retinopathy
- Consider potential secondary causes of hypertension, e.g. renovascular disease, hyperaldosteronism, or pheochromocytoma (see [Table 17](#)). Also, carefully evaluate substance abuse (e.g. cocaine), drugs that may increase BP (e.g. cyclosporine, sympathomimetics), liquorice, etc. More detail on work-up of suspected secondary hypertension is provided elsewhere.⁴

Echocardiography is recommended in patients with ECG abnormalities, and should be considered when the result will influence clinical decision-making. Fundoscopy is recommended in grade 2 or 3 hypertension and in all patients with DM. The routine measurement of other biomarkers and use of vascular imaging are not recommended.^{548–551}

4.7.5. Treatment of hypertension

The treatment of hypertension involves lifestyle interventions for all patients and drug therapy for most patients.

4.7.5.1 Lifestyle interventions to lower blood pressure and/or reduce cardiovascular risk

Lifestyle interventions are indicated for all patients with high-normal BP or hypertension because they can delay the need for drug treatment or complement the BP-lowering effect of drug treatment. Moreover, most lifestyle interventions have health benefits beyond their effect on BP. Lifestyle is discussed extensively in [section 4.3](#).

4.7.5.2 Initiation of drug treatment

Drug treatment decisions in CVD prevention are mostly based on absolute CVD risk, risk modifiers, comorbidities, estimated benefit of treatment, frailty, and patient preferences. The same is true for hypertension. Drug treatment of grade 1 hypertension (SBP 140–159 mmHg) has level A evidence for reducing CVD risk. In younger patients, however, the absolute 10-year CVD risk is often low, and lifetime benefit of treatment should be considered and communicated before instituting treatment ([Figure 6](#) and [section 3.2.3.6](#)). In many such cases, the absolute lifetime benefit per 10-mmHg

Table 17 Patient characteristics that should raise the suspicion of secondary hypertension.

Characteristics
Younger patients (<40 years) with grade 2 hypertension or onset of any grade of hypertension in childhood
Acute worsening of hypertension in patients with previously documented chronically stable normotension
Resistant hypertension (BP uncontrolled despite treatment with optimal or best-tolerated doses of three or more drugs including a diuretic, and confirmed by ABPM or HBPM)
Severe (grade 3) hypertension or a hypertension emergency
Presence of extensive HMOD
Clinical or biochemical features suggestive of endocrine causes of hypertension or CKD
Clinical features suggestive of OSA
Symptoms suggestive of pheochromocytoma or family history of pheochromocytoma

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ABPM = ambulatory blood pressure monitoring; BP = blood pressure; CKD = chronic kidney disease; HBPM = home blood pressure monitoring; HMOD = hypertension-mediated organ damage; OSA = obstructive sleep apnoea. Adapted from ⁴

reduction in SBP is at least moderate to high [[Figure 15](#) (lifetime benefit calibrated in low-to-moderate CVD risk countries)]. Also, the presence of HMOD mandates treatment of grade 1 hypertension. For grade 2 hypertension or higher (SBP >160 mmHg), treatment is recommended, because not only is the lifetime benefit of reducing BP almost universally high in such patients, there is also the importance of reducing the risk of HMOD resulting in other morbidities such as renal disease, haemorrhagic cerebrovascular disease, and HF.

4.7.5.3 Blood pressure treatment targets

When drug treatment is used, the aim is to control BP to target within 3 months. Evidence now suggests that the BP targets in the previous iteration of this guideline² were too conservative, especially for older patients. In line with the stepwise approach ([section 3.2.3.1](#)), it is now recommended that the first step in all treated patients should achieve a treated SBP <140 mmHg and diastolic BP (DBP) <80 mmHg.^{552,554} The recommended ultimate SBP treatment target range for younger patients (18–69 years) is 120–130 mmHg, although some patients may safely achieve lower treated SBP levels than this and, if they are well tolerated, there is no need to back-titrate treatment.^{552,554–556} The ultimate target SBP for patients aged ≥70 years is <140 mmHg and down to 130 mmHg if tolerated.^{552,554,557,580} This change in the BP target range for older people compared with the 2016 ESC prevention guidelines² is supported by evidence that these treatment targets are safely achieved in many older patients and are associated with significant reductions in the risk of major stroke, HF, and CV death.^{557,580} It also takes into account that the even lower SBP in the intensively treated group in SPRINT (Systolic Blood Pressure Intervention Trial) (mean 124 mmHg) probably reflects a conventional office SBP range of 130–139 mmHg.⁵⁵⁵ It is recognized, however, that the evidence supporting more strict targets is less strong for very old people (>80 years) and those who are frail. Also, in these older and especially frail

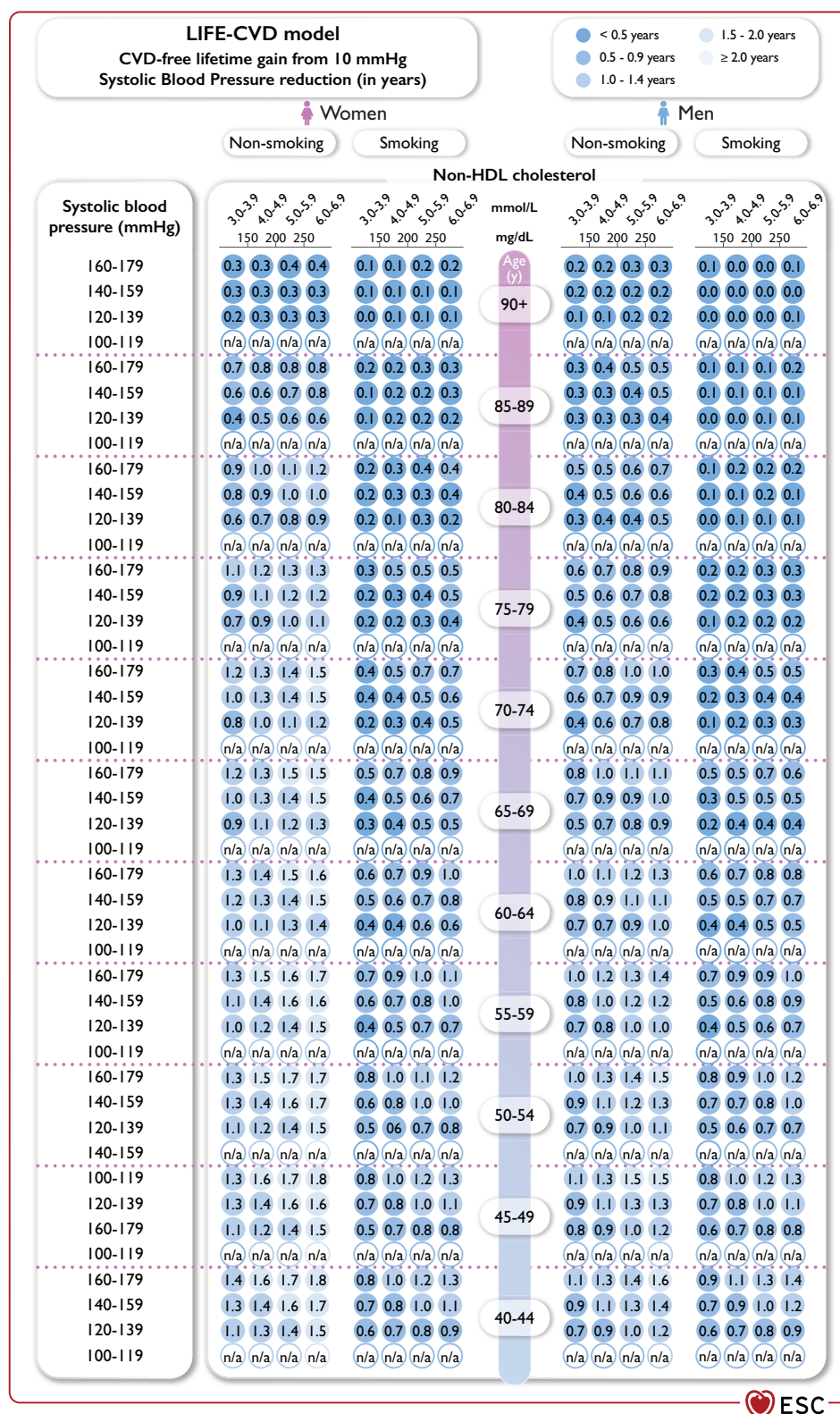


Figure 15 Lifetime benefit from lowering systolic blood pressure by 10 mmHg for apparently healthy persons, based on the following risk factors: age, sex, current smoking, systolic blood pressure, non-high-density lipoprotein cholesterol. The model is currently validated for low- and moderate-risk countries. The lifetime benefit is expressed as 'years of median life expectancy free from myocardial infarction or stroke' gained from 10 mmHg SBP lowering. The lifetime benefit is calculated by estimating lifetime CVD risk with the LIFE-CVD model multiplied by the HR (0.80) from a meta-analysis of the effect of BP lowering. For 20 mmHg SBP lowering, the average effect is almost twice as large, etc. For individualized estimations of lifetime benefit, this table can be used or the electronic version of LIFE-CVD, assessable via the ESC CVD risk app or <https://u-prevent.com/>. BP = blood pressure; CVD = cardiovascular disease; ESC = European Society of Cardiology; HDL-C = high-density lipoprotein cholesterol; HR = hazard ratio; LIFE-CVD = LIFETIME-perspective CardioVascular Disease; N/A = not applicable; SBP = systolic blood pressure.

Table 18 Recommended office blood pressure target ranges. The first step in all groups is a reduction to systolic blood pressure <140 mmHg. The subsequent optimal goals are listed below.

Age group	Office SBP treatment target ranges (mmHg)				
	Hypertension	+ DM	+ CKD	+ CAD	+ Stroke/TIA
18 – 69 years	120 – 130	120 – 130	<140 – 130	120 – 130	120 – 130
≥70 years	<140 mmHg, down to 130 mmHg if tolerated <i>Lower SBP acceptable if tolerated</i>				
DBP treatment target (mmHg)	<80 for all treated patients				

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CAD = coronary artery disease; CKD = chronic kidney disease; DBP = diastolic blood pressure; DM = diabetes mellitus; SBP = systolic blood pressure; TIA = transient ischaemic attack.

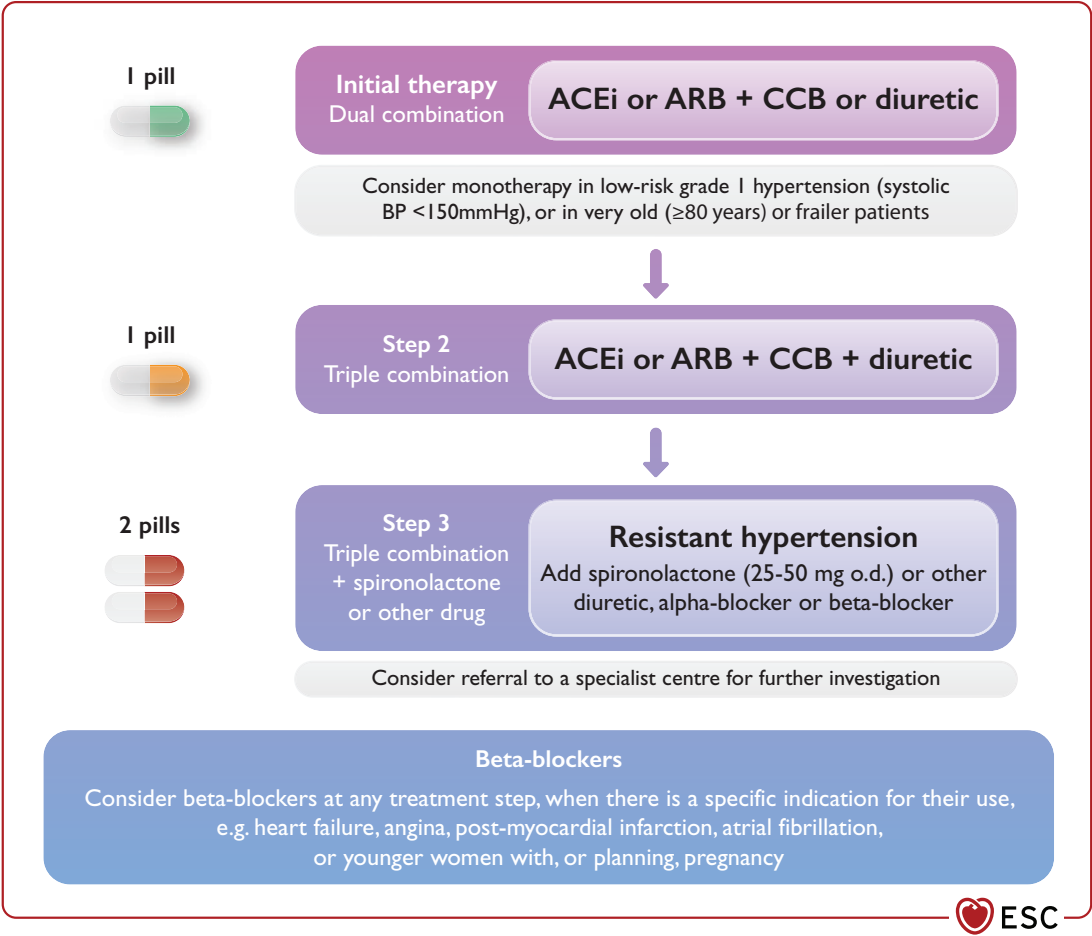


Figure 16 Core drug treatment strategy for hypertension. This algorithm is appropriate for most patients with hypertension-mediated organ damage, diabetes mellitus, cerebrovascular disease, and peripheral artery disease. ACE = angiotensin-converting enzyme; AF = atrial fibrillation; ARB = angiotensin receptor blocker; BP = blood pressure; CCB = calcium channel blocker; HF = heart failure; o.d. = *omni die* (once a day).

patients, it may be difficult to achieve the recommended target BP range due to poor tolerability or adverse effects, and high-quality measurement and monitoring for tolerability and adverse effects is especially important in these groups.⁵⁸⁰

Compared to previous ESC/ESH Hypertension Guidelines,⁴ we changed the cut-off for identifying who is 'older' from 65 to 70 years for reasons of consistency with other parts of the current guidelines. Although a single age cut-off is provided, it is important to stress that biological age influences this threshold in clinical practice. For

example, a very fit 75-year-old person may qualify for a treatment policy normally reserved for those <70 and, vice versa, a very frail 65-year-old person should sometimes be considered 'older'.

BP targets for patient subgroups with various comorbidities are shown in Table 18.

4.7.5.3.1. Blood pressure targets according to ambulatory and home blood pressure monitoring. There are no outcome-based trials that have used ABPM or HBPM to guide treatment. Therefore,

ABPM and HBPM BP targets are extrapolated from observational data. A treated office SBP of 130 mmHg likely corresponds to a 24-h SBP of 125 mmHg and home SBP <130 mmHg.⁴

4.7.5.4 Drug treatment of hypertension

The most important driver of benefit is the magnitude of BP lowering. Single-drug therapy will rarely achieve optimal BP control.

Initial therapy with a combination of two drugs should be considered usual care for hypertension.^{560–563,565,581} The only exceptions would be patients with a baseline BP close to the recommended target, who might achieve that target with a single drug, or very old (>80 years) or frail patients who may better tolerate a more gentle reduction of BP. Initial combination therapy, even low-dose combination therapy, is more effective at lowering BP than monotherapy,^{560,561,565} and will reduce BP faster and reduce heterogeneity in response.^{560,565} Moreover, initial combination therapy does not increase risk of adverse effects.^{560–563,565} Initiating therapy with two drugs will also help overcome treatment inertia where patients remain on one drug long term despite inadequate BP control.⁵⁶²

Single-pill strategy to treat hypertension: poor adherence to BP-lowering medication is a major cause of poor BP control rates, and is directly related to the number of pills.⁵⁸¹ Single-pill combination therapy (if available) is the preferred strategy. This strategy will control BP in most patients.^{560–565}

Recommended drug therapy and treatment algorithm: five major classes of BP-lowering drug therapy have shown benefit in reducing CV events; angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), beta-blockers, calcium channel blockers (CCBs), and thiazide or thiazide-like diuretics.⁵⁸² A recommended treatment algorithm based on best available evidence, pragmatic considerations (e.g. combination pill availability), and pathophysiological reasoning is shown in Figure 16.⁴ A combination of an ACE inhibitor or ARB with a CCB or thiazide/thiazide-like diuretic is the preferred initial therapy for most patients with hypertension.^{566–569} For those in whom treatment requires escalation to three drugs, a combination of an ACE inhibitor or ARB with a CCB and a thiazide/thiazide-like diuretic should be used.^{563,570,571} Beta-blockers should be used when there is a specific indication (e.g. angina, post myocardial infarction, arrhythmia, HFrEF, or as an alternative to an ACE inhibitor or ARB in women of child-bearing potential).⁵⁸² Combinations of an ACE inhibitor and an ARB are not recommended because of no added benefit on outcomes and increased risk of harm.^{575,576}

Specific modifications to the treatment algorithm are recommended for patients with CHD, CKD, HF, and AF.⁴

4.7.6. Resistant hypertension

Resistant hypertension is defined as BP being uncontrolled despite treatment with optimal or best-tolerated doses of three or more drugs including a diuretic, and confirmed by ABPM or HBPM. The prevalence of resistant hypertension is likely to be <10% of treated hypertensive patients. Spironolactone is the most effective drug for lowering BP in resistant hypertension when added to existing treatment; however, the risk of hyperkalaemia is increased in patients with

CKD and eGFR <45 mL/min/m² and blood potassium levels >4.5 mmol/L.^{555,572} Potassium-binding drugs reduce the risk of hyperkalaemia.⁵⁷³ When spironolactone is not tolerated, amiloride, alpha-blockers, beta-blockers, or centrally acting drugs, such as clonidine, have evidence supporting their use.^{555,572,574} Renal denervation and device-based therapy may be considered for specific cases, and are discussed in the 2018 ESC/ESH hypertension guidelines.⁴

4.7.7. Management of hypertension in women

The diagnosis and treatment of hypertension in women is similar to that in men, except for women of child-bearing potential or during pregnancy, because of potential adverse effects of some drugs on the foetus, especially in the first trimester. In addition, the effect of oral contraceptive pills on the risk of developing or worsening hypertension should be considered.⁴

4.7.8. Duration of treatment and follow-up

Treatment of hypertension is usually maintained indefinitely because cessation of treatment usually results in a return of BP to pretreatment levels. In some patients with successful lifestyle changes, it may be possible to gradually reduce the dose or number of drugs. After BP is stable and controlled, visits should be scheduled at least annually, and include the control of other risk factors, renal function, and HMOD, as well as reinforce lifestyle advice. When there is a loss of BP control in a previously well-controlled patient, non-compliance with therapy should be considered. Self-measurement of BP using HBPM helps engage the patient in their own management and can improve BP control. HBPM is essential to monitor BP control in patients with a significant ‘white-coat effect’ or masked hypertension. Supervision of patient follow-up increasingly involves nurses and pharmacists and is likely to become increasingly supported by telemedicine and app-based technologies.

4.8. Diabetes mellitus

Recommendations for the treatment of patients with diabetes mellitus

Recommendations	Class ^a	Level ^b
Screening		
When screening for DM in individuals with or without ASCVD, assessment of HbA1c (which can be done non-fasting) or fasting blood glucose should be considered. ⁵⁸³	IIa	A
Lifestyle		
Lifestyle changes including smoking cessation, a low saturated fat, high-fibre diet, aerobic PA, and strength training are recommended. ⁵⁸⁴	I	A
Reduction in energy intake is recommended to patients, to help achieve lower body weight or prevent or slow weight gain. ⁵⁸⁴	I	B
For those motivated to try, considerable weight loss with use of low-calorie diets followed by food reintroduction and weight-maintenance phases early after diagnosis can lead to DM remission and should be considered. ^{585,586}	IIa	A

Continued

Glycaemia targets		
A target HbA1c for the reduction of CVD risk and microvascular complications of DM of <7.0% (53 mmol/mol) is recommended for the majority of adults with either type 1 or type 2 DM. ^{587,588}	I	A
For patients with a long duration of DM and in old or frail adults, a relaxing of the HbA1c targets (i.e. less stringent) should be considered. ⁵⁸⁸	IIa	B
A target HbA1c of ≤6.5% (48 mmol/mol) should be considered at diagnosis or early in the course of type 2 DM in persons who are not frail and do not have ASCVD. ^{587,588}	IIa	B
Treatment of hyperglycaemia and ASCVD/cardiorenal risks		
Metformin is recommended as first-line therapy, following evaluation of renal function, in the majority of patients without previous ASCVD, CKD, or HF. ⁵⁸⁹	I	B
In persons with type 2 DM with ASCVD, metformin should be considered, unless contraindications are present. ^{5,590–592}	IIa	B
Avoidance of hypoglycaemia and excessive weight gain should be considered. ^{559,588,593}	IIa	B
In persons with type 2 DM and ASCVD, the use of a GLP-1RA or SGLT2 inhibitor with proven outcome benefits is recommended to reduce CV and/or cardiorenal outcomes. ^{590–592}	I	A
In patients with type 2 DM and TOD, ^c the use of an SGLT2 inhibitor or GLP-1RA with proven outcome benefits may be considered to reduce future CV and total mortality. ^{594–597}	IIb	B
In patients with type 2 DM and CKD, the use of an SGLT2 inhibitor is recommended to improve ASCVD and/or cardiorenal outcomes. ^{598,599}	I	A
In patients with type 2 DM and HFREF, use of an SGLT2 inhibitor with proven outcome benefits is recommended to lessen HF hospitalizations and CV death. ^{600,601}	I	A
In patients with type 2 DM but without ASCVD, HF, or CKD, use of an SGLT2 inhibitor or GLP-1RA should be considered based on estimated future risks (e.g. with the ADVANCE risk score or DIAL model) for adverse CVD or cardiorenal outcomes from risk factor profiles. ⁶⁰²	IIa	B

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ACR = albumin-to-creatinine ratio; ADVANCE = Action in Diabetes and Vascular disease: preterAx and diamicroN-MR Controlled Evaluation; ASCVD = atherosclerotic cardiovascular disease; CKD = chronic kidney disease; CV = cardiovascular; DIAL = Diabetes lifetime-perspective prediction; DM = diabetes mellitus; eGFR = estimated glomerular filtration rate; GLP-1RA = glucagon-like peptide-1 receptor agonist; HbA1c = glycated haemoglobin; HF = heart failure; HFREF = heart failure with reduced ejection fraction; PA = physical activity; SGLT2 = sodium-glucose cotransporter 2; TOD = target organ damage.

^aClass of recommendation.

^bLevel of evidence.

^cSee Table 4 for details.

4.8.1. Key risk factor concepts and newer paradigms

Except for glucose management, prevention of ASCVD follows the same principles as for people without type 2 DM. Achieving BP and LDL-C targets is particularly important. More recently, trial evidence has shown that drugs in the sodium-glucose cotransporter 2 (SGLT2) inhibitor or glucagon-like peptide-1 receptor agonist (GLP-1RA) classes lower ASCVD, HF, and renal risks independently of baseline HbA1c and whether patients are on metformin. Such benefits are most evident in those with existing ASCVD, HF, or CKD, but appear to extend to groups at elevated risk. This has led to newer treatment algorithms.

4.8.1.1 Lifestyle intervention

Lifestyle management is a first priority for ASCVD prevention and management of DM. Most persons with DM are obese, so weight control is crucial. Several dietary patterns can be adopted, where the predominance of fruits, vegetables, wholegrain cereals, and low-fat protein sources is more important than the precise proportions of total energy provided by the major macronutrients. Salt intake should be restricted. Specific recommendations include limiting saturated and trans fats and alcohol intake, monitoring carbohydrate consumption, and increasing dietary fibre. A Mediterranean-type diet, where fat sources are derived primarily from monounsaturated oils, is protective against ASCVD. More detail is provided in section 4.3.2.

A combination of aerobic and resistance exercise training is effective in preventing the progression of type 2 DM and for the control of glycaemia. Smokers should be offered cessation support (see section 4.5). Lifestyle intervention lowers future microvascular and macrovascular risks as well as mortality in the longer term.⁶⁰³ Intensive lifestyle changes with low-calorie diets and mean weight losses in the region of 10 kg leads to remission of type 2 DM in around 46% of cases at 1 year and 36% by 2 years.⁵⁸⁵ In those with prediabetes, other ASCVD risk factors should be assessed both before (to incentivize improvements) and after lifestyle changes have taken place.⁶⁰⁴

4.8.1.2 Glycaemic control

The UKPDS⁵⁸⁷ established the importance of intensive glucose lowering with respect to CVD risk reduction in persons newly diagnosed with DM, with better evidence to support metformin, which correctly remains the first agent of choice for the majority of patients diagnosed with DM. Three trials were conducted to see if CV events could be reduced further with more intensive glycaemia treatment.^{559,588,593} However, there were unexpected increases in total and ASCVD deaths in the ACCORD (Action to Control Cardiovascular Risk in Diabetes) trial⁵⁵⁹ and a similar trend in VADT (Veterans Affairs Diabetes Trial).⁵⁹³ The results prompted concerns about pursuing tight glucose control, particularly in older people with DM and in those with existing ASCVD. Subsequent meta-analyses of relevant trials showed reductions in non-fatal AMI and CAD events, but no effect on stroke or total mortality.^{605,606} The meta-analyses suggested that CVD benefits for an average HbA1c reduction of 0.9% over 5 years were less than via treatment of cholesterol and BP. HbA1c targets should be personalized to individual characteristics and preferences.

Four trials of dipeptidyl peptidase-4 inhibitors^{607–610} in patients with DM and existing ASCVD or at high risk demonstrated non-inferiority (i.e. safety) but not superiority with respect to CVD risk.

There was, however, an increase in the rate of hospitalization for HF with saxagliptin in the SAVOR-TIMI 53 (Saxagliptin Assessment of Vascular Outcomes Recorded in Patients with Diabetes Mellitus Thrombolysis in Myocardial Infarction) trial.⁶⁰⁸

4.8.1.3 Newer diabetes mellitus drug classes: cardiovascular disease benefits

Recent trials from two classes of drugs (SGLT2 inhibitors and GLP-1RAs) have shown CVD benefits that appear independent of glycaemic control and, where examined, of baseline metformin use.^{596,597,611} Their results have recently been systematically meta-analysed (Supplementary Figures 1 – 4).^{590,591}

For SGLT2 inhibitors, three trials demonstrated the CV benefits of empagliflozin, canagliflozin, and dapagliflozin.^{611–613} Major adverse CV events (MACE) were reduced modestly, by 14%, with no clear effect on stroke and an unclear effect on myocardial infarction.⁵⁹⁰ However, reductions in incident HF hospitalization/CVD death by 24% and renal endpoints by 44% were seen.⁵⁹⁰ The MACE benefits were evident only in those with baseline ASCVD, but HF and renal benefits appeared to extend to those with type 2 DM with multiple risk factors. However, a more recent trial in people with type 2 DM and ASCVD showed ertugliflozin to be non-inferior to placebo with respect to MACE outcomes.⁶¹⁴ Whether the results represent a class effect is, therefore, not clear. Four further SGLT2 inhibitor trials demonstrated the benefit of canagliflozin⁵⁹⁸ and dapagliflozin⁵⁹⁹ in patients with CKD [with DAPA-CKD (Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease) showing similar benefits in people without DM], and dapagliflozin⁶⁰⁰ and empagliflozin⁶⁰¹ in patients with HFrEF, with both trials showing similar benefits in those without type 2 DM.

The specific pattern of trial results (e.g. early separation of curves for HF hospitalization) suggests that the benefits of SGLT2 inhibitors may relate more to cardiorenal haemodynamic effects than to atherosclerosis.⁶⁰⁰ Other than genitourinary infections, rates of adverse events (including diabetic ketoacidosis) were generally low. One trial showed an excess of amputations and fractures,⁶¹² but none of the other trials noted imbalances. Patients should be advised on the importance of genitourinary hygiene before being prescribed these medications.

GLP-1RAs reduce MACE, CV death, and all-cause mortality by around 12%, with around a 9% reduction in myocardial infarction and a 16% reduction in stroke.⁵⁹¹ Furthermore, HF is lowered by 9% and a composite renal outcome was lowered by 17%. The results cannot be explained by lowering of glucose levels and, in multiple SGLT2 inhibitor and GLP-1RA trials, subgroup analyses suggested that these benefits could be independent of metformin use.^{594–597} Most trials were conducted in patients with existing ASCVD or, in the REWIND (Researching Cardiovascular Events With a Weekly Incretin in Diabetes) trial, with a significant proportion of patients at high risk for CVD.⁶¹⁵ Side-effects of this class mainly include nausea and vomiting, which can lessen with gradual up-titration. Risks of hypoglycaemia can be reduced by lowering doses of sulphonylureas or insulin.

The largely positive results of these two classes of drugs (SGLT2 inhibitors and GLP-1RAs) have led to rapid changes in DM algorithms, but with some differences in interpretation.⁶⁰² Most DM guidelines, including those within the 2020 American Diabetes Association (ADA)/European Association for the Study of Diabetes

(EASD) consensus report,⁵⁹² recommend that metformin should be used as first-line treatment, while the ESC Guidelines⁵ recommended in 2019 that SGLT2 inhibitors and GLP-1RAs may be used without metformin in people with DM and CVD or at high risk of CVD, as reviewed.⁶⁰² A subset of the writing groups of the ADA/EASD consensus report and the ESC Guidelines⁶¹⁶ was convened as an expert panel. The expert panel emphasized the overall commonalities of approach and the need to ensure that people with type 2 DM, CVD, HF, or CKD are treated appropriately with an SGLT2 inhibitor or GLP-1RA. The panel concluded that this approach should be initiated independent of background therapy, glycaemic control, or individualized treatment goals.⁶¹⁶ The view of the ESC is that metformin should be considered, but is not mandatory first-line treatment in patients with ASCVD or evidence of TOD. Certainly, the initiation of metformin in such patients should not forego or delay the initiation of evidence-based SGLT2 inhibitors or GLP-1RAs. A risk score plus cost-effective analyses would be useful to determine which patients free from ASCVD or evidence of TOD may be recommended for these newer drugs. In all the above, there is no evidence of any sex interaction in benefits. Finally, people with type 2 DM should be involved in decision-making after explanation of the potential benefits and side-effects of the drugs.

4.8.2. Type 1 diabetes mellitus

The DCCT (Diabetes Control and Complications Trial) established the importance of tight glucose control to lessen the risks of both microvascular and macrovascular disease in both men and women with type 1 DM.⁶¹⁷ A 27-year follow-up of this trial showed that 6.5 years of intensive DM therapy was associated with a modestly lower all-cause mortality rate.⁶¹⁷ A glycaemic target for HbA1c of 6.5–7.5% (48–58 mmol/mol) appears to be a balanced approach for long-term care.

Recently, metformin was shown not to lower progression of carotid IMT in persons with type 1 DM considered to be at elevated CVD risk.⁶¹⁸ Its use is not recommended in type 1 DM for this indication. SGLT2 inhibitors improve metabolic control in type 1 DM and may complement insulin therapy in selected patients.

4.9. Antithrombotic therapy

Recommendations for antithrombotic therapy

Recommendations	Class ^a	Level ^b
Aspirin 75 - 100 mg daily is recommended for secondary prevention of CVD. ⁶¹⁹	I	A
Clopidogrel 75 mg daily is recommended as an alternative to aspirin in secondary prevention in case of aspirin intolerance. ⁶²⁰	I	B
Clopidogrel 75 mg daily may be considered in preference to aspirin in patients with established ASCVD. ^{620,621}	IIb	A
Concomitant use of a proton pump inhibitor is recommended in patients receiving antiplatelet therapy who are at high risk of gastrointestinal bleeding. ^{622,623}	I	A

Continued

In patients with DM at high or very high CVD risk, low-dose aspirin may be considered for primary prevention in the absence of clear contraindications. ^{5,624,625}	IIb	A
Antiplatelet therapy is not recommended in individuals with low/moderate CV risk due to the increased risk of major bleeding. ^{624,626–630}	III	A

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ASCVD = atherosclerotic cardiovascular disease; CV = cardiovascular; DM = diabetes mellitus.

^aClass of recommendation.

^bLevel of evidence.

4.9.1. Antithrombotic therapy in individuals without atherosclerotic disease

In 2009, a meta-analysis in patients with low CVD risk reported a 12% reduction in ASCVD with aspirin but a significant increase in major bleeding.⁶¹⁹ CVD risk reduction and bleeding risks were similar in men and women.⁶³¹ More contemporary primary prevention trials reported no or little benefit in patients without ASCVD and a consistent increase in bleeding.^{624,626,627} An updated meta-analysis did not show a reduction in all-cause or CV mortality with aspirin, but did show a lower risk of non-fatal myocardial infarction (RR 0.82) and ischaemic stroke (RR 0.87).⁶²⁸ Conversely, aspirin was associated with a higher risk of major bleeding (RR 1.50), intracranial bleeding (RR 1.32), and major gastrointestinal bleeding (RR 1.52), with no difference in the risk of fatal bleeding (RR 1.09). Bleeding risks were particularly increased in older persons. Other recent meta-analyses found very similar results.^{629,630} Overall, although aspirin should not be given routinely to patients without established ASCVD, we cannot exclude that in some patients at high or very high CVD risk, the benefits outweigh the risks.^{632,633} In patients with DM and no evident ASCVD, the ASCEND study reported a 12% risk reduction and a significant increase in major bleeding, but not in fatal or intracranial bleeding.⁶²⁴ A meta-analysis of aspirin for primary prevention in DM found a number needed to treat of 95 to prevent one major adverse ischaemic event in 5 years.⁶²⁵ Hence, as in patients without DM, aspirin may be considered if CVD risk is exceptionally high. Only one in four patients in the ASCEND study were being treated with a proton pump inhibitor. Wider use than this could potentially amplify the benefit of aspirin in primary prevention for patients at higher atherosclerotic risk.

In apparently healthy persons <70 years of age with (very) high CVD risk, further studies are needed. Until then, decisions in these high-risk persons should be made on a case-by-case basis, taking both ischaemic risk and bleeding risk into consideration.

4.9.2. Antithrombotic therapy in individuals with established atherosclerotic disease

In established atherosclerotic disease, aspirin is associated with significant reductions in serious vascular events, including stroke and coronary events, and a 10% reduction in total mortality.⁶¹⁹ These benefits outweigh the bleeding hazards.

In patients with previous myocardial infarction, stroke, or LEAD, clopidogrel showed a slight superiority for ischaemic events with respect to aspirin, with a similar safety profile.⁶²⁰

Subgroup analysis suggested a greater benefit of clopidogrel in patients with LEAD. A meta-analysis showed a clinically modest risk reduction with P2Y₁₂ inhibitor monotherapy (number needed to treat: 244), and no effect on all-cause or vascular mortality and major bleeding.⁶²¹ More guidance on antithrombotic treatment in the specific settings of CAD, cerebrovascular disease, and LEAD, including possible indications for dual pathway inhibition in patients with LEAD, is given in [section 6](#).

4.9.3. Proton pump inhibitors

Proton pump inhibitors reduce the risk of gastrointestinal bleeding in patients treated with antiplatelet drugs and may be a useful adjunctive therapy to improve safety.^{634,635} Proton pump inhibitors that specifically inhibit CYP2C19 (omeprazole or esomeprazole) may reduce the pharmacodynamic response to clopidogrel. Although this interaction has not been shown to affect the risk of ischaemic events, coadministration of omeprazole or esomeprazole with clopidogrel is not recommended.⁶²²

4.10. Anti-inflammatory therapy

Recommendation for anti-inflammatory therapy

Recommendation	Class ^a	Level ^b
Low-dose colchicine (0.5 mg o.d.) may be considered in secondary prevention of CVD, particularly if other risk factors are insufficiently controlled or if recurrent CVD events occur under optimal therapy. ^{85,86}	IIb	A

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CVD = cardiovascular; o.d. = omni die (once a day).

^aClass of recommendation.

^bLevel of evidence.

Acknowledging that the process of atherosclerosis has inflammatory components has led to the investigation of various anti-inflammatory therapies in recent years. The first study to examine the effects of reducing inflammation without impacting lipid levels was CANTOS (Canakinumab Antiinflammatory Thrombosis Outcome Study), in which the monoclonal antibody, canakinumab, provided proof-of-concept for anti-inflammatory therapy in high-risk patients.⁶³⁶ This particular drug was, however, not further developed for this indication because of the risk of fatal infections and high costs. Methotrexate was the second anti-inflammatory drug studied for this purpose, but was not proven effective in reducing CVD outcomes.⁶³⁷

In 2019, COLCOT (Colchicine Cardiovascular Outcomes Trial) reported a significant reduction (HR 0.77) in CVD outcomes with low-dose colchicine [0.5 mg o.d. (once a day)] in patients with a recent AMI. The more recent LoDoCo2 (second low-dose colchicine) trial reinforced these results in patients with chronic CAD (HR 0.69).⁸⁵ This study observed a trend towards increased non-CV mortality, which requires further attention.

The use of colchicine in daily practice remains to be established based on further clinical study data and experiences in daily practice. Nonetheless, the encouraging results justify consideration of low-dose colchicine in selected, high-risk patients.

4.11. Cardiac rehabilitation and prevention programmes

Recommendations for cardiac rehabilitation

Recommendations	Class ^a	Level ^b
Participation in a medically supervised, structured, comprehensive, multidisciplinary EBCR and prevention programme for patients after ASCVD events and/or revascularization, and for patients with HF (mainly HFrEF), is recommended to improve patient outcomes. ^{638–642}	I	A
Methods to increase CR and prevention referral and uptake should be considered (i.e. electronic prompts or automatic referrals, referral and liaison visits, structured follow-up by nurses or health professionals, and early programme initiation after discharge). ^{643–646}	IIa	B
Home-based CR, telehealth, and mHealth interventions may be considered to increase patient participation and long-term adherence to healthy behaviours. ^{647,648}	IIb	B

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ASCVD = atherosclerotic cardiovascular disease; CR = cardiac rehabilitation; EBCR = exercise-based cardiac rehabilitation; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; mHealth = mobile device-based healthcare.

^aClass of recommendation.

^bLevel of evidence.

CR is a comprehensive, multidisciplinary intervention not just including exercise training and PA counselling, but also education, risk factor modification, diet/nutritional counselling, and vocational and psychosocial support.³⁵⁸ Prevention and rehabilitation programmes after ASCVD events or revascularization reduce CV hospitalizations, myocardial infarction, CV mortality and, in some programmes, all-cause mortality.^{638,640–642} They may also reduce depressive/anxiety symptoms.⁶⁴⁹ In patients with chronic HF (mainly HFrEF), exercise-based cardiac rehabilitation (EBCR) may improve all-cause mortality, reduce hospital admissions, and improve exercise capacity and quality of life.^{639,650} CR is generally cost-effective.⁶⁵¹

Clinical trials and registries are highly heterogeneous, which influences national guidelines, legislation, and reimbursement.^{652,653} The results of recent reviews provide clinicians with minimal requirements for successful CR after ACS or coronary artery bypass graft:

- CR is a comprehensive multidisciplinary intervention^{466,649,654,655}
- CR is supervised and carried out by adequately trained health professionals, including cardiologists⁶⁴⁹
- CR starts as soon as possible after the initial CV event⁶⁴⁹
- EBCR includes aerobic and muscular resistance exercise, which should be individually prescribed based on pre-exercise screening and exercise testing⁶⁵⁶
- The dose of EBCR (number of weeks of exercise training × average number of sessions/week × average duration of session in minutes) exceeds 1000⁶³⁸
- The number of EBCR sessions needs to exceed 36⁶⁴¹
- During CR, all individually recognized CV risk factors need to be addressed and treated.⁶⁴²

Recently, the European Association of Preventive Cardiology (EAPC) proposed minimal and optimal standards for improvement of secondary prevention through CR programmes in Europe.⁶⁵⁷

Although exercise training prescription should adopt the FITT (frequency, intensity, time duration, and type of exercise) model, inter-clinician variance and disagreement exists.⁶⁵⁸ To optimize exercise training, the EAPC has introduced a digital, interactive decision support tool; the EXPERT (EXercise Prescription in Everyday practice & Rehabilitation Training) Tool (<https://www.escardio.org/Education/Practice-Tools/CVD-prevention-toolbox/expert-tool>).⁶⁵⁹ No single exercise component is a significant predictor of mortality; only adherence to the full intervention improves outcome.⁶⁶⁰

Despite proven benefits, rates of referral, participation, and implementation are low.^{653,660,661} Uptake seems lower in women, but a variety of other intrapersonal, interpersonal, clinical, logistical, health system, and CR programme-related factors affect participation and adherence.⁶⁶² CR enrolment is higher if trained nurses or allied healthcare providers intervene face-to-face, whereas adherence may be higher when remote interventions are implemented (i.e. home-based).⁶⁴³ Nurse-coordinated programmes can increase effectiveness.^{644–646} Home-based CR with or without telemonitoring may increase participation and appear similarly effective as centre-based CR.⁶⁴⁷ Telehealth interventions are more effective than no intervention,⁶⁴⁸ but may also complement conventional CR. Also, mobile device-based healthcare (mHealth) delivery through smartphones may be as effective as traditional centre-based CR, showing significant improvements in health-related quality of life.⁶⁶³ These novel interventions may support the patient to maintain long-term healthy behaviours after specialized CR programmes.⁶⁶⁴

5. Policy interventions at the population level

Recommendations for policy interventions at the population level

Recommendations	Class ^a	Level ^{b,c}
Policies and population approaches to PA, diet, smoking and tobacco use, and alcohol in governmental restrictions and mandates, media and education, labelling and information, economic incentives, schools, worksites, and community settings follow different levels of recommendations (see specific tables in the supplementary material for section 5).		
Putting in place measures to reduce air pollution, including reducing PM emission and gaseous pollutants, reducing the use of fossil fuels, and limiting carbon dioxide emissions, are recommended, to reduce CVD mortality and morbidity.	I	C

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CVD = cardiovascular disease; PA = physical activity; PM = particulate matter.

^aClass of recommendation.

^bLevel of evidence.

^cLevel of evidence applies less well to policy interventions, and the type of empirical evidence varies widely across the separate approaches suggested.

5.1. Population-level approaches to the prevention of cardiovascular disease

Population level approaches to CVD prevention centre around upstream measures requiring broad public-health interventions targeting lifestyle and promoting monitoring of CVD. These measures are designed to address populations and are intended to shift the population attributable risk. This is based on a prevention paradox described by Geoffrey Rose in 1981.⁶⁶⁵ The population attributable risk depends on the RR and on the prevalence of a risk factor in the general population. If the prevalence of a significant RR factor is low, then the population attributable risk may be modest. Conversely, if a low-impact RR factor is common, the population attributable risk may be high. This prevention approach following the Geoffrey Rose paradigm^{665,666} states that small shifts in the risk of disease across a whole population consistently lead to greater reductions in disease burden than does a large shift in high-risk individuals only.^{667,668} In other words, many people exposed to a small risk may generate more disease than a few exposed to a conspicuous risk. This population-wide approach—as opposed to strategies targeting high-risk individuals—has major advantages at the population level whilst sometimes having only a modest benefit at the individual level, because it addresses the CV health of a large number of individuals over the entire life course. It should be noted that high-risk and population-level prevention strategies are not mutually exclusive and must therefore coexist.

Prevalence of high-risk conditions and incidence rates of CVD vary across countries. Many of their underlying causes are known, and they are closely related to dietary habits, PA, smoking, alcohol, employment, social deprivation, and the environment. The objective of population approaches to prevention of CVD is to control the underlying determinants of CV health and, in this way, reduce population incidence rates. The population approach may bring numerous benefits, such as narrowing the gap in health inequalities, preventing other conditions such as cancer, pulmonary diseases, and type 2 DM, and saving costs from the avoided CV events and early retirement due to health problems.

Individual behaviour is enacted in an environment with hierarchical levels, which encompass individual choice, family influence, cultural and ethnic grouping, workplace, healthcare, and policy at the regional, state and global levels (e.g. EU policies and international trade agreements). The aim of this section of the guidelines is to provide evidence-based suggestions for the most effective interventions to reduce CVD risk at the population level, improve CVD health, and promote healthy choices at the community, regional, and global level. Health challenges cannot be solved by the healthcare systems alone and require political support. To advance this cause, the WHO has been organizing Global Conferences on Health promotion since 1990.

5.2. Specific risk factor interventions at the population level

Population-level interventions aim to alter the societal environment, modify certain social determinants of health, and provide incentives to encourage changes in individual behaviour and exposure to risk factors. Social determinants of health include socioeconomic status (education, occupation, and income), wealth inequalities, neighbourhood and urban design, and social networks, to name but a few. Healthcare

professionals play an important role in advocating evidence-based population-level interventions. By modifying the general context, one can induce healthy decisions as a default in entire populations (all age groups and particularly vulnerable ones). The task for both national and local authorities is to create social environments that provide healthier defaults, taking health literacy into account.^{669,670} The evidence presented here builds on recent comprehensive reviews and individual studies, noting that it is rarely feasible to use an RCT to evaluate population-level interventions (in contrast to individual-level interventions).^{671,672} The importance of heart disease in women has become apparent and sex differences in CVD prevention have prompted sex-specific awareness campaigns with the aim of reducing sex disparities in research and clinical care. While interpreting this section, it is important to recognize that there are often vested interests, which may influence policy decisions on health promotion.

The [supplementary material](#) for this section presents evidence for population-level strategies dealing with specific risk factor interventions for PA ([section 5.2.1](#)), diet ([section 5.2.2](#)), smoking and tobacco use ([section 5.2.3](#)), and alcohol consumption ([section 5.2.4](#)). Lifestyle changes at the population level take time, may be expensive, and need to be sustained over time. Furthermore, the benefits may be slow to manifest; however, they persist over the long term and improve health-related quality of life and well-being.

5.2.1. Physical activity

Please see the [supplementary material section 3.1](#).

5.2.2. Diet

Please see the [supplementary material section 3.2](#).

5.2.3. Smoking and tobacco use

Please see the [supplementary material section 3.3](#).

5.2.4. Alcohol

Please see the [supplementary material section 3.4](#).

5.3. Environment, air pollution, and climate change

Air pollution contributes to mortality and morbidity. It specifically increases the risk of respiratory and CV diseases, notably CAD, HF, cardiac arrhythmias and arrest, cerebrovascular disease, and venous thromboembolism.^{158,673,674} Loss of life-expectancy due to ambient air pollution has been estimated at 2.9 years, accounting for an estimated global excess mortality of 8.8 million/year.¹⁵⁹ Plausible mechanisms by which air pollution is linked to CVD include promoting atherosclerosis, inflammation, thrombosis, systemic vascular dysfunction, myocardial fibrosis, epigenetic changes, and interactions with traditional risk factors.¹⁵⁸

Important sources of fine particles are road traffic, power plants, and industrial and residential heating using oil, coal, and wood. Main components of outdoor air pollution include airborne PM (ranging in size from coarse particles 2.5–10 µm, fine particles <2.5 µm (PM_{2.5}), and ultrafine particles <0.1 µm in diameter) and gaseous pollutants such as ozone, nitrogen dioxide, volatile organic compounds, carbon monoxide, and sulphur dioxide, produced primarily by fossil fuel combustion.^{158,675} Up to one-third of Europeans living in urban areas

are exposed to levels exceeding EU air-quality standards. The EU Commission released a policy package to be implemented by 2030, with measures to reduce harmful emissions from traffic, energy plants, and agriculture.

Indoor air pollution and exposure to noise must also be highlighted. Household air pollution, such as that produced from burning biomass, accounts for over 3 million deaths worldwide.³⁸ It has been estimated by the WHO that 30% of the European population is exposed to nightly levels of noise exceeding 55 dB.¹⁶¹ These levels have been associated with hypertension, arteriosclerosis, CAD, CV mortality, and stroke. It should be noted that mitigating efforts to reduce noise exposure have not, as yet, proven to have a beneficial health effect.¹⁶¹

The extent to which environmental exposures in soil and water contribute to CVD has also been established.¹⁵⁷ Interventions to reduce this pollution are required, including factory regulations and drinking water controls.¹⁵⁷

Patient organizations and health professionals have an important role in supporting education and policy initiatives. Information on patients' behaviour during smog peaks is needed. Economic incentives, such as reduced taxes on electric and hybrid cars, can contribute to the improvement of air quality as well as incentives encouraging the use of public transport. Urban design promoting the construction of new houses and schools in areas remote from highways and polluting industries needs to be urged.

'Clean air' legislation aimed at promoting decreased particle emissions, and promotion of public transport should also be encouraged. The urgency of accepting what might appear as 'comfort sacrifices' for distant health benefits, and the transitory high costs of reorganizing entire sections of industry, probably remain a major dilemma to the population-based approach. An example of such legislation is the European Green Deal, by which the EU aims to be climate neutral by 2050.

5.3.1.Climate change

Climate change resulting from the increasing use of fossil fuels, as a major source of both air pollution and 'greenhouse' gases, is becoming a major public health and environmental concern. Societal measures to reduce such fuels, and transfer towards renewable sources, are becoming urgent to reduce air pollution and climate change.⁶⁷⁶ The impact of diet, notably long-term non-sustainable meat-based food production chains, as well as the impact of sedentary lifestyles on climate-altering variables, will also need to be addressed by policy makers.

5.4. Implications for public health policy and advocacy at the governmental and non-governmental level

Please see the [supplementary material section 3.5](#).

6. Risk management of disease-specific cardiovascular disease

This section addresses CVD prevention in specific clinical contexts. A significant number of patients already have such comorbidities, which put them at additional risk. The general principles of lifestyle modification and treatment of major risk factors are outlined in [section 4](#). In this section, only disease-specific aspects are added.

6.1.Coronary artery disease

Disease-specific acute management of coronary syndromes is covered in detail in recent guidelines.^{677–680}

As for antithrombotic therapy, dual antiplatelet therapy (DAPT) for 12 months, preferably with prasugrel or ticagrelor, is the standard antithrombotic treatment after ACS.^{681–683} There are conflicting data as to whether prasugrel is preferable to ticagrelor.^{684,685} A 6-month duration of DAPT after ACS is generally too short,⁶⁸⁶ but may be considered in selected patients at high bleeding risk.

In patients with chronic coronary syndromes (CCS) undergoing elective PCI, the standard duration of DAPT is 6 months, but shortening this to 1–3 months is an option when bleeding risk is very high.⁶²² Clopidogrel is the P2Y₁₂ inhibitor of choice, but prasugrel and ticagrelor may be considered after complex interventions.⁶²²

Prolonged DAPT (>12 months) following PCI for either ACS or CCS is an option for patients who tolerate DAPT well and have features of high ischaemic risk.^{687,688} In patients with stable CAD, dual-pathway inhibition with low-dose rivaroxaban (2.5 mg *b.i.d.*) and aspirin improved CV outcomes at the price of more major bleeding events than aspirin alone.⁸³

Based on the above, and in line with the CCS Guidelines,⁶²² adding a second antithrombotic drug (P2Y₁₂ inhibitor or low-dose rivaroxaban) to aspirin for long-term secondary prevention should be considered for patients who are at high ischaemic risk and do not have a high risk of bleeding. It may also be considered in patients who are at moderate ischaemic risk and without a high risk of bleeding, but the benefits are lower.⁶²² More details on antithrombotic treatment options are found in the ESC Guidelines for CCS.⁶²²

Recommendations for patients with coronary artery disease

Recommendations	Class ^a	Level ^b
Aspirin 75–100 mg daily is recommended for patients with a previous myocardial infarction or revascularization. ⁶¹⁹	I	A
Aspirin 75–100 mg daily may be considered in patients without a history of myocardial infarction or revascularization, but with definitive evidence of CAD on imaging. ⁶²²	IIb	C
In ACS, DAPT with a P2Y ₁₂ inhibitor in addition to aspirin is recommended for 12 months, unless there are contraindications such as excessive risk of bleeding. ^{681–683}	I	A
In patients with CCS, clopidogrel 75 mg daily is recommended, in addition to aspirin, for 6 months following coronary stenting, irrespective of stent type, unless a shorter duration (1–3 months) is indicated due to risk or occurrence of life-threatening bleeding. ⁶²²	I	A
Adding a second antithrombotic drug (a P2Y ₁₂ inhibitor or low-dose rivaroxaban) to aspirin for long-term secondary prevention should be considered in patients with a high risk of ischaemic events and without high bleeding risk. ^{83,622,687–689}	IIa	A

Continued

Adding a second antithrombotic drug to aspirin for long-term secondary prevention may be considered in patients with a moderate risk of ischaemic events and without a high bleeding risk. ^{83,622,687–689}	IIb	A
ACE inhibitors (or ARB) are recommended if a patient has other conditions (e.g. HF, hypertension, or DM). ⁶²²	I	A
Beta-blockers are recommended in patients with LV dysfunction or systolic HF. ⁶²²	I	A
In patients with established ASCVD, oral lipid-lowering treatment with an ultimate LDL-C goal of <1.4 mmol/L (55 mg/dL) and a ≥50% reduction in LDL-C vs. baseline is recommended.	I	A

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ACE = angiotensin-converting enzyme; ACS = acute coronary syndromes; ARB = angiotensin receptor blocker; ASCVD = atherosclerotic cardiovascular disease; CAD = coronary artery disease; CCS = chronic coronary syndromes; DAPT = dual antiplatelet therapy; DM = diabetes mellitus; HF = heart failure; LDL-C = low-density lipoprotein cholesterol; LV = left ventricular.

^aClass of recommendation.

^bLevel of evidence.

The management of dyslipidaemia and hypertension in patients with CAD is discussed in [sections 4.6 and 4.7](#), respectively. For ACE inhibitors (or ARBs) and beta-blockers, see also the 2019 ESC Guidelines for diagnosis and management of CCS.⁶²²

6.2. Heart failure

The management of HF aims to improve mortality, hospitalization rate, and quality of life.⁶⁹⁰ To achieve this, multidisciplinary management programmes and structured follow-up with patient education, optimization of medical treatment, using telehealth facilities, lifestyle changes, psychosocial support, and improved access to care are fundamental.^{691–694}

Regarding the management of CVD risk factors, similar basic rules apply for those with and without HF. However, in HF, low cholesterol levels^{695,696} and low body weight are associated with increased mortality.^{697,698} Initiation of lipid-lowering therapy is not recommended in patients with HF without compelling indications for their use.³ Whereas unintentional weight loss is associated with a worse prognosis regardless of baseline BMI, the effects of intentional weight loss remain unclear.

Conversely, regular exercise training (particularly combined aerobic and resistance exercises) improves clinical status in all patients with HF^{650,699,700} and improves CVD burden and prognosis in HFrEF.^{700,701}

It is recommended to screen all patients with HF for both CV and non-CV comorbidities; if present, they should be treated.⁶⁹⁰ These diseases include CAD, hypertension, lipid disorders, DM, obesity, cachexia and sarcopenia, thyroid disorders, CKD, anaemia, iron deficiency, and sleep apnoea.⁶⁹⁰

For patients with symptomatic HFrEF, neurohormonal antagonists [ACE inhibitors,^{702–705} ARBs,⁷⁰⁶ angiotensin receptor neprilysin inhibitors (ARNIs),^{707–710} beta-blockers,^{711–717} and mineralocorticoid receptor antagonists (MRAs)^{718,719}] improve survival and reduce the risk of HF hospitalizations.⁶⁹⁰ These drugs also reduce the

risk of CV events in patients with symptomatic HFrEF.^{702–719} Importantly, these drugs should be up-titrated to the maximum tolerated doses, which may be different for men and women, particularly in patients recently discharged after HF hospitalization.^{690,720,721}

SGLT2 inhibitors (currently dapagliflozin and empagliflozin) added on top of neurohormonal blockade reduces the risk of CV death and worsening HF in patients with symptomatic HFrEF, with or without DM,^{600,601} and are recommended for all patients with symptomatic HFrEF already treated with an ACE inhibitor (or ARNI), a beta-blocker, and an MRA.

Recently, an oral soluble guanylate cyclase receptor stimulator (vericiguat), administered along with standard neurohormonal blockade in symptomatic patients with HFrEF with recent HF hospitalization, reduced the composite of death from any cause or HF hospitalization.⁷²²

Other drugs bring additional moderate benefits for selected patients with symptomatic HFrEF. Diuretics,^{723,724} ivabradine,^{725,726} and hydralazine^{727,728} should be considered, and digoxin⁷²⁹ may be considered as complementary therapies in specific patients with symptomatic HFrEF. Some of these therapies reduce CV morbidity and mortality (e.g. ivabradine).

Additionally, for selected patients with symptomatic HFrEF, there are indications for an implantable cardioverter defibrillator to reduce the risk of sudden death and all-cause mortality, and for cardiac resynchronization therapy to reduce morbidity and mortality (for details, see 2021 HF Guidelines).⁶⁹⁰

Recommendations regarding pharmacological and non-pharmacological interventions for patients with symptomatic (New York Heart Association class II–IV) heart failure with reduced ejection fraction (left ventricular ejection fraction <40%) with proven benefits on clinical outcomes, including cardiovascular morbidity and mortality.

Recommendations	Class ^a	Level ^b
It is recommended that patients with HF are enrolled in a comprehensive CR programme to reduce the risk of HF hospitalization and death. ^c 691–694	I	A
EBCR is recommended in stable symptomatic patients with HFrEF to reduce the risk of HF hospitalization. ^{700,701}	I	A
It is recommended to screen patients with HF for both CV and non-CV comorbidities which, if present, should be treated, provided safe and effective interventions exist, not only to alleviate symptoms but also to improve prognosis. ^c	I	A
An ACE inhibitor is recommended, in addition to a beta-blocker and an MRA, for patients with symptomatic HFrEF to reduce the risk of HF hospitalization and death. ^{702–705}	I	A
A beta-blocker is recommended, in addition to an ACE inhibitor (or an ARNI) and an MRA, for patients with stable, symptomatic HFrEF to reduce the risk of HF hospitalization and death. ^{711–717}	I	A

Continued

An MRA is recommended for patients with HFrEF already treated with an ACE inhibitor (or an ARNI) and a beta-blocker, to reduce the risk of HF hospitalization and death. ^{718,719}	I	A
Sacubitril/valsartan is recommended as a replacement for an ACE inhibitor to reduce the risk of HF hospitalization and death in patients with HFrEF. ^{707,730}	I	B
An ARB is recommended to reduce the risk of HF hospitalization or CV death in symptomatic patients with HFrEF who are unable to tolerate an ACE inhibitor and/or ARNI (patients should also receive a beta-blocker and an MRA). ⁷⁰⁶	I	B
Dapagliflozin or empagliflozin are recommended, in addition to optimal treatment of an ACE inhibitor (or ARNI), a beta-blocker, and an MRA, for patients with HFrEF to reduce the risk of HF hospitalization and death. ^{600,601,730}	I	A
Vericiguat may be considered in patients with symptomatic HFrEF who have experienced HF worsening despite treatment with an ACE inhibitor (or an ARNI), a beta-blocker, and an MRA, to reduce the risk of HF hospitalization or CV death. ⁷²²	IIb	B
Diuretics are recommended in patients with HFrEF with signs and/or symptoms of congestion to reduce the risk of HF hospitalization. ^{723,724}	I	C
Ivabradine should be considered in symptomatic patients with LVEF ≤35%, in sinus rhythm, and with a resting heart rate ≥70 bpm despite treatment with an evidence-based dose of a beta-blocker (or maximum tolerated dose below that), an ACE inhibitor (or an ARNI), and an MRA, to reduce the risk of HF hospitalization or CV death. ⁷²⁵	IIa	B
Ivabradine should be considered in symptomatic patients with LVEF ≤35%, in sinus rhythm, and with a resting heart rate ≥70 b.p.m. who are unable to tolerate or have contraindications for a beta-blocker to reduce the risk of HF hospitalization or CV death. Patients should also receive an ACE inhibitor (or ARNI) and an MRA. ⁷²⁶	IIa	C
Hydralazine and isosorbide dinitrate should be considered in self-identified black patients with LVEF ≤35% or with LVEF <45% combined with a dilated LV in NYHA class III–IV despite treatment with an ACE inhibitor (or ARNI), a beta-blocker, and an MRA, to reduce the risk of HF hospitalization and death. ⁷³¹	IIa	B
Hydralazine and isosorbide dinitrate may be considered in patients with symptomatic HFrEF who cannot tolerate ACE inhibitors, ARBs, or ARNIs (or if they are contraindicated), to reduce the risk of death. ⁷²⁸	IIb	B

Continued

Digoxin may be considered in patients with symptomatic HFrEF in sinus rhythm despite treatment with an ACE inhibitor (or ARNI), a beta-blocker, and an MRA, to reduce the risk of hospitalizations (all-cause and HF).⁷²⁹

IIb**B**

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ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor neprilysin inhibitor; b.p.m. = beats per minute; CR = cardiac rehabilitation; CV = cardiovascular; EBCR = exercise-based cardiac rehabilitation; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; LV = left ventricle; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; NYHA = New York Heart Association.

^aClass of recommendation.

^bLevel of evidence.

^cApplies to all patients with HF, regardless of LVEF.

For implantable cardioverter-defibrillator and cardiac resynchronization recommendations, see ⁶⁹⁰

6.3. Cerebrovascular diseases

Interventions for cerebrovascular diseases depend on the type of event, i.e. ischaemic or haemorrhagic.^{732,733} Ischaemic events are mainly caused by atherothrombosis, cardiac embolism, or small vessel disease.⁷³⁴ Other mechanisms (e.g. arterial dissection, patent foramen ovale, thrombophilia, inherited diseases) are relatively rare. Intracerebral haemorrhage is mostly caused by hypertensive angiopathy and/or cerebral amyloid angiopathy.⁷³⁵ Bleeding can be precipitated by surges in BP values, use of anticoagulants, or diseases impairing coagulation.^{733,735}

In patients with ischaemic stroke or TIA, antithrombotics prevent further vascular events. Cardioembolic ischaemia, which occurs mainly in AF, requires anticoagulation (see [sections 3.4.3 and 6.6](#)).^{736–742} In non-cardioembolic mechanism, platelet inhibitors are recommended.^{619,620,743–753}

In non-cardioembolic ischaemic stroke, aspirin is the most studied antithrombotic drug. Aspirin 75–150 mg/day reduces the risk of recurrent ischaemic stroke and serious vascular events.^{619,743} Clopidogrel shows slight superiority to aspirin.⁶²⁰ In patients with ischaemic stroke or TIA and ipsilateral carotid stenosis, ticagrelor added to aspirin compared to aspirin alone reduced the risk of stroke or death at 1 month, without an increase of severe bleeding.⁷⁵⁴ Adding aspirin to clopidogrel was associated with a non-significant reduction in major vascular events and an increased long-term bleeding risk.^{747–749} However, in patients with minor ischaemic stroke or TIA, a short course of DAPT with aspirin and clopidogrel is beneficial.^{750,751} Similarly, ticagrelor and aspirin vs. aspirin alone reduces stroke or death at 30 days after mild-to-moderate ischaemic stroke or TIA not treated with thrombolysis or thrombectomy. However, DAPT with ticagrelor and aspirin did not improve the incidence of disability and contributed to severe bleeding.⁷⁵⁵ DAPT with dipyridamole plus aspirin also showed superiority over aspirin alone.⁷⁴⁴ In patients with ischaemic stroke, however, dipyridamole plus aspirin vs. clopidogrel alone showed similar rates of recurrent stroke, including haemorrhagic stroke,⁷⁴⁵ but more major haemorrhagic events. In patients with non-cardioembolic ischaemic stroke, oral vitamin K antagonists are not superior to aspirin and carry a higher bleeding risk.^{752,753} In the absence of a definite cause of ischaemia and a presumed occult cardioembolic

source (e.g. embolic stroke of undetermined cause), neither dabigatran nor rivaroxaban are better than aspirin.^{756,757}

Recommendations for BP and lipid management are congruent to the general recommendations outlined in sections 4.6 and 4.7.4. In patients with either ischaemic or haemorrhagic cerebrovascular disease who have a BP of 140/90 mmHg or higher, lowering BP reduces the risk of recurrent stroke.^{758,759} Optimal BP targets in these patients are uncertain, as is the optimal drug regimen.⁷⁶⁰ Most evidence is available for ACE inhibitors, ARBs, and diuretics. Comorbidities may guide the choice of antihypertensive agent. In patients with recent lacunar stroke, the target SBP is <130 mmHg.⁷⁶¹

In patients with stroke (ischaemic or haemorrhagic) or TIA with an LDL-C level of 100–190 mg/dL, atorvastatin 80 mg/day reduced the overall incidence of strokes and CV events.⁷⁶² A recent trial supported an LDL-C target of <1.8 mmol/L (70 mg/dL).⁵⁰⁸

Evidence of cerebrovascular lesions (e.g. white matter hyperintensities, lacunes, non-lacunar ischaemia) in the absence of any stroke history is a relatively common finding at neuroimaging, especially in older patients. Silent cerebrovascular disease is a marker of increased risk of stroke.^{763,764} Arterial hypertension, DM, and cigarette smoking contribute to these lesions and should be attended to. There are no studies addressing the best treatment options for silent cerebral ischaemia.⁷⁶⁵

Recommendations for patients with cerebrovascular disease

Recommendations	Class ^a	Level ^b
In patients with a cerebrovascular event, improvement of lifestyle factors in addition to appropriate pharmacological management is recommended. ^{732,733,741}	I	A
In patients with ischaemic stroke or TIA, prevention with antithrombotics is recommended; choice of antithrombotic depends on the mechanism of event. Use of an antiplatelet is recommended for patients with non-cardioembolic ischaemic stroke or TIA, and use of an anticoagulant is recommended in patients with cardioembolic ischaemic stroke or TIA. ^{732,741}	I	A
In patients with non-cardioembolic ischaemic stroke or TIA, prevention with aspirin only, or dipyridamole plus aspirin, or clopidogrel alone is recommended. ^{620,743–745}	I	A
In patients with minor ischaemic stroke ^c or TIA, DAPT with aspirin and clopidogrel or with aspirin and ticagrelor, for 3 weeks after the acute event should be considered. ^{750,751,755}	IIa	A
In patients with stroke or TIA who have BP of 140/90 mmHg or higher, BP lowering is recommended. ^{757,766}	I	A

BP = blood pressure; DAPT = dual antiplatelet therapy; TIA = transient ischaemic attack.

^aClass of recommendation.

^bLevel of evidence.

^cMinor ischaemic stroke defined as score at National Institutes of Health Stroke Scale ≤3, or ≤5 depending on the trial.

6.4. Lower extremity artery disease

Symptomatic or asymptomatic LEAD (ABI ≤0.90) is associated with a doubling of the 10-year rate of coronary events, CV mortality, and total mortality.¹²⁵ Within 5 years of LEAD diagnosis, 20% develop AMI or stroke, and mortality is 10–15%.⁷⁶⁷

All LEAD patients require lifestyle improvement and pharmacological therapy. Smoking cessation increases walking distance and lowers amputation risk.² In patients with DM, glycaemic control improves limb outcomes.⁷⁶⁸ Statins provide modest improvements in walking distance, and lower the risk of adverse limb events.^{769,770} Combining a statin with ezetimibe⁷⁷¹ or a PCSK9 inhibitor also has beneficial effects.⁷⁷²

Platelet inhibitors are used to prevent limb-related and general CV events. The optimal antiplatelet strategy remains unclear.⁷⁷³ DAPT is currently recommended only after intervention (irrespective of the stent type) for at least 1 month.

In the COMPASS (Cardiovascular Outcomes for People Using Anticoagulation Strategies) trial, low-dose rivaroxaban added to aspirin in CVD patients with an ABI <0.90 reduced not only ASCVD events, but also major adverse limb events, including amputation (HR 0.54),

Recommendations for patients with lower extremity artery disease: best medical therapy

Recommendations	Class ^a	Level ^b
Smoking cessation is recommended in all patients with LEAD. ^{29,781}	I	B
Healthy diet and PA are recommended for all patients with LEAD.	I	C
In patients with intermittent claudication: • Supervised exercise training is recommended. ^{782–784}	I	A
• Non-supervised exercise training is recommended when supervised exercise training is not feasible or available.	I	C
Antiplatelet therapy is recommended in patients with symptomatic LEAD. ^c	I	C
In patients with LEAD and hypertension, it is recommended to control BP at <140/90 mmHg. ^{776,785,786}	I	A
In patients with LEAD and DM, strict glycaemic control is recommended. ⁷⁶⁸	I	A
ACE inhibitors or ARBs should be considered as first-line therapy in patients with PAD and hypertension. ^{d 575,787}	IIa	B
In patients with DM and chronic symptomatic LEAD without high bleeding risk, a combination of low-dose rivaroxaban (2.5 mg b.i.d.) and aspirin (100 mg o.d.) may be considered. ⁷⁷⁴	IIb	B

ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; b.i.d. = bis in die (twice a day); BP = blood pressure; CCB = calcium channel blocker; DM = diabetes mellitus; LEAD = lower extremity artery disease; o.d. = omni die (once a day); PA = physical activity; PAD = peripheral artery disease.

^aClass of recommendation.

^bLevel of evidence.

^cEvidence is not available for all sites. When evidence is available, recommendations specific for the vascular site are presented in corresponding sections.

^dCCBs should be proposed in black individuals.

albeit at the cost of higher major bleeding risk.⁷⁷⁴ These results, combined with similar benefits of rivaroxaban vs. aspirin monotherapy, suggest a benefit of anticoagulants in LEAD. However, further studies are needed. Optimal antithrombotic therapy is addressed in more detail in the 2017 ESC/European Society for Vascular Surgery (ESVS) Guidelines.⁷⁷⁵ Importantly, in patients with isolated asymptomatic LEAD (e.g. low ABI), antiplatelet treatment is not recommended.⁷⁷⁵

Recommendations for BP and lipid management are congruent to the general recommendations outlined in sections 4.6 and 4.7. Hypertension targets are based mainly on INVEST (INternational Verapamil-SR/Trandolapril Study).⁷⁷⁶ An SBP below 110–120 mmHg may increase CV events in patients with LEAD.⁷⁷⁶ ACE inhibitors and ARBs reduce CV events in patients with LEAD,^{575,777} and are preferred (as monotherapy or as part of a combination drug regimen).⁷⁷⁸ Beta-blockers are not contraindicated in mild-to-moderate LEAD as they do not affect walking capacity or adverse limb events,⁷⁷⁹ and significantly reduce coronary events.⁷⁸⁰ Nevertheless, beta-blockers should be carefully considered in critical limb-threatening ischaemia.

6.5. Chronic kidney disease

Severe CKD is associated with a very high risk of CVD and is considered a CAD risk equivalent (see section 3.2). As GFR declines, non-traditional risk factors emerge and non-atherosclerotic CVD event risk increases.²⁰⁴ Trials often exclude patients with eGFR <30 mL/min/1.73 m². In patients on dialysis, coronary syndromes may present atypically, and angina equivalents—such as shortness of breath or fatigue—are frequent.⁷⁸⁸ Standard CVD risk management is effective in patients on dialysis, but unique haemodialysis-specific syndromes (i.e. intradialytic hypotension and myocardial stunning) associated with mortality complicate treatment and modify outcomes.

Risk classification of patients with various degrees of CKD is summarized in Table 4. Treatment with a statin or statin/ezetimibe combination is recommended in CKD patients with sufficiently high CVD risk, but not in those treated with kidney replacement therapy. This recommendation is built on evidence from SHARP (Study of Heart and Renal Protection), which demonstrated a reduction of major atherosclerotic events.⁵²⁵ Statins should be dosed according to a moderate-intensity regimen based on limited experience and risks associated with high-intensity regimens.⁵⁴³ Subgroup analysis of a recent study with a PCSK9 inhibitor has shown that the benefits may extend to those with earlier CKD stages (60–90 as well as 30–60 mL/min/1.73 m²).⁷⁸⁹

Treatment with an ACE inhibitor or an ARB is recommended in patients with DM, hypertension, and albuminuria. These medications should be titrated to the maximum tolerated dose (Kidney Disease Improving Global Outcomes grading 1B).

Individualized HbA1c targets, ranging from 6.5% to <8.0% in patients with DM and non-dialysis-dependent CKD, are recommended in parallel. The role of SGLT2 inhibitors and GLP-1RAs in CKD associated with DM is addressed in section 4.8. Dapagliflozin has shown promising reno- and cardioprotective effects,⁵⁹⁹ and more studies investigating SGLT2 inhibitors in CKD patients without DM are ongoing.⁷⁹⁰

Overall, the management of CAD in CKD patients must be informed by the modification of its clinical presentation in CKD, as well as comorbidity and risks of treatment side-effects. Treatment of established risk factors is often suboptimal in patients with CKD.

Recommendations in patients with chronic kidney disease: best medical therapy^a

Recommendations	Class ^b	Level ^c
Treatment with an ACE inhibitor or an ARB is recommended in patients with DM, hypertension, and albuminuria. These medications should be titrated to the highest approved dose that is tolerated.	I	B
An SGLT2 inhibitor with proven outcome benefits should be considered for the prevention of renal deterioration and mortality in patients with CKD. ⁵⁹⁹	IIa	B
Combination treatment with ACE inhibitors and ARBs is not recommended.	III	C

ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; CKD = chronic kidney disease; DM = diabetes mellitus; SGLT2 = sodium-glucose cotransporter 2.

^aRecommendations on CKD management in patients with DM are found in section 4.8.

^bClass of recommendation.

^cLevel of evidence.

6.6. Atrial fibrillation

The simple ‘Atrial fibrillation Better Care’ (ABC) holistic pathway (‘A’ = Anticoagulation/Avoid stroke; ‘B’ = Better symptom management; ‘C’ = Cardiovascular and Comorbidity optimization) streamlines integrated care of patients with AF.²¹⁵ The ABC pathway lowers risk of all-cause death and the composite of stroke, major bleeding, CV death, or first hospitalization,⁷⁹¹ and lowers rates of CV events^{792,793} and health-related costs.⁷⁹⁴

The ‘C’ component of the ABC pathway refers to identification and management of concomitant diseases, cardiometabolic risk factors, and unhealthy lifestyle factors. Therapy of underlying conditions improves rhythm control in persistent AF and HF.²¹⁶ In obese patients, weight reduction prevents AF recurrences and symptoms.^{795–802} Given that hypertension precipitates AF, treatment of hypertension is mandatory. Alcohol excess is a risk factor for incident AF,^{803,804} and abstinence reduced AF recurrences in regular drinkers.⁷⁹⁸ Many studies have demonstrated beneficial effects of moderate exercise/PA.^{805–807} The incidence of AF appears, however, to be increased in elite athletes, mainly related to endurance sports.^{808–811} Patients should be encouraged to practise moderate-intensity exercise and remain physically active to prevent AF incidence or recurrence, but avoid excessive endurance exercise. CR is a universally recommended programme for patients with ACS and/or revascularization, and for patients with HF.^{639,640,655} The benefits of EBCR are more uncertain in patients with AF, but CR remains recommended in patients with the aforementioned indications.⁸¹² Continuous PAP may improve rhythm control and attenuate AF recurrences in OSA patients.^{813–816} Intensive glycaemic control does not affect the rate of new-onset AF.⁸¹⁷ Optimal glycaemic control during the 12 months before AF ablation does, however, reduce AF recurrence after ablation.⁸¹⁸ All patients with HF and AF should receive guideline-adherent HF therapy.⁸¹⁹

Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with atrial fibrillation²¹⁵

Recommendations	Class ^a	Level ^b
Identification and management of risk factors and concomitant diseases are recommended to be an integral part of treatment. ⁷⁹⁵	I	B
Modification of unhealthy lifestyle and targeted therapy of intercurrent conditions is recommended to reduce AF burden and symptom severity. ^{216,795–802}	I	B
Attention to good BP control is recommended in AF patients with hypertension to reduce AF recurrences and risk of stroke and bleeding. ^{800,801}	I	B
In obese patients with AF, weight loss together with management of other risk factors should be considered to reduce AF incidence, AF progression, AF recurrences, and symptoms. ^{795–797}	IIa	B
Advice and management to avoid alcohol excess should be considered for AF prevention and in AF patients considered for oral anticoagulant therapy. ^{798,803,804}	IIa	B
PA should be considered to help prevent AF incidence or recurrence, with the exception of excessive endurance exercise, which may promote AF. ^{805–812}	IIa	C
Optimal management of OSA may be considered to reduce AF incidence, AF progression, AF recurrences, and symptoms. ^{813–816}	IIb	C

AF = atrial fibrillation; BP = blood pressure; OSA = obstructive sleep apnoea; PA = physical activity.

^aClass of recommendation.

^bLevel of evidence.

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6.7. Multimorbidity

The older adult population is growing fast and survival after acute CVD has improved,⁸²⁰ leading to an increasing number of older patients with CVD and multimorbidity.^{821,822} This development is associated with high healthcare costs,^{823,824} worse outcome measures, higher readmission rates,⁸²⁵ and mortality.⁸²⁶

Up to 70% of patients aged ≥70 years have at least one CVD and two-thirds also develop non-CVD comorbidities. Multimorbidity is important in patients with CVD.⁸²³

The prevailing CV conditions in patients aged >60 years are hypertension, hyperlipidaemia, ischaemic heart disease, arrhythmia, DM, and CAD.⁸²³ Other frequent comorbidities include anaemia and arthritis. Low vision, back and neck problems, osteoarthritis, COPD, depression, and cancer are the most common non-CV comorbidities in CVD patients. Most studies have found no sex differences in the number of comorbidities. However, men have more CVD comorbidities and women have more non-CVD comorbidities (in particular more depression).^{822,826,827}

So far, guidance for the treatment of CVD has focused mainly on single CVDs. In multimorbid patients, application of a single guideline for one CVD is often not feasible as therapeutic competition is highly prevalent (22.6%)⁸²⁰ and treatment for one condition can worsen a coexisting condition. The challenges for managing CVD and multimorbidity are disease-disease, disease-drug, and drug-drug interactions.⁸²⁰ Further, pharmacokinetics can be different in patients with comorbidities, and life expectancy has to be taken into account when starting a new medication. A value-based approach should always be discussed and proposed when possible.⁸²⁰ The incremental benefit of medication when added to an already complex regimen is often uncertain.⁸²⁸ Moreover, care for multimorbid CVD patients is often fragmented and given by multiple providers, complicating decision-making and adherence to recommended treatment.⁸²⁰

Multimorbid CVD patients have been underrepresented in most clinical trials that underlie the guidelines. Trials including patients with multimorbidity and endpoints that matter to patients, pragmatic trials, and the use of registries and big data could help elucidate how to optimize treatment and care for patients with CVD and multimorbidity.⁸²⁰

There is a plea for a paradigm shift from disease-focused to patient-centred care for multimorbid CVD patients, with a central place for patients' overarching goals of care.⁸²⁸ 'What matters to you?' should be the central question, instead of 'what is the matter?'.

Patient-centred care should include assessment of patients' preferences, interpretation of the evidence and its application to the specific patient, consideration of overall prognosis, including life expectancy, functional status, and quality of life, and clinical feasibility. Adherence to treatment, the occurrence of adverse drug events, the economic burden, and the stress experienced by caregivers should be taken into account when optimizing therapies and care plans where adherence to essential medication is emphasized and non-essential drugs are stopped.⁸²⁸ Furthermore, advanced care planning should be initiated early. Multidisciplinary teams and close collaboration between primary care workers and specialists is needed. Finally, automated decision support systems for multimorbidity and CVD could help in aligning the relevant evidence and making adequate decisions.⁸²⁹

7. Key messages

Risk factors and risk classification

- The major risk factors for ASCVD are cholesterol, BP, cigarette smoking, DM, and adiposity.
- Risk factors are treated in a stepwise approach to reach the ultimate treatment goals in apparently healthy people, patients with established ASCVD, and patients with DM.
- 10-year CVD risk is estimated in apparently healthy people aged 40–69 years with SCORE2, and in people aged ≥70 years with SCORE2-OP.
- Age-specific 10-year CVD risk thresholds—together with consideration of risk modifiers, frailty, comorbidities, lifetime CVD risk, treatment benefit, polypharmacy, and patient preferences—guide treatment decisions for lipid and BP treatment.
- There are various options of communicating the (residual) CVD risk, and this should be tailored to the individual patient.

Risk modifiers

- Psychosocial stress is associated with risk of ASCVD.
- Current risk scores may under- or overestimate CVD risk in differing ethnic minority groups.
- CAC scoring is the best-established imaging modality to improve CVD risk stratification.
- Frailty is a functional risk factor of both CV and non-CV morbidity and mortality.
- Frailty assessment is not a method to determine eligibility for any particular treatment, but rather serves to build an individualized care plan with predefined priorities.
- Family history should be enquired about routinely, and a positive family history of premature ASCVD should be followed by comprehensive CVD risk assessment.
- Current data does not support the use of genomic risk scores in CVD risk assessment in primary prevention.
- ASCVD development and prognosis are linked to social gradients.
- Air pollution is strongly associated with ASCVD.
- Additional circulating and urine biomarkers should not be routinely measured.
- Assess CVD risk in persons with obesity.

Clinical conditions

- CKD is an independent risk factor for ASCVD, and ASCVD is the leading cause of death in CKD.
- A short-term reduction in albuminuria by approximately 30% upon starting RAAS inhibition is associated with improved CV and kidney outcomes.
- Similarly, SGLT2 inhibitors are associated with long-term benefits in CV and renal risks.
- AF is associated with an increased risk of death and an increased risk of CVD.
- Ischaemic HF constitutes the most advanced clinical manifestation of atherosclerosis within the myocardium.
- The diagnosis of overt HF, as well as asymptomatic presentation with LV dysfunction, increases the risk of CVD events (myocardial infarction, ischaemic stroke, CV death).
- There is an overlap between cancer and CV risk factors; CV risk in patients with cancer depends on both the CV toxicity of treatments and patient-related factors.
- Signs or symptoms of cardiac dysfunction should be monitored before, periodically during, and after treatment.
- Exercise should be strongly advised, in particular aerobic exercise, to prevent cardiotoxicity.
- COPD is a major risk factor for CVD, especially ASCVD, stroke, and HF.
- COPD patients are prone to arrhythmias (AF and ventricular tachycardia) and sudden cardiac death.
- All COPD patients should be investigated for CVD.
- Common COPD medications are usually safe in terms of CV adverse events.
- Chronic inflammatory conditions increase CVD risk.
- Infection with HIV is associated with an increased risk of LEAD and CAD.
- There is an association between influenza and periodontitis infections and ASCVD.

- Migraine, particularly migraine with aura, is an independent risk factor for stroke and ischaemic cardiac disease.
- The risk of ischaemic stroke in subjects with migraine with aura is magnified by the use of combined hormonal contraceptives and cigarette smoking.
- Non-restorative sleep and a sleep duration that varies significantly up or down from the optimum of 7 h are associated with increased CV risk.
- Mental disorders are common in the general population (12-month prevalence of 27%) and are associated with excess mortality.
- The onset of CVD increases the risk of mental disorders by 2.2-fold, leading to a worse prognosis.
- Some mental disorders—even symptoms of anxiety and depression—are associated with the development of CVD and with a worse prognosis in those with existing CVD (CHD, arterial hypertension, AF, HF).
- Excess mortality is mainly caused by behaviour-dependent risk factors (e.g. smoking addiction) and an impaired capacity for self-care (e.g. treatment adherence).
- NAFLD is associated with other cardiometabolic risk factors.
- Patients with NAFLD should be evaluated for other cardiometabolic risk factors.
- Sex-specific conditions:
 - Preeclampsia and pregnancy-related hypertension are associated with a higher risk of CVD.
 - Polycystic ovary syndrome confers a significant risk for future development of DM.
 - ED is associated with future CV events and mortality in men.
 - CVD risk should be assessed in men with ED.
 - Asking about ED should be a standard procedure in routine CV risk assessment in men.

Risk factors and interventions at the individual level

- Regular PA is a mainstay of ASCVD prevention.
- Aerobic PA in combination with resistance exercise and the reduction of sedentary time are recommended for all adults.
- A healthy diet lowers the risk of CVD and other chronic diseases.
- A shift from a more animal- to plant-based food pattern may reduce CVD.
- Achieving and maintaining a healthy weight through lifestyle changes has favourable effects on risk factors (BP, lipids, glucose metabolism) and lowers CVD risk.
- When changes in diet and PA—as well as other conventional, non-invasive interventions—are unsuccessful, bariatric surgery should be considered for high-risk individuals.
- Anti-obesity medications with protective ASCVD effects may also be considered.
- Patients with mental disorders have sharply increased lifestyle risks that need recognition and treatment.
- Mental healthcare improves stress symptoms and quality of life, reduces the risk of suicide, and may improve CV outcomes.
- The treatment of ASCVD patients with mental disorders requires interdisciplinary cooperation and communication.
- Stopping smoking rapidly reduces CVD risk and is the most cost-effective strategy for ASCVD prevention.

- There is strong evidence for medication-assisted interventions: NRT, bupropion, varenicline, and drugs in combination. The most effective are assistance using drug therapy and follow-up support.
- Lower is better: the effect of LDL-C on the risk of CVD appears to be determined by both the baseline level and the total duration of exposure to LDL-C.
- Lowering LDL-C with statins, ezetimibe, and—if needed and cost-effective—PCSK9 inhibitors, decreases the risk of ASCVD proportionally to the absolute achieved reduction in LDL-C.
- When LDL-C goals according to level of risk cannot be attained, aim to reduce LDL-C by $\geq 50\%$ and then strive to reduce other risk factors as part of a shared decision-making process with the patient.
- When hypertension is suspected, the diagnosis should be confirmed by repeated office BP measurement at different visits, or ABPM or HBPM.
- Lifestyle interventions are indicated for all patients with hypertension and can delay the need for drug treatment or complement the BP-lowering effect of drug treatment.
- BP-lowering drug treatment is recommended in many adults when office BP is $\geq 140/90$ mmHg and in all adults when BP is $\geq 160/100$ mmHg.
- BP treatment goals are lower than in the previous ESC CVD prevention guidelines for all patient groups, including independent older patients.
- Wider use of single-pill combination therapy is recommended to reduce poor adherence to BP treatment.
- A simple drug treatment algorithm should be used to treat most patients, based on combinations of a renin–angiotensin system (RAS) blocker with a CCB or thiazide/thiazide-like diuretic, or all three. Beta-blockers may also be used where there is a guideline-directed indication.
- Many patients with hypertension will be at sufficient risk to benefit from statin therapy for primary prevention. Antiplatelet therapy is indicated for secondary prevention.
- A multifactorial approach, including lifestyle changes, is critical in persons with type 2 DM.
- Management of hyperglycaemia reduces the risk of microvascular complications and, to a lesser extent, the risk of CVD. Glycaemic targets should be relaxed in older adults and frail individuals.
- New antihyperglycaemic drugs are particularly important for persons with type 2 DM with existing ASCVD and (heightened risk of) HF or renal disease, broadly irrespective of glycaemia levels.

Type 1 diabetes mellitus

- Intensive management of hyperglycaemia in DM reduces the risk of micro- and macrovascular complications and premature mortality; a target of 6.5–7.5% (48–58 mmol/mol) HbA1c is recommended.
- Metformin is not recommended in type 1 DM to lower CVD risk.
- Dapagliflozin has been recommended for use in type 1 DM, although there is an increased risk of diabetic ketoacidosis with such therapies.

- Targeting other risk factors, in particular smoking, BP, and cholesterol levels, remains an important means to lower CVD risk in type 1 DM.
- All patients with established ASCVD require some form of antithrombotic therapy.
- Anti-inflammatory therapy is a promising strategy in CVD prevention.
- Patients after ACS and/or coronary artery bypass graft/PCI, or with chronic HFrEF, should participate as early as possible in structured, multidisciplinary EBCR and prevention programmes.
- EBCR and prevention programmes must comply with certain quality standards and be individualized to each patient's profile.
- Participation and long-term adherence to these programmes has to be encouraged and enhanced. Telerehabilitation and mHealth may help towards achieving this target.

Population-level approaches to cardiovascular disease prevention

Physical activity

- A significant percentage of the worldwide population, in particular the European population, shows high levels of sedentary behaviour and physical inactivity.
- The percentage of those exercising at a regular level is greater in men than in women.
- Global progress to increase PA has been slow, largely due to lack of awareness and investment.
- The optimal dose of different types of PA for CVD and general prevention is still controversial and subjected to frequent updates. Increasing moderate-to-vigorous PA and reducing sitting time, however, is beneficial and any level of PA is considered better than none.
- PA for health promotion should be implemented by physicians in the same way as drug prescription and should also be promoted by other healthcare professionals.
- Population-based interventions are effective in promoting PA for groups based on age, sex, and race, for high-, middle-, and low-income populations, and for different environments (e.g. kindergarten, school, gyms, companies, and worksites in general).
- Daily PA at school should be practised for at least 3 h/week, and preferably for 60 minutes per day.
- Population-based approaches are complementary to individual-centred interventions.
- Diet
- Structural measures such as changes in agricultural supply chain and food industry, product reformulation, limitations on (digital) marketing to children, taxes on unhealthy foods/nutrients, and consumer-friendly nutrition labelling will improve healthy food choices.
- Healthy environments in the community, on public transport, at schools, and in workplaces will stimulate a healthier lifestyle.
- The WHO Global Action Plan for the Prevention and Control of Non-Communicable Diseases 2013–2020 extended to 2025 recommends to develop goals in global, regional, and national agendas. Within the 10 voluntary targets to reach in 2025 is a 30% relative reduction in mean population intake of sodium/salt.⁸³⁰

Smoking and tobacco use

- Adolescence is the most vulnerable period for the uptake of smoking, with lifelong consequences.
- Previous prevention campaigns reduced tobacco use in girls much less than in boys.
- Teenagers should be informed that smoking is not helpful in weight control.
- High taxes on all tobacco products is the most effective policy measure to reduce smoking uptake by the young.
- There should be restrictions on smokeless tobacco due to strong evidence of harm.
- Also, restrictions on e-cigarettes due to evidence of harm.
- Plain packaging is effective in reducing the attractiveness of tobacco products.
- There should be restrictions on advertising, promotion, and sponsorship by the tobacco industry.
- A goal would be to make a common European decision to achieve a smoking-free Europe by 2030.

Alcohol

- Alcohol intake is associated with increased CV mortality, and alcohol use is the leading risk factor for premature death and disability among people aged 15–49 years.
- The interventions for addressing the harmful use of alcohol are cost-effective, with a good return (i.e. increasing alcoholic beverage minimum unit pricing and excise taxes, restricting access to alcoholic beverages, and implementing comprehensive restrictions and bans on advertising and the promotion of alcoholic beverages).
- Healthcare providers may inquire about alcohol intake in every medical evaluation and should inform patients that alcohol is energy-dense: it provides 7 kcal/g and no nutrients.

Environment, air pollution, and climate change

- Air pollution contributes to mortality and morbidity, and specifically increases the risk of respiratory and CV diseases.
- Environmental exposure has taken on new urgency, as air pollution, in addition to its health effects, has also been ascribed as a major contributor to climate changes, notably through the burning of fossil fuels leading to increasing emissions of carbon dioxide.

Risk management of disease-specific cardiovascular disease

Coronary artery disease

- Multidimensional prevention is crucial for short- and long-term outcomes in CAD.

Heart failure

- Patients with HF benefit from multidisciplinary care management programmes.
- Several neurohormonal antagonists, as well as novel molecules, improve clinical outcomes in symptomatic patients with HFrEF.

Cerebrovascular diseases

- Ischaemic events are mainly caused by atherothrombosis, cardioembolism, or small vessel disease, whereas intracerebral

haemorrhage is mostly caused by hypertensive angiopathy or cerebral amyloid angiopathy.

- Platelet inhibitors are recommended for non-cardioembolic events and anticoagulants for cardioembolic events.
- In patients with a previous stroke or TIA and high BP, BP lowering reduces the recurrence risk.
- In patients with stroke or TIA, statins prevent CVD and cerebrovascular events.
- Lower extremity artery disease
- LEAD is associated with an increased CVD risk.
- Antiplatelet therapy (alone or in combination with low-dose oral anticoagulation) reduces the risk of adverse limb events and overall CVD risk in patients with LEAD.
- Smoking cessation and control of other CVD risk factors improve prognosis.

Chronic kidney disease

- Hypertension, dyslipidaemia, and DM are prevalent among individuals with CKD and require a high-risk treatment strategy approach.
- Risk management includes lifestyle, smoking cessation, nutrition, sufficient RAAS blockade, target BP control, lipid management, and—in established CVD—aspirin.
- A high value is placed on self-management education programmes and team-based integrated care in patients with DM, CKD, and CVD.

Atrial fibrillation

- Holistic management of patients with AF improves prognosis and reduces health-related costs.
- Comprehensive risk-factor modification and targeting underlying conditions reduce AF burden and recurrence.

Multimorbidity

- The number of patients with multiple CV and non-CV comorbidities is rapidly increasing.
- Therapeutic competition should be considered in multimorbid patients, as the treatment of one condition might worsen a coexisting condition.
- A paradigm shift from disease-focused to patient-centred care for multimorbid CVD patients is recommended.

8. Gaps in evidence

CVD risk classification

- Country-specific risk algorithms for patients with established CVD and people with DM.
- Formal comparison of effectiveness and cost-effectiveness of CVD risk-guided treatment vs. treatment guided by risk factor level.
- Comparison of the precision of competing risk-adjusted CVD risk models vs. standard CVD risk models.
- Incorporating potential risk markers into conventional risk models, such as socioeconomic status and ethnicity.
- Comparison of treatment benefit-guided strategy vs. risk-guided strategy in reducing risk factor levels and CVD risk.

- Management of CVD risk in older people (>85 years) with marked fragility, for whom no data currently exist.
- Comparison of different methods for the estimation of lifetime CVD risk and lifetime benefit of risk factor treatment.

Risk modifiers

Psychosocial factors

- More evidence that psychosocial factors improve risk prediction beyond the classical risk-factor models.

Ethnicity

- Whether recalibration of factors for ethnicity are homogeneous in various European countries.
- Risks associated with other ethnic backgrounds.

Frailty

- Consensus on a clinically orientated screening tool for frailty to be applied across the spectrum of ASCVD.
- Quantitative contribution of frailty to the global CVD risk-prediction scheme.
- At which degree of frailty treatment of specific risk factors should be less aggressive.

Family history

- Disentangle the role and (genetic, socioeconomic, etc.) mechanisms of family history on CVD risk.

Genetics

- The potential of polygenic risk scores to complement existing risk scores.

Socioeconomic determinants

- More evidence from different risk regions that the inclusion of socioeconomic factors improves risk prediction beyond classical risk factor models in both men and women.

Environmental exposure

- Whether air pollution reclassifies risk in individual patients.

Biomarkers

- Added value of biomarkers in risk classification.

Clinical conditions

Chronic kidney disease

- Identification of a good biomarker, besides albuminuria, and perhaps the use of CAC score to subclassify CV risk in CKD.
- Early and precise identification of progressive CKD with novel biomarkers that are more sensitive than eGFR and albuminuria.

Atrial fibrillation

- Evaluate the effect of interventions aimed at reducing outcomes beyond stroke.
- Is AF a causal factor for increased CVD morbidity and mortality?
- Stroke risk prediction for low-risk AF patients.

- Emerging evidence suggests that stroke can occur in patients with AF even after sinus rhythm is restored.

Heart failure

- It remains unknown whether patients with HFrEF of ischaemic origin should have different target LDL-C levels than those recommended for secondary prevention in individuals without HF.

Cancer

- RCTs using preventive therapy to demonstrate a clear effect on prevention of CV events.

Chronic obstructive pulmonary disease

- Although common pathophysiological pathways between CVD and COPD are probable, they remain to be clarified.

Inflammatory conditions

- The optimal way of integrating information on chronic inflammatory conditions into CVD risk assessment.
- The effect of modern anti-inflammatory drugs on CV risk [e.g. anti-tumour necrosis factor (TNF), interleukin (IL)-1, IL-17, IL-23 biologics].

Infections

- Large-scale studies to assess the efficacy of influenza vaccination or periodontitis treatment in preventing CVD.
- The association of infection with HIV and total CVD risk.

Migraine

- There are no data that allow reliable identification of subgroups of migraineurs at particular high risk (e.g. active migraine, high-frequency auras, young subjects, women).
- The role of comorbid factors (e.g. patent foramen ovale, thrombophilic factors) is unclear, and at the moment there is no indication to screen or to manage for these factors.

Sleep disorders

- There is lack of evidence that the inclusion of sleep improves risk prediction.
- Trials are needed that target the complex pathways linking sleep disturbances with CVD.

Mental disorders

- The precise mechanism by which mental disorders increase CVD remains uncertain.
- How the consideration of mental disorders improves CV risk models.

Non-alcoholic fatty liver disease

- Whether NAFLD increases CV risk beyond traditional risk factors.

Sex-specific conditions

- The degree to which increased CVD risk associated with several of the female-specific conditions occurs independently of

conventional CVD risk factors, although data in women are still underpowered compared to men.

- Information on whether female-specific conditions improve risk classification.
- There are insufficient data to draw conclusions on a possible increased risk of hypertension or DM with premature menopause.
- Studies on the specificities of CVD disease in the transgender population are scarce.

Erectile dysfunction

- The benefit of routine screening for ED and the most effective tool to assess it are still unclear.
- The benefit of assessment of subclinical vascular disease in men with ED and low-to-intermediate CVD risk is unclear.

Risk factors and interventions at the individual level

Physical activity and exercise

- Knowledge of the relative importance of the various characteristics of aerobic PA and resistance exercise, or their combination, on all-cause mortality, CV incidence, and mortality.
- Understanding how sex, age, weight, race/ethnicity, occupation, and socioeconomic status may modify associations between PA and health outcomes.
- Implementation of strategies to achieve long-term adherence to PA.
- Evaluation of the effects of eHealth tools in promoting PA.

Nutrition

- Effective strategies to encourage people to change their diet and to enjoy and maintain a healthy diet.

Body weight

- Knowledge and implementation of effective lifestyle and medication-assisted strategies to achieve weight loss and maintain a long-term healthy weight.

Mental healthcare and psychosocial interventions

- The effectiveness of mental healthcare for the prevention of major CVD events.
- How to implement effective CVD prevention measures in this high-risk population of patients with mental disorders.

Smoking intervention

- A better understanding of how to incorporate effective smoking cessation into clinical practice.

Lipids

- Direct empirical evidence for the stepwise approach to treatment intensification from RCTs. The feasibility and effects of reaching LDL-C levels <1.4 mmol/L (55 mg/dL) needs further investigation, especially in primary care.
- Particularly among people at low-to-moderate CVD risk, older people, and for newer interventions, more evidence of the effects of lipid-modifying treatments on overall mortality is needed in the form of long-term post-trial follow-up in RCTs.

- The cost-effectiveness of using lifetime CVD risk and more precise CVD risk scores to target interventions needs further investigation.
- The value of triglycerides or HDL-C values as a target for therapy.
- Whether lipoprotein(a) lowering against background statin, ezetimibe and PCSK9i therapy can reduce the risk of ASCVD.
- Whether functional foods and food supplements with a lipid-lowering effect can safely reduce the risk of CVD.

Blood pressure

- What is the incremental benefit, over CVD risk calculators, of measures of HMOD in reclassifying the CV risk of patients with hypertension?
- Direct empirical evidence for the stepwise approach to treatment intensification from RCTs.
- What are the benefits of BP treatment for patients with BP in the high-normal range?
- More data on the benefits of BP treatment in very old people and the influence of frailty.
- Effect of single-pill vs. multidrug treatment strategies on adherence to treatment, BP control, and clinical outcomes.
- Effectiveness of antihypertensive treatment in preventing cognitive dysfunction or dementia.
- Efficacy and cost-effectiveness of invasive procedures and devices for the treatment of hypertension.
- Sex-specific BP treatment thresholds for men and women.

Diabetes mellitus

- More work is needed to develop risk scores for both MACE and HF in type 2 DM.
- Whether combined SGLT2 inhibitor and GLP-1RA treatments lower MACE or other outcomes beyond either drug alone requires testing.
- Longer-term safety of newer classes of drug is required.

Antithrombotic therapy

- The role of antithrombotic therapy in primary prevention in (very) high-risk individuals remains to be established.

Cardiac rehabilitation and prevention programmes

- The effect and the optimal delivery of EBCR in women, older/frail patients, patients with cardiac implantable electronic devices, after heart transplantation or valve replacement, and in patients with AF, stroke, HFpEF, LEAD, or multiple comorbidities.
- Alternative and cost-effective models of CR need to ensure participation globally, including low- and middle-income countries.
- Large RCTs investigating the long-term effects of home-based telerehabilitation and mHealth are needed.

Environment, air pollution, and climate change

- Individual-level exposure studies are needed to better specify the effect of mitigating measures.

Risk management of disease-specific cardiovascular disease

Coronary artery disease

- The efficacy and safety of aspirin or other antithrombotic therapy in patients without clinical manifestations of CAD—but with atherosclerotic disease identified on imaging, such as CCTA—requires further assessment.
- The optimal long-term antithrombotic therapy in patients at high risk of ischaemic events is uncertain.
- Clinical studies comparing the efficacy and safety of P2Y₁₂ inhibitors vs. low-dose rivaroxaban or other factor Xa inhibitors, in combination with aspirin, are warranted to determine which subgroups will derive greater clinical benefit with each strategy.

Heart failure

- For patients with HFpEF, no specific pharmacotherapy or device implantation has been shown to modify the risk of any CV outcome.
- Lower dosage of HF treatments in women with HFrEF needs to be addressed, since women were underrepresented in many HF trials.

Cerebrovascular disease

- The optimal selection of patient for a short course of DAPT.
- The optimal antihypertensive regimen and target BP.
- The optimal target level of LDL-C.
- Optimal treatment for patients with silent cerebrovascular disease.

Lower extremity artery disease

- The optimal type and potency of antithrombotic therapy in patients with different manifestations of symptomatic or asymptomatic LEAD are partly unclear.

Chronic kidney disease

- Few CVD trials have a focus on patients with CKD, particularly those with advanced CKD.
- Additional prospective studies focusing on diagnosis, prevention, and treatment of CAD and CVD are needed in CKD.

Atrial fibrillation

- The effects of various CV risk factors and comorbidities in AF.
- Optimal treatment of OSA and its effect on AF progression and symptoms.

Multimorbidity

- The effect of different clusters or combinations of CV and non-CV comorbidities on CV outcomes.
- Optimal, pragmatic treatment strategies in patients with CV and non-CV comorbidities, with particular focus on treatment adherence and therapeutic competition.

9. 'What to do' and 'what not to do' messages from the guidelines

Recommendations	Class ^a	Level ^b
Recommendations for cardiovascular disease risk assessment		
Systematic global CVD risk assessment is recommended in individuals with any major vascular risk factor (i.e. family history of premature CVD, FH, CVD risk factors such as smoking, arterial hypertension, DM, raised lipid level, obesity, or comorbidities increasing CVD risk).	I	C
Systematic CVD risk assessment in men <40 years of age and women <50 years of age with no known CV risk factors is not recommended.	III	C
Recommendations for cardiovascular disease risk estimation		
In apparently healthy people <70 years of age without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorders, estimation of 10-year fatal and non-fatal CVD risk with SCORE2 is recommended.	I	B
In apparently healthy people ≥70 years of age without established ASCVD, DM, CKD, genetic/rarer lipid or BP disorder, estimation of 10-year fatal and nonfatal CVD risk with SCORE2-OP is recommended.	I	B
Patients with established CVD and/or DM and/or moderate-to-severe renal disease and/or genetic/rarer lipid or BP disorders are to be considered at high or very high CVD risk.	I	A
A stepwise treatment-intensification approach aiming at intensive risk factor treatment is recommended for apparently healthy people at high or very high CVD risk, as well as patients with established ASCVD and/or DM, with consideration of CVD risk, treatment benefit of risk factors, risk modifiers, comorbidities, and patient preferences.	I	B
Treatment of ASCVD risk factors is recommended in apparently healthy people without DM, CKD, genetic/rarer lipid or BP disorders who are at very high risk (SCORE2 ≥7.5% for age under 50; SCORE2 ≥10% for age 50–69; SCORE2-OP ≥15% for age ≥70 years).	I	C
Recommendation for cardiovascular disease risk communication		
An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended.	I	C
Recommendations for risk modifiers		
The routine collection of other potential modifiers, such as genetic risk scores, circulating or urinary biomarkers, or vascular tests or imaging methods (other than CAC scoring or carotid ultrasound for plaque determination), is not recommended.	III	B

Continued

Recommendations for cardiovascular disease risk assessment in specific clinical conditions		
In all CKD patients, with or without DM, appropriate screening for ASCVD and kidney disease progression, including monitoring changes in albuminuria is recommended.	I	C
It is recommended to monitor cardiac dysfunction using imaging techniques and circulating biomarkers before, periodically during, and after cancer treatment.	I	B
Screening for CV risk factors and optimization of the CV risk profile is recommended in patients on treatment for cancer.	I	C
It is recommended that all COPD patients be investigated for ASCVD and ASCVD risk factors.	I	C
In patients with CVD, obesity, and hypertension, regular screening for non-restorative sleep is indicated (e.g. by the question: 'how often have you been bothered by trouble falling or staying asleep, or sleeping too much?').	I	C
If there are significant sleep problems, which are not responding within 4 weeks to sleep hygiene, referral to a specialist is recommended.	I	C
It is recommended that mental disorders with either significant functional impairment or decreased use of healthcare systems be considered as influencing total CVD risk.	I	C
It is recommended for adults of all ages to strive for at least 150–300 min a week of moderate-intensity or 75–150 min a week of vigorous-intensity aerobic PA, or an equivalent combination thereof, to reduce all-cause mortality, CV mortality, and morbidity.	I	A
It is recommended that adults who cannot perform 150 min of moderate-intensity PA a week should stay as active as their abilities and health condition allow.	I	B
It is recommended to reduce sedentary time to engage in at least light activity throughout the day to reduce all-cause and CV mortality and morbidity.	I	B
Performing resistance exercise, in addition to aerobic activity, is recommended on 2 or more days per week to reduce all-cause mortality.	I	B
Recommendations for nutrition and alcohol		
A healthy diet is recommended as a cornerstone of CVD prevention in all individuals.	I	A
It is recommended to adopt a Mediterranean or similar diet to lower risk of CVD.	I	A
It is recommended to replace saturated with unsaturated fats to lower the risk of CVD.	I	A
It is recommended to reduce salt intake to lower BP and risk of CVD.	I	A
It is recommended to choose a more plant-based food pattern, rich in fibre, that includes whole grains, fruits, vegetables, pulses, and nuts.	I	B
It is recommended to restrict alcohol consumption to a maximum of 100 g per week.	I	B
It is recommended to eat fish, preferably fatty, at least once a week and restrict (processed) meat.	I	B
It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake.	I	B
Recommendations for body weight		
It is recommended that overweight and obese people aim for a reduction in weight to reduce BP, dyslipidaemia, and risk of type 2 DM, and thus improve their CVD risk profile.	I	A
While a range of diets are effective for weight loss, it is recommended that a healthy diet in regard to CVD risk is maintained over time.	I	A
Recommendations for mental healthcare and psychosocial interventions at the individual level		
Patients with mental disorders need intensified attention and support to improve adherence to lifestyle changes and drug treatment.	I	C
In ASCVD patients with mental disorders, evidence-based mental healthcare and interdisciplinary cooperation are recommended.	I	B
In patients with HF and major depression, SSRIs, SNRIs, and tricyclic antidepressants are not recommended.	III	B
Recommendations for smoking intervention strategies		
All smoking of tobacco should be stopped, as tobacco use is strongly and independently causal of ASCVD.	I	A
Smoking cessation is recommended regardless of weight gain, as weight gain does not lessen the ASCVD benefits of cessation.	I	B
Recommendations on low-density lipoprotein cholesterol goals		
A stepwise treatment-intensification approach is recommended for apparently healthy people at high or very high CVD risk, as well as patients with established ASCVD and/or DM with consideration of CVD risk, treatment benefit, risk modifiers, comorbidities, and patient preferences.	I	C

Continued

Recommendations for pharmacological low-density lipoprotein cholesterol lowering for those <70 years of age		
It is recommended that a high-intensity statin is prescribed up to the highest tolerated dose to reach the LDL-C goals set for the specific risk group.	I	A
In patients with established ASCVD, lipid-lowering treatment with an ultimate LDL-C goal of <1.4 mmol/L (55 mg/dL) and a ≥50% reduction in LDL-C vs. baseline is recommended.	I	A
If the goals are not achieved with the maximum tolerated dose of a statin, combination with ezetimibe is recommended.	I	B
For secondary prevention patients not achieving their goals on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor is recommended.	I	A
For very-high-risk FH patients (that is, with ASCVD or with another major risk factor) who do not achieve their goals on a maximum tolerated dose of a statin and ezetimibe, combination therapy including a PCSK9 inhibitor is recommended.	I	C
Statin therapy is not recommended in premenopausal female patients who are considering pregnancy or are not using adequate contraception.	III	C
Recommendation for drug treatments of patients with hypertriglyceridaemia		
Statin treatment is recommended as the first drug of choice for reducing CVD risk in high-risk individuals with hypertriglyceridaemia [triglycerides >2.3 mmol/L (200 mg/dL)].	I	A
Recommendations for the treatment of dyslipidaemias in older people (≥70 years)		
Treatment with statins is recommended for older people with ASCVD in the same way as for younger patients.	I	A
It is recommended that the statin is started at a low dose if there is significant renal impairment and/or the potential for drug interactions.	I	C
Recommendation for the treatment of dyslipidaemias in diabetes mellitus		
In patients with type 2 DM at very high risk (e.g. with established ASCVD and/or severe TOD) intensive lipid-lowering therapy, ultimately aiming at ≥50% LDL-C reduction and an LDL-C of <1.4 mmol/L (55 mg/dL) is recommended.	I	A
In patients with type 2 DM >40 years at high risk, lipid-lowering treatment with an ultimate LDL-C goal of ≥50% LDL-C reduction and an LDL-C of <1.8 mmol/L (70 mg/dL) is recommended.	I	A
Recommendations for lipid management in patients with moderate-to-severe chronic kidney disease (Kidney Disease Outcomes Quality Initiative stages 3–5)		
The use of statins or statin/ezetimibe combination is recommended in patients with non-dialysis-dependent, stage 3–5 CKD.	I	A
In patients with dialysis-dependent CKD who are free of ASCVD, commencing statin therapy is not recommended.	III	A
Recommendations for the clinical management of hypertension		
Classification of BP		
It is recommended that BP should be classified as optimal, normal, high-normal, or grades 1–3 hypertension, according to office BP.	I	C
Diagnosis of hypertension		
It is recommended to base the diagnosis of hypertension on:		
• Repeated office BP measurements, on more than one visit, except when hypertension is severe (e.g. grade 3 and especially in high-risk patients) or	I	C
• Out-of-office BP measurement with ABPM and/or HBPM when feasible.	I	C
Assessment of HMOD		
To evaluate for the presence of HMOD, measurement of serum creatinine, eGFR, electrolytes, and ACR is recommended for all patients. A 12-lead ECG is recommended for all patients, and echocardiography is recommended for those with ECG abnormalities or signs/symptoms of LV dysfunction. Fundoscopy or retinal imaging is recommended for patients with grades 2 or 3 hypertension and all hypertensive patients with DM.	I	B
Thresholds for initiation of drug treatment of hypertension		
For grade 1 hypertension, treatment initiation based on absolute CVD risk, estimated lifetime benefit, and the presence of HMOD is recommended.	I	C
For patients with grade 2 hypertension or higher, drug treatment is recommended.	I	A
Office BP treatment targets		
It is recommended that the first objective of treatment is to lower BP to <140/90 mmHg in all patients, and that subsequent BP targets are tailored to age and specific comorbidities.	I	A

Continued

In treated patients aged 18–69 years, it is recommended that SBP should ultimately be lowered to a target range of 120–130 mmHg in most patients.	I	A
In treated patients aged ≥70 years, it is recommended that SBP should generally be targeted to <140 and down to 130 mmHg if tolerated.	I	A
In all treated patients, DBP is recommended to be lowered to <80 mmHg.	I	A
Treatment of hypertension: lifestyle interventions		
Lifestyle interventions are recommended for people with high-normal BP or higher.	I	A
Treatment of hypertension: drug treatment		
It is recommended to initiate antihypertensive treatment with a two-drug combination in most patients, preferably as a single-pill combination. Exceptions are frail older patients and those with low-risk, grade 1 hypertension (particularly if SBP <150 mmHg).	I	B
It is recommended that the preferred combinations include a RAS blocker (i.e. an ACE inhibitor or ARB) with a CCB or diuretic, but other combinations of the five major classes can be used (ACE inhibitor, ARB, beta-blocker, CCB, thiazide/thiazide-like diuretic).	I	A
It is recommended, if BP remains uncontrolled with a two-drug combination, that treatment be increased to a three-drug combination, usually a RAS blocker with a CCB and a diuretic, preferably as a single-pill combination.	I	A
It is recommended, if BP is not controlled by a three-drug combination, that treatment should be increased by the addition of spironolactone, or if not tolerated, other diuretics such as amiloride or higher doses of other diuretics, an alpha-blocker or beta-blocker, or clonidine.	I	B
The combination of two RAS blockers is not recommended.	III	A
Recommendations for the treatment of patients with diabetes mellitus		
Lifestyle		
Lifestyle changes including smoking cessation, a low saturated fat, high-fibre diet, aerobic PA, and strength training are recommended.	I	A
Reduction in energy intake is recommended to patients, to help achieve lower body weight or prevent or slow weight gain.	I	A
Glycaemia target		
A target HbA1c for the reduction of CVD risk and microvascular complications of DM of <7.0% (53 mmol/mol) is recommended for the majority of adults with either type 1 or type 2 DM.	I	A
Treatment of hyperglycaemia and ASCVD/cardiorenal risks		
Metformin is recommended as first-line therapy, following evaluation of renal function, in the majority of patients without previous ASCVD, CKD, or HF.	I	B
In persons with type 2 DM and ASCVD, the use of a GLP-1RA or SGLT2 inhibitor with proven outcome benefits is recommended to reduce CV and/or cardiorenal outcomes.	I	A
In patients with type 2 DM and CKD, the use of an SGLT2 inhibitor is recommended to improve CVD and/or cardiorenal outcomes.	I	A
In patients with type 2 DM and HFrEF, use of an SGLT2 inhibitor with proven outcome benefits is recommended to lessen HF hospitalizations and CV death.	I	A
Recommendations for antithrombotic therapy		
Aspirin 75–100 mg daily is recommended for secondary prevention of CVD.	I	A
Clopidogrel 75 mg daily is recommended as an alternative to aspirin in secondary prevention in case of aspirin intolerance.	I	B
Concomitant use of a proton pump inhibitor is recommended in patients receiving antiplatelet therapy who are at high risk of gastrointestinal bleeding.	I	B
Antiplatelet therapy is not recommended in individuals with low/moderate CV risk due to the increased risk of major bleeding.	III	A
Recommendations for cardiac rehabilitation		
Participation in a medically supervised, structured, comprehensive, multidisciplinary EBCR and prevention programme for patients after ASCVD events and/ or revascularization, and for patients with HF (mainly HFrEF), is recommended to improve patient outcomes.	I	A
Recommendation for policy interventions at the population level		
Putting in place measures to reduce air pollution, including reducing PM emission and gaseous pollutants, reducing the use of fossil fuels, and limiting carbon dioxide emissions, are recommended, to reduce CVD mortality and morbidity.	I	C

Continued

Recommendations for patients with coronary artery disease		
Aspirin 75–100 mg daily is recommended for patients with a previous myocardial infarction or revascularization.	I	A
In ACS, DAPT with a P2Y ₁₂ inhibitor in addition to aspirin is recommended for 12 months, unless there are contraindications such as excessive risk of bleeding.	I	A
In patients with CCS, clopidogrel 75 mg daily is recommended, in addition to aspirin, for 6 months following coronary stenting, irrespective of stent type, unless a shorter duration (1–3 months) is indicated due to risk or the occurrence of life-threatening bleeding.	I	A
ACE inhibitors (or ARB) are recommended if a patient has other conditions (e.g. HF, hypertension, or DM).	I	A
Beta-blockers are recommended in patients with LV dysfunction or systolic HF.	I	A
In patients with established ASCVD, oral lipid-lowering treatment with an ultimate LDL-C goal of <1.4 mmol/L (55 mg/dL) and a ≥50% reduction in LDL-C vs. baseline is recommended.	I	A
Recommendations regarding pharmacological and nonpharmacological interventions for patients with symptomatic (New York Heart Association class II–IV) heart failure with reduced ejection fraction (left ventricular ejection fraction <40%) with proven benefits on clinical outcomes, including cardiovascular morbidity and mortality		
It is recommended that patients with HF are enrolled in a comprehensive CR programme to reduce the risk of HF hospitalization and death.	I	A
EBCR is recommended in stable symptomatic patients with HFrEF to reduce the risk of HF hospitalization.	I	A
It is recommended to screen patients with HF for both CV and non-CV comorbidities, which, if present, should be treated, provided safe and effective interventions exist, not only to alleviate symptoms but also to improve prognosis.	I	A
An ACE inhibitor is recommended, in addition to a beta-blocker and an MRA, for patients with symptomatic HFrEF to reduce the risk of HF hospitalization and death.	I	A
A beta-blocker is recommended, in addition to an ACE inhibitor (or an ARNI) and an MRA, for patients with stable, symptomatic HFrEF to reduce the risk of HF hospitalization and death.	I	A
An MRA is recommended for patients with HFrEF already treated with an ACE inhibitor (or an ARNI) and a beta-blocker, to reduce the risk of HF hospitalization and death.	I	A
Sacubitril/valsartan is recommended as a replacement for an ACE inhibitor to reduce the risk of HF hospitalization and death in patients with HFrEF.	I	B
An ARB is recommended to reduce the risk of HF hospitalization or CV death in symptomatic patients with HFrEF who are unable to tolerate an ACE inhibitor and/or ARNI (patients should also receive a beta-blocker and an MRA).	I	B
Dapagliflozin or empagliflozin are recommended, in addition to optimal treatment of an ACE inhibitor (or ARNI), a beta-blocker, and an MRA, for patients with HFrEF to reduce the risk of HF hospitalization and death.	I	A
Diuretics are recommended in patients with HFrEF with signs and/or symptoms of congestion to reduce the risk of HF hospitalization.	I	C
Recommendations for patients with cerebrovascular disease		
In patients with a cerebrovascular event, improvement of lifestyle factors in addition to appropriate pharmacological management is recommended.	I	A
In patients with ischaemic stroke or TIA, prevention with antithrombotics is recommended; choice of antithrombotic depends on the mechanism of event. Use of an antiplatelet is recommended for patients with non-cardioembolic ischaemic stroke or TIA, and use of an anticoagulant is recommended in patients with cardioembolic ischaemic stroke or TIA.	I	A
In patients with non-cardioembolic ischaemic stroke or TIA, prevention with aspirin only, or dipyridamole plus aspirin, or clopidogrel alone is recommended.	I	A
In patients with stroke or TIA who have BP of 140/90 mmHg or higher, BP lowering is recommended.	I	A
Recommendations for patients with lower extremity artery disease: best medical therapy		
Smoking cessation is recommended in all patients with LEAD.	I	B
Healthy diet and PA are recommended for all patients with LEAD.	I	C
In patients with intermittent claudication:	I	A
• Supervised exercise training is recommended	I	A
• Non-supervised exercise training is recommended when supervised exercise training is not feasible or available.	I	C
Antiplatelet therapy is recommended in patients with symptomatic LEAD.	I	C
In patients with LEAD and hypertension, it is recommended to control BP at <140/90 mmHg.	I	A
In patients with LEAD and DM, strict glycaemic control is recommended.	I	A

Continued

Recommendations in patients with chronic kidney disease: best medical therapy		
Treatment with an ACE inhibitor or an ARB is recommended in patients with DM, hypertension, and albuminuria. These medications should be titrated to the highest approved dose that is tolerated.	I	B
Combination treatment with ACE inhibitors and ARBs is not recommended.	III	C
Recommendations for lifestyle interventions and management of risk factors and concomitant diseases in patients with atrial fibrillation		
Identification and management of risk factors and concomitant diseases are recommended to be considered an integral part of treatment.	I	B
Modification of unhealthy lifestyle and targeted therapy of intercurrent conditions is recommended to reduce AF burden and symptom severity.	I	B
Attention to good BP control is recommended in AF patients with hypertension to reduce AF recurrences and risk of stroke and bleeding.	I	B

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ABPM = ambulatory blood pressure monitoring; ACE = angiotensin-converting enzyme; ACR = albumin-to-creatinine ratio; ACS = acute coronary syndromes; AF = atrial fibrillation; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor neprilysin inhibitor; ASCVD = atherosclerotic cardiovascular disease; BP = blood pressure; CAC = coronary artery calcium; CCB = calcium channel blocker; CCS = chronic coronary syndromes; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; CR = cardiac rehabilitation; CV = cardiovascular; CVD = cardiovascular disease; DAPT = dual antiplatelet therapy; DBP = diastolic blood pressure; DM = diabetes mellitus; EBCR = exercise-based cardiac rehabilitation; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; FH = familial hypercholesterolaemia; GLP-1RA = glucagon-like peptide-1 receptor agonist; HbA1c = glycated haemoglobin; HBPM = home blood pressure monitoring; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; HMOD = hypertension-mediated organ damage; LDL-C = low-density lipoprotein cholesterol; LEAD = lower extremity artery disease; LV = left ventricular; MRA = mineralocorticoid receptor antagonist; PA = physical activity; PCSK9 = proprotein convertase subtilisin/kexin type 9; PM = particulate matter; RAS = renin-angiotensin system; SBP = systolic blood pressure; SCORE2 = Systematic Coronary Risk Estimation 2; SCORE2-OP = Systematic Coronary Risk Estimation 2-Older Persons; SGLT2 = sodium-glucose cotransporter 2; SNRI = serotonin-noradrenaline reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; TIA = transient ischaemic attack; TOD = target organ damage.

10. Quality indicators

Quality indicators (QIs) are tools that may be used to evaluate care quality, including that of processes of care and clinical outcomes.⁷³⁰ They may also serve as a mechanism for enhancing adherence to guideline recommendations, through quality assurance endeavours and benchmarking of care providers.⁸³¹ As such, the role of QIs in driving quality improvement is increasingly recognized and attracts interest from healthcare authorities, professional organizations, payers, and the public.⁸³²

The ESC recognizes the need for measuring and reporting the quality and outcomes of CV care. One aspect of this is the development and implementation of QIs for CVD. The methodology by which the ESC QIs are developed has been published⁸³² and, to date, a suite of QIs for an initial tranche of CV conditions has been produced.^{833,834} To facilitate quality improvement initiatives, the disease-specific ESC QIs are included in corresponding ESC Clinical Practice Guidelines.^{215,680} This is further enhanced by way of their integration into the EORP (EURObservational Research Programme) and the EuroHeart (European Unified Registries On Heart Care Evaluation and Randomized Trials) project.⁸³⁵

For CVD prevention, QIs are available for specific conditions, such as the management of high BP⁸³⁶ and secondary lipid prevention.⁸³⁷ However, a comprehensive set of QIs that encompasses the depth and breadth of CVD prevention is lacking. Such a set may evaluate the adoption of, and adherence to, the guideline recommendations provided in this document, and may be applied retrospectively to assess the delivery of evidence-based care. Thus, and in line with other ESC Clinical Practice Guidelines, the process of developing and defining QIs for CVD prevention has been initiated during the writing of this guideline and the results will be published in a separate document.

11 Supplementary data

[Supplementary data](#) with additional Supplementary Figures, Tables, and text complementing the full text are available on the European Heart Journal website and via the ESC website at <https://www.escardio.org/guidelines>.

12. Author information

Author/Task Force Member Affiliations: **Yvo M. Smulders**, Internal Medicine, Amsterdam University Medical Center, Amsterdam, Netherlands; **David Carballo**, Cardiology, Geneva University Hospitals, Geneva, Switzerland; **Konstantinos C. Koskinas**, Cardiology, Bern University Hospital – INSELSPIITAL, Bern, Switzerland; **Maria Bäck**, Unit of Physiotherapy, Department of Health, Medicine and Caring Sciences, Linköping University, Linköping, Sweden, and Department of Molecular and Clinical Medicine, Institute of Medicine, Sahlgrenska Academy, University of Gothenburg, and Sahlgrenska University Hospital, Gothenburg, Sweden; **Athanase Benetos**, Geriatric Department CHRU de Nancy and ISERM DCAC, Université de Lorraine, Nancy, France; **Alessandro Biffi**, Cardiology, FIMS and EFSMA, Rome, Italy; **José-Manuel Boavida**, APDP – Diabetes Portugal, IDF-E International Diabetes Federation – Europe, Lisbon, Portugal; **Davide Capodanno**, Cardiothoracic, Vascular and Transplants, Policlinico “G. Rodolico-San Marco”, University of Catania, Catania, Italy; **Bernard Cosyns**, Cardiology, Centrum voor Hart en vaatziekte (CHVZ) Universitair Ziekenhuis Brussel, Brussels, Belgium; **Carolyn A. Crawford**, (Northern Ireland), ESC Patient Forum, Sophia Antipolis, France; **Constantinos H. Davos**, Cardiovascular Research Laboratory, Biomedical Research Foundation, Academy of

Athens, Athens, Greece; **Ileana Desormais**, INSERM, Univ. Limoges, CHU Limoges, IRD, U1094 Tropical Neuroepidemiology, GEIST, Limoges, France; **Emanuele Di Angelantonio**, Department of Public Health and Primary Care, University of Cambridge, Cambridge, UK; **Oscar H. Franco Duran**, ISPM Institute of Social and Preventive Medicine, University of Bern, Bern, Switzerland; **Sigrun Halvorsen**, Department of Cardiology, Oslo University Hospital Ullevaal, Oslo, Norway; **F. D. Richard Hobbs**, NDPCHS, University of Oxford, Oxford, UK; **Monika Hollander**, Julius Center for Health Sciences and Primary Care, UMC Utrecht, Utrecht University, Utrecht, Netherlands; **Ewa A. Jankowska**, Department of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland; **Matthias Michal**, Department of Psychosomatic Medicine and Psychotherapy, University Medical Center Mainz, Mainz, Germany; **Simona Sacco**, Department of Applied Clinical and Biotechnological Sciences, University of L'Aquila, L'Aquila, Italy; **Naveed Sattar**, Institute of Cardiovascular and Medical Sciences, University of Glasgow, Glasgow, UK; **Lale Tokgozoglu**, Cardiology, Hacettepe University, Ankara, Turkey; **Serena Tonstad**, Preventive Cardiology, Oslo University Hospital, Oslo, Norway; **Konstantinos P. Tsoufis**, First Cardiology Clinic, Medical School, National and Kapodistrian University, Hippokraton Hospital, Athens, Greece; **Ineke van Dis**, European Heart Network, Brussels, Belgium; **Isabelle C. van Gelder**, Cardiology, University of Groningen, University Medical Center Groningen; Groningen, Netherlands; **Christoph Wanner**, Department of Nephrology, University Würzburg, Germany; **Bryan Williams**, Institute of Cardiovascular Science, University College London, London, UK

13. Appendix

ESC Scientific Document Group

Includes Document Reviewers and ESC National Cardiac Societies.

Document Reviewers: Guy De Backer (CPG Review Coordinator) (Belgium), Vera Regitz-Zagrosek (CPG Review Coordinator) (Germany), Anne Hege Aamodt (Norway), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Christian Albus (Germany), Riccardo Asteggiano (Italy), Magnus Bäck (Sweden), Michael A. Borger (Germany), Carlos Brotons (Spain), Jelena Čelutkienė (Lithuania), Renata Cifkova (Czech Republic), Maja Cikes (Croatia), Francesco Cosentino (Italy), Nikolaos Dargatzis (Germany), Tine De Backer (Belgium), Dirk De Bacquer (Belgium), Victoria Delgado (Netherlands), Hester Den Ruijter (Netherlands), Paul Dendale (Belgium), Heinz Drexel (Austria), Volkmar Falk (Germany), Laurent Fauchier (France), Brian A. Ference¹ (United Kingdom), Jean Ferrières (France), Marc Ferrini (France), Miles Fisher² (United Kingdom), Danilo Fliser (Germany), Zlatko Fras (Slovenia), Dan Gaita (Romania), Simona Giampaoli (Italy), Stephan Gielen (Germany), Ian Graham (Ireland), Catriona Jennings (Ireland), Torben Jorgensen (Denmark), Alexandra Kautzky-Willer (Austria), Maryam Kavousi (Netherlands), Wolfgang Koenig (Germany), Aleksandra Konradi (Russia), Dipak Kotecha (United Kingdom), Ulf Landmesser (Germany), Madalena Lettino (Italy), Basil S. Lewis (Israel), Aleš Linhart (Czech Republic), Maja-Lisa Løchen (Norway), Konstantinos Makrilakis (Greece), Giuseppe Mancina (Italy), Pedro Marques-Vidal

(Switzerland), John William McEvoy (Ireland), Paul McGreavy (United Kingdom), Bela Merkely (Hungary), Lis Neubeck (United Kingdom), Jens Cosedis Nielsen (Denmark), Joep Perk (Sweden), Steffen E. Petersen (United Kingdom), Anna Sonia Petronio (Italy), Massimo Piepoli (Italy), Nana Goar Pogossova (Russia), Eva Irene Bossano Prescott (Denmark), Kausik K. Ray (United Kingdom), Zeljko Reiner (Croatia), Dimitrios J. Richter (Greece), Lars Rydén (Sweden), Evgeny Shlyakhto (Russia), Marta Sitges (Spain), Miguel Sousa-Uva (Portugal), Isabella Sudano (Switzerland), Monica Tiberi (Italy), Rhian M. Touyz (United Kingdom), Andrea Ungar (Italy), W.M. Monique Verschuren (Netherlands), Olov Wiklund (Sweden), David Wood (United Kingdom/Ireland), Jose Luis Zamorano (Spain).

ESC National Cardiac Societies actively involved in the review process of the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice:

Algeria: Algerian Society of Cardiology, Naima Hammoudi; **Armenia:** Armenian Cardiologists Association, Parounak Zelveian; **Austria:** Austrian Society of Cardiology, Peter Siostrzonek; **Azerbaijan:** Azerbaijan Society of Cardiology, Elman Alakbarov; **Belarus:** Belorussian Scientific Society of Cardiologists, Olga Pavlova; **Belgium:** Belgian Society of Cardiology, Johan De Sutter; **Bosnia and Herzegovina:** Association of Cardiologists of Bosnia and Herzegovina, Mirza Dilić; **Bulgaria:** Bulgarian Society of Cardiology, Nina Gotcheva; **Croatia:** Croatian Cardiac Society, Bosko Skoric; **Cyprus:** Cyprus Society of Cardiology, Hera Heracleous Moustra; **Czech Republic:** Czech Society of Cardiology, Renata Cifkova; **Denmark:** Danish Society of Cardiology, Ann Bovin; **Egypt:** Egyptian Society of Cardiology, Bassem Zarif; **Estonia:** Estonian Society of Cardiology, Margus Viigimaa; **Finland:** Finnish Cardiac Society, Anna-Mari Hekkala; **France:** French Society of Cardiology, Serge Kownator; **Georgia:** Georgian Society of Cardiology, Zurab Pagava; **Germany:** German Cardiac Society, Ulf Landmesser; **Greece:** Hellenic Society of Cardiology, Harry Grassos; **Hungary:** Hungarian Society of Cardiology, Eszter Szabados; **Iceland:** Icelandic Society of Cardiology, Karl Andersen; **Ireland:** Irish Cardiac Society, John William McEvoy; **Israel:** Israel Heart Society, Barak Zafrir; **Italy:** Italian Federation of Cardiology, Francesco Barilla; **Kosovo (Republic of):** Kosovo Society of Cardiology, Pranvera Ibrahim; **Kyrgyzstan:** Kyrgyz Society of Cardiology, Erkin Mirrakhimov; **Latvia:** Latvian Society of Cardiology, Iveta Mintale; **Lebanon:** Lebanese Society of Cardiology, Samir Arnaout; **Lithuania:** Lithuanian Society of Cardiology, Rimvydas Šlapikas; **Luxembourg:** Luxembourg Society of Cardiology, Cristiana Banu; **Malta:** Maltese Cardiac Society, Mark Abela; **Moldova (Republic of):** Moldavian Society of Cardiology, Victor Rudi; **Montenegro:** Montenegro Society of Cardiology, Aneta Boskovic; **Morocco:** Moroccan Society of Cardiology, Mohamed Alami; **Netherlands:** Netherlands Society of Cardiology, Hareld M.C. Kemps; **North Macedonia:** North Macedonian Society of Cardiology, Marijan Bosevski; **Norway:** Norwegian Society of Cardiology, Erik Ekker Solberg; **Poland:** Polish Cardiac Society, Tomasz Zdrojewski; **Portugal:** Portuguese Society of Cardiology, Carlos Rabaçal; **Romania:** Romanian Society of Cardiology, Dan Gaita; **Russian Federation:** Russian Society of Cardiology, Yury Belenkov; **San Marino:** San Marino Society of

Cardiology, Luca Bertelli; **Serbia:** Cardiology Society of Serbia, Vojislav Giga; **Slovakia:** Slovak Society of Cardiology, Daniel Pella; **Slovenia:** Slovenian Society of Cardiology, Zlatko Fras; **Spain:** Spanish Society of Cardiology, Regina Dalmau; **Sweden:** Swedish Society of Cardiology, Anna Kiessling; **Switzerland:** Swiss Society of Cardiology, Otmar Pfister; **Syrian Arab Republic:** Syrian Cardiovascular Association, Yassin Bani Marjeh; **Tunisia:** Tunisian Society of Cardiology and Cardio-Vascular Surgery, Salem Abdessalem; **Turkey:** Turkish Society of Cardiology, Oner Ozdogan; **Ukraine:** Ukrainian Association of Cardiology, Elena Nesukay; **United Kingdom of Great Britain and Northern Ireland:** British Cardiovascular Society, Riyaz Patel; **Uzbekistan:** Association of Cardiologists of Uzbekistan, Guzall Mullabayeva.

ESC Clinical Practice Guidelines Committee (CPG):

Colin Baigent (Chairperson) (United Kingdom), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Sotiris Antoniou (United Kingdom), Elena Arbelo (Spain), Riccardo Asteggiano (Italy), Andreas Baumbach (United Kingdom), Michael A. Borger (Germany), Jelena Čelutkienė (Lithuania), Maja Cikes (Croatia), Jean-Philippe Collet (France), Volkmar Falk (Germany), Laurent Fauchier (France), Chris P. Gale (United Kingdom), Sigrun Halvorsen (Norway), Bernard Jung (France), Tiny Jaarsma (Sweden), Aleksandra Konradi (Russia), Konstantinos C. Koskinas (Switzerland), Dipak Kotecha (United Kingdom), Ulf Landmesser (Germany), Basil S. Lewis (Israel), Aleš Linhart (Czech Republic), Maja-Lisa Løchen (Norway), Jens Cosedis Nielsen (Denmark), Steffen E. Petersen (United Kingdom), Eva Irene Bossano Prescott (Denmark), Amina Rakisheva (Kazakhstan), Marta Sitges (Spain), Rhian M. Touyz (United Kingdom)

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